NILAMBARA ELECTRICALS

Complete User Manual

Repair Shop Management System

Version 1.0 | Generated 21 October 2026

Employee and administrator operating guide

482/A/3, Colombo Road, Kaduwela, Sri Lanka.

Currency: LKR | Timezone: Asia/Colombo

Based on the existing implementation, the original BRD and isolated test evidence. Business-owner acceptance remains pending.

Contents

Select a chapter to navigate. PDF footers show physical page numbers.

1. Introduction
2. Getting started
3. Dashboard
4. Customer and company management
5. Repair receipt management
6. Repair job management
7. Quotation management
8. Invoice management
9. Payment management
10. Credit management
11. Stock and inventory
12. Supplier and purchase management
13. POS system
14. Reports
15. Employee management and permissions
16. System settings
17. Print template management
18. Backup and restore
19. Common errors and troubleshooting
20. Full business workflow
21. Terminology and navigation tips
22. Current verification and limitations
23. System relationship diagrams
24. Real application screenshot atlas

Document control	Value
Version / generation	1.0 / 21 October 2026
Scope	Detailed implemented workflows for all six roles
Basis	Original BRD (20 pages, 27 sections); source/route/schema inspection; fresh Step 13 JUnit and native checks
Author / approval	Codex documentation runner / business-owner approval pending
Verification boundary	Automated scenario evidence and real fixture screenshots; no production deployment or live business-data writes
Companion sources	documentation/sources/; documentation/evidence/; documentation/screenshots/

1. Introduction

The system supports appliance/motor repairs, spare-parts retail, shared inventory, suppliers, billing, credit and administrative controls for Nilambara Electricals, 482/A/3, Colombo Road, Kaduwela, Sri Lanka. Currency is LKR; timezone is Asia/Colombo. Intended users are Administrator, Receptionist, Technician, Cashier, Storekeeper and Manager. Browser access, a running PHP application and the existing MySQL-compatible server are required. This manual describes implementation observed on 20 October 2026; owner acceptance is not implied. Training and screenshots use isolated test fixtures, never genuine business records.

2. Getting started

On the existing computer, the operator starts the existing MySQL server and Laravel development server. From the actual application folder, set PHPRC to the repository .tools/php.ini and run php artisan serve --host=127.0.0.1 --port=8000 --no-reload. Open http://127.0.0.1:8000/login (or your operator-provided address). The documentation screenshots used an isolated server on port 8087; do not substitute that temporary port for the shop address. Enter your assigned username and password; use the sidebar and page breadcrumbs to navigate. Logout with the sign-out action. Current configured session lifetime is 120 minutes of inactivity; end-to-end wall-clock expiry was not exercised in this task. Disabled accounts lose access. There is no general self-service account settings/password recovery screen; an administrator manages linked user details/passwords. A forbidden action requires permission review. Do not share logins.

```
cd D:\xampp\htdocs\Nilambara_Electricals\nilambara-repair-management
$env:PHPRC = (Resolve-Path ..\.tools\php.ini).Path
php artisan serve --host=127.0.0.1 --port=8000 --no-reload
```

3. Dashboard

Open Dashboard and choose Today, Yesterday, This Week, This Month, This Year or Custom with valid dates, then Apply. Jobs received counts the selected received-date cohort. Active excludes Delivered, Cancelled and Unable to Repair. Pending Inspection, Under Repair, Waiting for Parts, Completed, Ready for Collection and Delivered count the current states of jobs received in that cohort; these are not necessarily jobs whose status changed during the period. Technicians are scoped to assigned jobs. Repair/POS sales sum finalized documents by finalized date. Cash-term sales are not the same as cash actually collected. Payments collected are net receipts after relevant reversals; advances/credit collections are separate categories. Outstanding customer credit is positive cumulative account balance at period end. Low/out stock uses active parts/default locations and reorder thresholds. Dashboard stock valuation currently displays Incomplete; use the dedicated Valuation report for known-cost detail. Recent jobs, invoices and POS sales reflect the corresponding selected cohort/date. Visible cards depend on grants.

Operational reference	Implementation
Required inputs	Period or custom date range
Validation / restrictions	Custom end date cannot precede start. Cards/links are permission filtered.
Effect / balance changes	Reads period/cohort statistics; it does not create financial or stock transactions.
Navigation	/dashboard
Screenshot reference	dashboard; see chapter 24
Evidence	Step11DashboardTest|DashboardTest|Step12AuditTest; Step 13 test-case appendix

3.1. Actions in Business dashboard

- Date filters and repair metrics: Select Today, Yesterday, This Week, This Month, This Year or Custom; choose both custom dates; Apply. Received-date cohort job counts, with assigned-only technician data.
- Sales, credit and stock indicators: Review visible cards and recent documents; compare the same period in reports. Finalized sales, net collections, customer account credit and low-stock indicators; cards depend on grants.

4. Customer and company management

Open Customers and search first. To register an individual choose Individual and enter customer name, address, valid mobile and registration date. To register a company choose Company and enter company name, address, mobile and date; contact person/email are optional. Customer code is generated. Save and inspect the profile. Search name/mobile/code/company, use the filters and pagination, then Edit to correct contacts. Activate requires update access; deactivate requires its separate grant and preserves references. Verified pre-system opening balances may be entered only at registration by authorized financial staff; editing cannot rewrite them. Customer

profiles link job, invoice/payment and statement histories. In Statement select dates and review brought-forward balance, account movements and closing balance before print/PDF. Required fields, invalid email/phone and unauthorized financial fields return validation errors.

Operational reference	Implementation
Required inputs	Type; name for individual or company name for company; address; mobile; registration date. Contact person, alternate phone, email and notes are optional.
Validation / restrictions	Valid contact format and email; financial opening balance needs customers.statement. Opening balance and generated ID are immutable after creation.
Effect / balance changes	Creates/edits the customer master; nonzero opening balance posts one ledger entry. Deactivation preserves references.
Navigation	/customers; /customers/create; /customers/{customer}/deactivate; /customers/{customer}/edit; /customers/{customer}/statement
Screenshot reference	customers; see chapter 24
Evidence	CustomerManagementTest; Step 13 test-case appendix

4.1. Actions in Customer and company management

- Register individual: Choose Individual; enter name, address, mobile and registration date; review and Save. Generated CUS reference and individual master.
- Register company: Choose Company; enter company name, address, mobile and registration date; optional contact person; Save. Company master with searchable company name.
- Search and profile: Search name, customer code, company or mobile; open the matching profile. Paginated matches and linked history; search excludes inactive suggestions.
- Edit customer: Open the correct profile; Edit; correct contacts; save without changing opening balance. Updated master with immutable ID/opening history.
- Activate or deactivate customer: Review pending work; use Deactivate, or Activate when authorized, on the saved customer. Status change preserving transaction references.
- Opening account balance: On first registration only, enter a verified pre-system balance; otherwise leave zero. One nonzero opening ledger entry, not cash received.
- Customer statement and PDF: Open Statement; choose dates; review opening, movements and closing; print or PDF. Read-only statement with template/history; no new payment.

5. Repair receipt management

Open New repair receipt. Select an active customer and receiving employee; enter unique manual Job Number, job type, item name, brand or model and nonnegative labour cost. Record reported fault, accessories, physical condition, visible damage and special notes accurately even where optional. Brand/model cannot both be absent.

Enter an advance only when received and select its method; verify method-specific acceptance in the saved payment. Save or Save/Print. The transaction creates receipt, job, initial Received history and job card together; failure rolls the transaction back. Automatic JOB-00001-style reference comes from the database ID and is separate from the immutable staff-entered Job Number. Search receipts by text/date/type/status/receiver. Give the A5 receipt to the customer; subsequent copies use Duplicate. A5 PDF is available on saved documents and creates print/audit history.

Operational reference	Implementation
Required inputs	Active customer; unique manual Job Number; job type; item name; brand or model; labour cost; active receiving employee. Fault/accessories/condition/notes and advance are optional.
Validation / restrictions	Brand is required without model and model without brand. Labour cost is nonnegative. A positive advance needs a supported payment method.
Effect / balance changes	Creates receipt, repair job, initial Received status and job card atomically, with any advance. Automatic JOB reference differs from immutable manual number.
Navigation	/repair-receipts; /repair-receipts/create; /repair-receipts/{receipt}; /repair-receipts/{receipt}/print
Screenshot reference	receipt-create; see chapter 24
Evidence	RepairReceiptAndJobManagementTest|StockPurchasingWorkflowTest; Step 13 test-case appendix

5.1. Actions in Repair receipt management

- Customer selection and appliance intake: Select active customer and receiver; enter Job Number, type, appliance, brand/model and labour cost; record fault/condition; Save. Receipt, job and job card created together.
- Automatic and manual identifiers: Read automatic JOB reference and manual Job Number separately; use either appropriate search field. Unique immutable manual number alongside database-derived reference.
- Intake advance: Record only money actually received and its method while receiving; verify saved receipt before retry. Advance/payment allocation included atomically with intake.
- Receipt list and filters: Filter by text/date/status/job type/receiver; open the actual receiving record. Paginated intake records and linked job history.
- Original, duplicate and A5 PDF: Open saved receipt; Preview/Print or PDF; use duplicate action for later copies. A5 customer document, duplicate marker and print audit.

6. Repair job management

Open Repair Jobs, Job Status or My Assigned Jobs. Inspect the customer/item/intake and histories before acting. A manager uses Assign to select an active technician; only assigned technicians can inspect/repair that record under the policy. Record inspection findings, repair description and actual costs; selling charge is separate.

Choose the actual current status and add notes. An issued quotation must be approved before Under Repair or issue of parts. Completion and delivery fill timestamps when absent; cancellation requires a reason. Printing A6 job cards is a separate permission; preview/PDF the saved card. The 17 statuses below are actual enum values and practical meanings. The implementation does not enforce a complete sequential 17-step transition graph; do not infer an automatic payment/delivery approval from a status label.

Operational reference	Implementation
Required inputs	Existing job; action-specific technician, inspection/repair text, costs, status and notes
Validation / restrictions	Technicians are restricted to assigned jobs. Entering Under Repair with an issued quotation requires approval. Cancellation requires its reason.
Effect / balance changes	Assignments, inspections and status changes retain history. Completion/delivery timestamps are recorded; a status alone is not a new receipt/payment.
Navigation	/repair-jobs; /repair-jobs/{job}/assign; /repair-jobs/{job}/inspection; /repair-jobs/{job}/job-card; /repair-jobs/{job}/labour-charge; /repair-jobs/{job}/repair; /repair-jobs/{job}/status
Screenshot reference	job-detail; see chapter 24
Evidence	RepairReceiptAndJobManagementTest|QuotationApprovalEnforcementTest|StockPurchasingWorkflowTest; Step 13 test-case appendix

6.1. Actions in Repair job management

- Job list, details and assigned jobs: Open Repair Jobs or My Assigned Jobs; search the reference and open the record. Policy-filtered job details and links.
- Technician assignment: Open job Assign; select an active technician; save the assignment. Assignment history and technician link; no stock/payment.
- Inspection: On an accessible assigned job, enter the inspection details and save. Retained inspection record and audit.
- Repair details and costs: Record actual repair description, parts/labour costs and completion data on the job. Repair evidence and actual costs; cost is distinct from customer price.
- Status and status history: Select the actual current status; add notes/reason as required; save and review history. Previous/new status and actor retained; Under Repair approval gate.
- Completion and delivery: Mark Repair Completed when work is complete; record Delivered only on actual handover. Completion/delivery timestamps filled when absent; no extra receipt.
- Unquoted labour selling charge: For an unquoted unbilled job, enter the labour selling charge using its dedicated form. Billable job charge; no silent quoted/billed price change.
- A6 job card: Open saved job card; print A6 or download PDF; inspect signatures and writing area. Internal card and print/export history.

6.2. All 17 implemented status labels

Status	Practical meaning
Received	Item accepted at intake; automatic initial history.
Pending Inspection	Waiting for inspection; assign/queue work.
Under Inspection	Technician is examining the item; record findings.
Quotation Pending	Estimate needs preparation; quotation itself does not use stock.
Waiting for Customer Approval	Issued estimate awaits actual customer decision; approval gates repair/issue.
Customer Contact Failed	Customer could not be contacted; record attempts/notes.
Advance Payment Pending	Awaiting agreed advance; status does not create payment.
Waiting for Spare Parts	Parts not available; requests do not reserve stock.
Under Repair	Repair in progress; issued quotation must have approval.
Price Modification	Price discussion/change required; use revision/approval rather than silent repricing.
Re – Assembling	Item being reassembled; actual enum uses an en dash in its label.
Repair Completed	Repair work completed; completion timestamp recorded when missing.
Ready for Collection	Ready for customer pickup; verify invoice/collection separately.
Customer Contact Failed After Ready	Unable to reach customer after ready; record contact attempts.
Delivered	Actual handover recorded; delivery timestamp filled when missing.
Cancelled	Job cancelled with required reason; history retained.
Unable to Repair	Repair cannot be completed; record findings and resolve any financial/stock consequences separately.

7. Quotation management

Open Quotations > Create. Select the correct job, quotation date, valid-until date, proposed repair and labour selling charge. Add at least one part/description line with positive quantity; selected parts supply defaults, while a manual line needs description/unit price. Review decimal line totals, parts total and labour total; Save draft, Preview or Issue. Editable drafts may be revised; issued changes use Create revision so older versions/decisions remain. Preview or download A4/A5 PDF, with duplicate output when appropriate. An authorized staff member

records actual Approved/Rejected decision with date/time, contact name and in-person/telephone/email/other method. This records communication, not an electronic signature or verified identity. Approval transfers selling totals without stock/payment. Rejection follows the implemented status handling and retains decision history.

Operational reference	Implementation
Required inputs	Repair job; quotation date; valid-until date; proposed repair; labour charge; at least one estimate line with positive quantity and price/part
Validation / restrictions	Validity cannot precede quotation date. Draft/issued/approved states constrain edits. Approval/rejection requires contact name, decision date not in future and method (in person, telephone, email or other).
Effect / balance changes	Drafts/issuance preserve document snapshots and history. Approval transfers agreed selling charges; no quotation posts stock or receives cash.
Navigation	/quotations/create; /quotations/{quotation}/decision; /quotations/{quotation}/edit; /quotations/{quotation}/issue; /quotations/{quotation}/print; /quotations/{quotation}/revisions
Screenshot reference	quotation-approval; see chapter 24
Evidence	QuotationManagementTest|QuotationApprovalEnforcementTest |StockPurchasingWorkflowTest; Step 13 test-case appendix

7.1. Actions in Quotation management and decisions

- Create estimates and draft: Select job; enter date, validity, proposed repair, labour and at least one part/description line; Save draft. Decimal line totals and draft history; no stock deduction.
- Edit draft: Open editable draft; update allowed estimate lines/dates; Save. Validated estimate revision while draft is editable.
- Issue quotation: Review customer/job, terms and total; issue the draft for customer decision. Issued snapshot and waiting-approval workflow; no payment.
- Customer approval: On issued quotation record Approved, actual date, contact name and communication method. Approval history and agreed job selling totals; no stock/payment.
- Customer rejection: Record Rejected with the actual contact/method and explanatory notes. Append-only decision and implementation-specific job status.
- Revision and history: Use Create revision when an issued estimate needs change; inspect prior versions and obtain a new decision. Old quotation/decisions retained; revised estimate not silently approved.
- A4/A5 print, duplicate and PDF: Preview the saved quotation; choose A4/A5 PDF or marked duplicate as available. Quoted values and print history; no stock or collection.

8. Invoice management

Open Create Invoice. Choose active customer and 1-30 distinct eligible jobs from that same customer; already-billed or ineligible/unapproved work is refused. Review job-wise parts/labour/approved quotation amounts and advances. Enter invoice date, cash/credit terms and optional due date, collector, vehicle, delivery information/date and notes. Preview or save draft, then Issue when reviewed. A draft reserves job links; editing may append jobs but cannot remove original links or silently reprice them. Issue posts the customer liability and allocates eligible advances once. Printing is A5 with marked duplicate and PDF options under policy permissions. Amount due is derived from invoice amount and net allocations; changing terms does not receive money. Invoice delivery metadata is separate from job Delivered status.

Operational reference	Implementation
Required inputs	Customer; 1-30 distinct eligible jobs for that customer; invoice date; cash/credit terms; draft/issue/preview intent. Due date, collection/vehicle/delivery details and notes are optional.
Validation / restrictions	Jobs cannot belong to different customers or be billed twice. Approved quotation/billable job state is checked. Issued invoices cannot use draft editing; linked draft jobs cannot be removed.
Effect / balance changes	Draft reserves job links; issuance posts customer liability and allocates eligible advances once. Job amounts and original histories are retained.
Navigation	/invoices; /invoices/create; /invoices/{invoice}; /invoices/{invoice}/cancel; /invoices/{invoice}/issue; /invoices/{invoice}/print
Screenshot reference	invoice-create; see chapter 24
Evidence	InvoiceManagementTest|PaymentManagementTest|StockPurchasingWorkflowTest; Step 13 test-case appendix

8.1. Actions in Invoice management

- Create and select multiple jobs: Choose customer, select 1-30 distinct eligible same-customer jobs, set date/terms and review job-wise charges. Draft links reserve jobs against duplicate billing.
- Advances and invoice totals: Review line totals, linked job advances, paid and outstanding before taking money. Derived balance; allocations prevent counting advances twice.
- Draft edit and issue: Review draft; append only eligible jobs if required; issue the invoice once. Issued liability and snapshot; earlier draft links cannot be removed.
- Delivery information: Record collector, vehicle, delivery date/information when known in the invoice form. Saved delivery metadata, separate from a job's status history.
- A5 print, duplicate and PDF: Use the issued invoice's Preview/Print, duplicate or PDF action; verify customer/job totals. A5 document and history; export additionally needs policy permission.

- Authorized cancellation: Inspect financial state; use Cancel only if permitted and provide a reason of at least ten characters. Retained original and cancellation history; paid allocations constrain action.

9. Payment management

Open Record payment. Select Invoice payment or Advance. For an invoice choose the saved invoice; for an advance select the customer and optional job, not an invoice. Enter actual positive two-decimal amount and payment date. Cash, bank transfer and cheque have their displayed reference/bank/date fields; cheque date belongs only to cheque payments. Receiver is the signed-in active employee unless receive-for-others is granted. Save once and review allocations. Partial payments may repeat as money is received. Multiple payments can settle an invoice; the UI receives against one invoice per entry, so a multi-invoice allocation form is Not Implemented even though allocation services support multiple records. Print/PDF/duplicate the saved payment receipt. Authorized Refund and Accounting reversal are distinct; retain original, provide reason and record an actual refund method where appropriate. Reversals release allocations and affect collection/account history without deleting the original sale.

Operational reference	Implementation
Required inputs	Invoice or customer/job advance target; positive two-decimal amount; payment date; method; method-specific bank/cheque references; notes if needed
Validation / restrictions	Invoice and advance targets cannot be mixed. Cannot overpay target. Receiver defaults to active signed-in employee; receiving for others requires separate permission.
Effect / balance changes	Posts one receipt and allocation/ledger movement atomically. Partial payments update derived balance. Refund and accounting reversal have distinct effects.
Navigation	/payments/create; /payments/{payment}; /payments/{payment}/reverse
Screenshot reference	payment-create; see chapter 24
Evidence	PaymentManagementTest|LatestPurchaseCostTest|StockPurchasingWorkflowTest; Step 13 test-case appendix

9.1. Actions in Payment management

- Cash receipt: Choose Invoice payment, saved invoice, actual cash amount and date; submit once. Payment, allocation and ledger entry atomically.
- Bank transfer receipt: Select Bank transfer and enter actual amount/date and required reference details. Saved transfer payment and allocation; no invented funds.
- Cheque receipt: Choose Cheque and complete cheque date/reference/bank fields as shown. Recorded cheque method; recording is not a bank-clearance guarantee.
- Customer or job advance: Select Advance and correct customer; optionally target a job; do not also select an invoice. Available advance allocated later without another cash receipt.

- Partial and repeated invoice settlements: Record the genuine partial amount; verify outstanding; repeat only when another payment occurs. One invoice can have many payments. UI settles one invoice per entry.
- Payment receipt and history: Open saved payment, inspect allocations/reversals, then print, duplicate or download where permitted. Saved payment receipt and append-only history.
- Refund and accounting reversal: Open original payment; choose the action matching refund versus bookkeeping error; enter reason/method and confirm. Original retained, allocations released and ledger adjusted; collections versus sales remain distinct.

10. Credit management

Open Credit Management to inspect finalized credit invoices with outstanding balances, due dates and settlement history. Collect chooses the existing invoice and payment form; receive the actual partial/full amount and confirm updated outstanding. Do not issue another invoice merely to collect an old debt. Cancelled invoices are excluded. Authorized users review customer statements and Customer Credit/Overdue reports by date/customer, then export if granted. Cashier credit access does not imply permission for statements, reversals or financial analysis.

Operational reference	Implementation
Required inputs	Customer, date range, outstanding invoice and actual collection amount/method
Validation / restrictions	Requires credit access; collection also requires payment permissions. Cancelled invoices are excluded.
Effect / balance changes	Lists existing debt and settlement history. Collecting credit allocates a payment, not a second sale.
Navigation	/credit-invoices; /credit-invoices/{invoice}/collect; /reports
Screenshot reference	credit; see chapter 24
Evidence	PaymentManagementTest|ReportsTest; Step 13 test-case appendix

10.1. Actions in Credit management

- Outstanding credit invoices: Open Credit Management; filter and review actual outstanding invoices. Unpaid/partial finalized credit; cancelled invoices excluded.
- Credit collection: Select outstanding invoice; enter actual settlement in the payment form and inspect updated balance. Payment allocation against existing sale, not new revenue.
- Credit statements and reports: Choose Customer Credit/Overdue or Customer Statement; apply appropriate customer/dates. Authorized debt summaries; export is separately restricted.

11. Stock and inventory

Under Stock & Purchasing open Inventory > Register part. Enter unique SKU, name, unit/precision, selling price, location and reorder thresholds; barcode/category/brand/preferred supplier are optional. Register permitted references if needed. Opening quantity posts once; provide opening cost only when known. Edit allowed details later, but unit is locked after stock movements. Stock is received through purchase GRNs. Technicians request parts against accessible jobs; a request neither reserves nor deducts stock. Authorized issuers check approval and availability, then issue; POS and repairs use the same locked balance. Before billing an unissued request can be cancelled or an entire unused resalable issue returned. Partial issue return is Not Implemented. For corrections/damage use signed quantity/location/reason; damaged stock is removed. Physical Count records expected/actual and posts the difference. Check movement history and Low/Out report/dashboard alerts. No negative available stock is permitted.

Operational reference	Implementation
Required inputs	SKU/name/unit/price/location; optional barcode/category/brand/supplier/opening cost; movement quantity and reason for stock controls
Validation / restrictions	Unique SKU and nonnull barcode; unit precision; no negative available stock. Unit cannot change after movements. Issue must be authorized and job open/unbilled.
Effect / balance changes	Opening/GRN movements increase stock; issues/POS/damage decrease it. Requests and quotations do not reserve/deduct stock. Counts post only actual-minus-expected.
Navigation	/inventory; /inventory/create; /inventory/{part}; /inventory/{part}/edit; /inventory/{part}/{operation}; /purchases/{purchase}/receive; /reports/{report}; /stock-issues; /stock-issues/{usage}/cancel; /stock-issues/{usage}/issue
Screenshot reference	inventory; see chapter 24
Evidence	StockPurchasingWorkflowTest|LatestPurchaseCostTest|POSManagementTest; Step 13 test-case appendix

11.1. Actions in Stock and inventory

- Register parts, SKU and barcode: Register part with unique SKU/name/unit/location and price; optionally barcode/category/brand/preferred supplier. Part master; opening quantity/cost creates movement only when entered.
- Edit part and references: Open the correct part, edit permitted details, or create an allowed unit/location/brand/category reference. Updated master; unit frozen after movement; reference grants checked.
- Opening stock and known cost: At registration record actual opening quantity once and known unit cost only if supported by records. Opening movement; unknown cost stays unknown rather than zero.
- Stock receiving: Receive the actual purchase line quantity/location through a GRN. Positive source-linked movement and latest cost source.

- Request repair parts: Open Part Requests, choose accessible job, part, location, quantity and notes; submit. Request only, with no reservation/deduction.
- Approve and issue repair parts: Check request, available stock and customer approval; issue once. Negative movement, cost snapshot and protected job selling charges.
- Cancel request or full unused issue: Before billing cancel unissued request, or physically receive the entire unused resalable issued quantity and give a reason. Original retained; full linked return reverses issue. Partial issue return unavailable.
- Correction and damage: Open part; choose correction/damaged, correct location and signed quantity/reason; confirm physical facts. Correction/damage movement; negative availability refused.
- Physical stock count: Count location physically; enter actual count and reason, then review expected-versus-actual difference. Count history and difference movement only.
- Low stock and movements: Search part; inspect movements; open Stock report or dashboard alerts for low/out states. Read-only movement history and threshold alerts.
- Latest purchase valuation: Open Valuation report with inventory/purchase grants; inspect known and unknown costs. Latest posted location purchase cost, known opening fallback; old return cannot replace newer cost.

12. Supplier and purchase management

Register supplier with name/address/mobile, verified opening balance, credit days and active state; add optional contacts/terms. Search/edit contacts while preserving immutable opening history. Supplier detail shows entries, balance, purchases and payments; the Supplier Statement report provides date-filtered account reporting. Create a purchase draft with supplier/date/terms, supplier reference, item quantities/unit costs, discount and other costs. Draft posts neither payable nor stock. Management finalizes once to post liability. Receive actual deliveries one line/location per GRN submission; partial deliveries are supported and overreceiving is rejected. Print/PDF purchase and GRN from saved records. Authorized supplier payments target either purchase or opening balance, reject overpayment and support full reversal. Supplier returns remove received/unreturned/available stock; only explicitly agreed return credit reduces liability. A supplier account credit is not a cash refund. Historical repeated-part purchase lines require source reconciliation before return.

Operational reference	Implementation
Required inputs	Name, address, mobile, opening balance on creation, credit days, active state; optional contact/telephone/email/terms/notes
Validation / restrictions	Opening balance cannot be changed by editing. Financial operations require purchases.approve; separate opening/purchase targets prevent over-allocation.
Effect / balance changes	Keeps supplier master and liability history separate from customer accounts. Reversal retains original payment.
Navigation	/suppliers/create; /suppliers/{supplier}; /suppliers/{supplier}/edit; /suppliers/{supplier}/opening-payments
Screenshot reference	supplier-detail; see chapter 24
Evidence	StockPurchasingWorkflowTest; Step 13 test-case appendix

12.1. Actions in Supplier management

- Register supplier: Enter name/address/mobile/opening balance/credit days/status and optional contact details; Save. SUP master and one nonzero opening ledger entry.
- Edit and search supplier: Search name/code/mobile; open Edit; correct contacts/terms without changing opening balance. Master metadata updated, original financial opening retained.
- Supplier accounts and statements: Review supplier balance/account entries, purchases and payments; use authorized Supplier Statement report for selected dates. Existing liability history, not a customer ledger.
- Opening balance settlement: Open supplier; enter actual opening settlement amount/method/reference; save and verify balance. Separate opening allocation; overpayment blocked.

Operational reference	Implementation
Required inputs	Supplier, purchase date, cash/credit terms, discount/other costs, 1-100 part/quantity/unit-cost lines; supplier reference/due date optional
Validation / restrictions	Purchase date not future; supplier reference unique per supplier; receiving cannot exceed remaining ordered quantity. GRN selects one line/location per submission.
Effect / balance changes	Draft has no stock/payable. Finalize posts payable once. Actual partial GRN receipts post stock and latest purchase costs. Supplier return credit must be explicitly agreed.
Navigation	/purchases/create; /purchases/{purchase}/document/{format}; /purchases/{purchase}/finalize; /purchases/{purchase}/payments; /purchases/{purchase}/receive; /purchases/{purchase}/returns; /supplier-payments/{payment}/reverse
Screenshot reference	grn; see chapter 24
Evidence	StockPurchasingWorkflowTest|PrintGapCompletionTest; Step 13 test-case appendix

12.2. Actions in Purchase and GRN management

- Purchase draft with items: Select supplier/date/terms, add part quantities and unit costs; enter known discount/extras; Save draft. No stock/payable until authorized finalization.
- Purchase finalization: Review supplier invoice/reference and totals; finalize once with appropriate authority. Posts supplier payable once; still no stock receipt.
- GRN and partial receiving: Choose one purchase line and location; enter delivered quantity; Save; repeat for other lines/deliveries. Partial source-linked receiving; no overreceiving or duplicate posting.
- Supplier payment: Open finalized purchase; enter genuine amount/method/reference; save and check outstanding. Separate supplier payment/allocation reduces liability.
- Supplier payment reversal: Select original supplier payment; enter correction reason and reverse when authorized. Full original retained; allocation released and liability restored.

- Supplier return and agreed credit: Select source line/location, physically returned quantity, reason and explicitly agreed supplier credit where authorized. Stock decreases; only entered agreed account credit reduces payable, not cash refund.
- Purchase and GRN documents: Open saved purchase/GRN; choose browser document or PDF and duplicate as needed. Actual purchase/received quantity and protected print history.

13. POS system

Open POS. Search SKU/barcode/product/brand/category, or focus the barcode field and scan. Add product, change quantity or remove a line. Review location availability and unit precision. A discount requires its dedicated permission. Cash checkout needs tendered amount; verify change. Sale, total-only payment and stock deduction post atomically. Repeated identical protected retries return the same sale; changed payload cannot reuse the operation key. Bank transfer/cheque creates a pending draft with references; an authorized manager verifies actual receipt before stock/payment posting. A recorded cheque is not claimed cleared. Use POS sales history to open saved sales and print/reprint/download the configured receipt. Authorized Returns/Refunds reference original sale quantities, distinguish resalable/damaged goods and preserve the original. Resalable goods restore stock at original cost; damaged goods do not become saleable.

Operational reference	Implementation
Required inputs	Barcode/search; cart quantities; method; cash tender or transfer/cheque reference; cheque date when needed; optional customer, discount and notes
Validation / restrictions	Positive sale total, sufficient stock and unit precision. Discounts/returns/refunds/verification each need permission. Changed payload cannot reuse an operation key.
Effect / balance changes	Cash checkout atomically posts sale, payment and stock. Bank/cheque stays pending until manager verification. Resalable returns restore original recorded cost.
Navigation	/pos; /pos/cart/add; /pos/checkout; /pos/returns; /pos/sales; /pos/sales/{sale}/print; /pos/sales/{sale}/verify
Screenshot reference	pos; see chapter 24
Evidence	POSManagementTest|LatestPurchaseCostTest; Step 13 test-case appendix

13.1. Actions in Point of sale

- Product search and scanner input: Search SKU/barcode/name/brand/category or focus barcode input and scan; inspect matching product. Available product choices; no stock posting until checkout.
- Cart quantity controls: Add chosen part, change quantity or remove item; check precision, stock and total. Session cart; server revalidates at checkout.
- Permitted discounts: Enter authorized discount; review nonnegative net sale before checkout. Discount audit and exact decimal calculation.

- Cash checkout: Choose Cash, enter actual tender and confirm change; complete checkout once. Atomic finalized sale, total-only payment and negative stock; tender/change retained.
- Pending bank/cheque verification: Cashier creates transfer/cheque sale with references; manager verifies actual receipt on saved sale. Before verification: draft, no stock/payment. After: final posting.
- Sales history: Open POS sales history, filter and inspect the saved sale/verification state. Policy-protected saved sales and line history.
- Receipt print and PDF: Open saved sale; Print, reprint or PDF; use configured thermal width. Receipt and duplicate history with Sinhala-capable font.
- Returns and refunds: Find original sale, enter actual returned quantity and resalable/damaged classification; record authorized refund. Original retained; resalable stock restored at original cost; exact cents preserved.

14. Reports

Open Business Reports or the relevant report category. Select report, dates and applicable customer/employee/supplier/item/category/status/payment/sort filters, then review. Repair reports cover jobs/technicians. Sales cover repair/POS/invoice, daily/monthly/yearly, cash/credit, cashier and discounts/refunds. Collection reports measure net money movements separately from issued sales. Credit reports show customer/overdue debt and statements. Stock reports cover balances, movements, repair/POS use, adjustments, purchase history and valuation. Supplier reports cover purchases, outstanding and statements. Financial reports also show expenses and recorded-cost profit estimates. PDF exports use A4 landscape and at most 1000 rows; XLSX/CSV allow 50000, with exports separately permission checked. Latest purchase cost was owner-selected. Unknown/mismatched costs, purchase-wide freight/discount allocation and ambiguous refund treatment keep estimates incomplete. Do not call these audited net income.

Operational reference	Implementation
Required inputs	Report type; applicable date/customer/employee/supplier/item/category/status/method filters; allowlisted sort/direction
Validation / restrictions	End date cannot precede start. Financial reports need financial access. Export permission is separate. PDF limit 1000 rows; XLSX/CSV limit 50000 rows.
Effect / balance changes	Reports read saved transactions. Latest purchase valuation and recorded-cost profit are estimates; missing cost/freight policy keeps affected values incomplete.
Navigation	/reports/{report}; /reports/{report}/export/{format}
Screenshot reference	reports; see chapter 24
Evidence	ReportsTest|Step11ReportsTest|Step12AuditTest|LatestPurchaseCostTest; Step 13 test-case appendix

14.1. Actions in Reports and analysis

- Repair and technician reports: Choose Repair/Technicians, set dates/status/employee and review saved jobs. Read-only repair cohorts and technician work.
- Sales and collection reports: Choose sales, invoice/POS, cashier, discount/refund or cash collection report using permitted grants. Issued sales distinguished from net cash collections.
- Financial and credit reports: Choose financial/customer/supplier debt report; review time basis and actual entries. Account/expense/collection summaries; restricted financial data.
- Inventory and purchase reports: Choose Stock/Movements/Repair Parts/POS Parts/Purchases; filter item/category/supplier/location as available. Shared movement, receiving and purchasing evidence.
- Recorded-cost profit estimate: Open Profit Estimate; review incomplete flags before interpreting margin. Recorded issued/POS costs, labour and expenses; unresolved extras/refunds stay incomplete.
- PDF, Excel and CSV export: Apply filters; select PDF, XLSX or CSV; narrow range if export limit is reached. A4 landscape PDF or spreadsheet/CSV download with formula-safe text.

14.2. Actual report catalogue

repair, technicians, sales, collections, advances, customer-credit, overdue-credit, supplier-outstanding, purchases, expenses, stock, stock-movements, repair-parts, pos-parts, adjustments, purchase-history, supplier-purchases, valuation, profit-estimate, daily-sales, monthly-sales, yearly-sales, repair-income, invoice-sales, cash-credit-sales, cashier-sales, discounts, refunds, cash-summary, cash-collections, customer-statement, supplier-statement

15. Employee management and permissions

Open Employees to search/register/edit staff. Employee identity/reference is distinct from the linked User login; staff may have no account. Optional account creation needs employee, user-create and role-management grants. Enter a unique username and strong confirmed password; authentication uses username, not email. Administrator manages passwords, account status, roles and overrides. Deactivating employment disables linked login and invalidates sessions while preserving history. Reactivating employee does not silently enable a separately disabled user. Roles & Permissions configures current grants; explicit individual deny wins over inherited grants, while Inherit removes an override. Self-change, final-admin and administrator-override protections apply. Use the current role snapshot in Feature Guidance; customized operational roles may differ from fresh-fixture screenshots.

Operational reference	Implementation
Required inputs	Employee identity/contact/employment fields; optional account username/password/role subject to grants
Validation / restrictions	Employee and username identifiers are unique; account creation also needs users.create and roles.manage. Strong password rules apply; final administrator is protected.
Effect / balance changes	Employee may exist without login. Deactivation preserves history, disables linked access and invalidates sessions.

Operational reference	Implementation
Navigation	/admin/users/{user}; /employees/create; /employees/{employee}/edit
Screenshot reference	employees; see chapter 24
Evidence	EmployeeManagementTest|CreateUserCommandTest; Step 13 test-case appendix

15.1. Actions in Employee and account administration

- Employee registration: Enter employee details; create a linked account only when user/role permissions are also held. Employee may exist without credentials; generated immutable reference.
- Employee editing and activation: Edit actual contact/employment data; deactivate/activate through authorized controls. History retained; linked inactive account sessions invalidated.
- Account passwords and status: Open linked account; administrator resets a strong password or reviews enablement, without exposing credentials. Hashed password and audit; no public recovery screen.

Operational reference	Implementation
Required inputs	Role grants or explicit allow/deny/inherit overrides for an eligible different user
Validation / restrictions	Seeded roles cannot be deleted from UI; administrator cannot lose essential access. Self role/override changes and administrator overrides are restricted.
Effect / balance changes	Actual role grants plus overrides determine access. Explicit deny wins. Technician assignment is a separate record policy.
Navigation	/admin/roles/{role}/edit; /admin/users/{user}/permissions; /admin/users/{user}/role
Screenshot reference	roles; see chapter 24
Evidence	EmployeeManagementTest; Step 13 test-case appendix

15.2. Actions in Roles and permission controls

- Role assignment: Choose another eligible user's role in account administration; review least required access. Role association, last-admin and self-change protections.
- Role grant customization: Open Roles & Permissions; inspect current grants, adjust permitted role and save. Actual grant matrix, not an assumption from seeded defaults.
- Individual permission overrides: Open eligible user; choose Allow, Deny or Inherit per permission; review effective access. Deny overrides grants; admin/self restrictions preserved.

15.3. Current operational role guidance

The following grants were read from the operational role relations, not inferred from the default seeder. Individual override count at inspection: 0. Subsequent administrator customization can change access. Some actions need additional record policy or several permissions together.

Administrator

Manage accounts/roles/settings/templates/audit and protected backups. Preserve the final active administrator. Restore is CLI-only and requires explicit target approval.

Current permission grants (98): audit.view, backups.configure, backups.create, backups.download, backups.restore, backups.view, credit.collect, credit.reports, credit.view, customers.create, customers.deactivate, customers.statement, customers.update, customers.view, dashboard.view, employees.create, employees.deactivate, employees.update, employees.view, inventory.adjust, inventory.create, inventory.issue, inventory.stock-count, inventory.update, inventory.view, invoices.cancel, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, job-cards.print, job-cards.view, jobs.assign, jobs.cancel, jobs.complete, jobs.create, jobs.inspect, jobs.repair, jobs.update, jobs.update-status, jobs.view, payments.create, payments.export, payments.print, payments.receive, payments.receive-for-others, payments.reverse, payments.view, permissions.manage, pos.discount, pos.payment.verify, pos.receipt.print, pos.refund, pos.reports.view, pos.return, pos.sale, pos.sale.cancel, pos.sales.view, pos.view, print-templates.reset, print-templates.update, print-templates.view, purchases.approve, purchases.create, purchases.receive, purchases.view, quotations.approve, quotations.cancel, quotations.create, quotations.issue, quotations.print, quotations.revise, quotations.update, quotations.view, receipts.create, receipts.print, receipts.reprint, receipts.view, reports.credit, reports.export, reports.financial, reports.inventory, reports.repair, reports.sales, reports.view, roles.manage, roles.view, settings.update, settings.view, suppliers.create, suppliers.update, suppliers.view, users.create, users.disable, users.update, users.view

Receptionist

Search/register customers; receive repairs and print intake/card documents; prepare quotations where granted. No assumed authority for money, stock, financial statements or administration.

Current permission grants (19): customers.create, customers.update, customers.view, dashboard.view, invoices.view, job-cards.print, job-cards.view, jobs.create, jobs.view, quotations.create, quotations.issue, quotations.print, quotations.update, quotations.view, receipts.create, receipts.print, receipts.view, reports.repair, reports.view

Technician

Inspect and repair assigned jobs, update their status and request parts. Cannot use another technician's job by changing its URL. Stock issue and customer-money authority are separate.

Current permission grants (10): dashboard.view, job-cards.view, jobs.complete, jobs.inspect, jobs.repair, jobs.update-status, jobs.view, quotations.view, reports.repair, reports.view

Cashier

Create/issue invoices, receive actual collections and advances, collect credit and perform permitted POS sales/printing. No default refund/reversal, supplier or settings authority.

Current permission grants (23): credit.collect, credit.view, customers.view, dashboard.view, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, payments.create, payments.export, payments.print, payments.receive, payments.view, pos.receipt.print, pos.sale, pos.sales.view, pos.view, quotations.print, quotations.view, reports.sales, reports.view

Storekeeper

Maintain parts, request/issue/adjust/count stock and enter/receive purchases. Purchase approval, supplier administration and supplier money are management-only under current grants.

Current permission grants (14): dashboard.view, inventory.adjust, inventory.create, inventory.issue, inventory.stock-count, inventory.update, inventory.view, pos.view, purchases.create, purchases.receive, purchases.view, reports.export, reports.inventory, reports.view

Manager

Assign repairs, prepare/approve quotations, supervise invoices/collections/credit, suppliers/purchase approval and financial reports. No assumed settings/account/restore access.

Current permission grants (52): credit.collect, credit.reports, credit.view, customers.statement, customers.view, dashboard.view, invoices.cancel, invoices.create, invoices.export, invoices.issue, invoices.print, invoices.update-draft, invoices.view, jobs.assign, jobs.view, payments.create, payments.export, payments.print, payments.receive, payments.reverse, payments.view, pos.discount, pos.payment.verify, pos.receipt.print, pos.refund, pos.reports.view, pos.return, pos.sale.cancel, pos.sales.view, pos.view, purchases.approve, purchases.create, purchases.receive, purchases.view, quotations.approve, quotations.cancel, quotations.create, quotations.issue, quotations.print, quotations.revise, quotations.update, quotations.view, reports.credit, reports.export, reports.financial, reports.inventory, reports.repair, reports.sales, reports.view, suppliers.create, suppliers.update, suppliers.view

16. System settings

Open System Settings with settings.view/update. Verify business name/address/telephone, actual logo PNG/JPEG within 2 MB, three-letter currency, valid timezone, English/Sinhala default language, allowed date/time formats, paper and footer. Save and inspect new page/document output. Print-template overrides may supersede business defaults. Do not enter database secrets or OAuth tokens here. The configured real logo is used in these documents; no logo has been invented. Some UI labels remain English or unresolved translation keys; this limitation is recorded, not silently translated.

Operational reference	Implementation
Required inputs	Business name/address/telephone; PNG/JPEG logo; currency; timezone; language; date/time formats; paper/footer
Validation / restrictions	Currency has three ASCII letters, valid timezone, allowlisted formats. Logo JPEG/PNG at most 2 MB. Secrets/arbitrary configuration cannot be edited.
Effect / balance changes	Validated settings affect new pages/documents and create audit history; financial history is not rewritten.
Navigation	/settings
Screenshot reference	settings; see chapter 24
Evidence	SystemSupportTest; Step 13 test-case appendix

16.1. Actions in System settings

- Business identity and logo: Verify business details; upload actual PNG/JPEG logo within 2 MB; Save. Validated settings and audit, current configured logo only.
- Currency, language and formats: Choose allowed currency/timezone/language/date/time/paper/footer settings; review new output. New pages/documents reflect values; no historical amount rewriting.

Existing Sinhala UI labels (copied without new translation):

පාලන පුවරුව | පාරිභෝගිකයන්

17. Print template management

Open Print Templates. Choose the relevant existing document template and review paper, orientation, margins, font, branding/title/footer, duplicate label, table/signature settings. A5 receipt and A6 job card support browser/PDF output. Quotations support A4/A5, invoices A5, POS configured thermal width and payment receipts configured paper. Supplier purchase/GRN and customer statements use their saved templates. Changes/reset create a new version and retain older ones. Use Preview and a saved fixture; confirm long descriptions, Sinhala, page breaks and signatures before physical printing. Font is 7-16 pt; margin 0-30 mm; thermal width 40-120 mm; widths cannot exceed 100%. History records preparation/export, not physical printer success.

Operational reference	Implementation
Required inputs	Existing template, title/branding, allowed paper/orientation, font 7-16 pt, margins 0-30 mm, footer/duplicate marker
Validation / restrictions	Allowed paper depends on type; thermal width 40-120 mm. Column widths at most 100%; signature-area arrays valid. Local PNG/JPEG logo paths only.
Effect / balance changes	Editing/reset creates a version and retains older versions. Preparing print/PDF records history; history is not proof of physical printing.
Navigation	/print-templates; /print-templates/{template}/reset
Screenshot reference	print-templates; see chapter 24
Evidence	PrintTemplateManagementTest|PrintGapCompletionTest|StockPurchasingWorkflowTest; Step 13 test-case appendix

17.1. Actions in Print template management

- Receipt and A6 card templates: Open receipt/job-card template, adjust allowed paper/title/margins/fonts/footer and Preview. Versioned A5 receipt/A6 card settings and duplicate labels.
- Quotation and invoice templates: Review A4/A5 quotation or A5 invoice settings; preview long descriptions before use. Versioned customer document defaults; original financial values retained.

- POS and payment receipt templates: Review thermal width or payment receipt paper/font/footer and inspect a saved fixture output. Versioned receipt output; physical hardware acceptance separate.
- Template reset and history: Use Reset only for the intended template; inspect the new version and older history. New version of original defaults; prior versions retained.

18. Backup and restore

Open Backup Management with the relevant permissions. Current policy defaults are 02:00 (2:00 AM) Asia/Colombo, retention 30 days and at most 30 scheduled copies. Operational private destination is unset and host scheduler has not been activated in this work. Google Drive folder/email/OAuth remain blank at the owner's request; no real upload is claimed. After an approved private root outside application/web roots and client paths are configured, use Create backup, inspect history/size/checksum, Verify and authorized Download. SQL dumps need protected/encrypted storage; encrypted destination metadata does not encrypt the dump. Review failure/missed/cloud warnings; local success differs from off-computer success. CLI-only restore backup:restore ID is a separately approved recovery action requiring exact target, active administrator/password/restore grant, safety backup and maintenance mode. Failure stays in maintenance for investigation. Isolated recovery has been tested; operational restore has not been performed.

Operational reference	Implementation
Required inputs	Approved private backup root and MySQL clients; existing backup ID for verify/download/restore; authorized administrator/password for CLI restore
Validation / restrictions	Private root outside web/application; checksum/nonempty file; restore confirmation, exact target and fresh safety backup required. Restore is unavailable over HTTP.
Effect / balance changes	Default 02:00 Asia/Colombo, 30 days/30 scheduled copies. Operational destination and Drive credentials remain unset. Recovery test uses disposable data only.
Navigation	/admin/backups/configure; /admin/backups/{backup}/verify; /backups
Screenshot reference	backups; see chapter 24
Evidence	BackupManagementTest|Step11BackupTest|Step11RecoveryTest|SystemSupportTest; Step 13 test-case appendix

18.1. Actions in Backup and restore

- Manual backup and history: After approved private destination/client configuration, Create backup; inspect success/size/checksum/history. Protected nonempty SQL dump, or recorded failure. Current live destination is unset.
- Backup verification and download: Verify saved backup; download only if authorized and integrity passes. SHA-256 evidence; private authorized download, not public link.

- Schedule and retention policy: Review 02:00 Asia/Colombo and 30-day/30-copy policy; check host scheduler separately. Policy exists, but this task does not activate scheduler; Drive is blank.
- Backup failure alerts: Review failed/missed/cloud states; correct configuration through authorized operator and retry. Local/cloud statuses separate; local success does not imply Drive success.
- Authenticated CLI restore: With separately approved target/recovery point, run backup:restore ID; confirm exact DB/admin/password and safety copy. Maintenance lifecycle and audited import; only isolated recovery tested here.

19. Common errors and troubleshooting

Verified validation/security behavior: invalid login rejects credentials; repeated failure throttles; required/invalid fields return errors; duplicate Job Number is rejected; inactive/missing customer cannot be selected for intake; mixed/already-billed jobs cannot be invoiced; insufficient stock prevents posting; overpayment, mixed advance/invoice targets and unauthorized receiver fail; unauthorized routes return forbidden/redirect; changed/missing/corrupt backup refuses verification. Correct the displayed fields and inspect saved history before retrying after a timeout. General guidance, not physically verified: if a printer fails, check device/driver/paper/actual-size settings and preview without reposting the transaction; if a scanner fails, focus barcode input and compare typed barcode lookup. For backup failures review private path/client permissions and history with an administrator. For session expiration sign in again and confirm whether the original transaction saved. Capture document reference/time/error, never passwords, for support.

20. Full business workflow

Use an approved isolated UAT database for training. Register/search customer; receive appliance/advance; save automatic job/card; assign technician; record inspection; prepare/issue quotation; record actual customer approval; request/issue parts; perform repair and record actual costs/status; mark completed; invoice eligible same-customer jobs; allocate advances once; collect remaining/partial amount; record delivery; reconcile reports. Separate POS workflow: search/scan > cart > permitted discount > cash checkout or pending transfer verification > receipt > any authorized return/refund. Separate supplier workflow: supplier > draft purchase > authorized finalize > actual partial GRNs > supplier payment > any authorized reversal/return credit. These flows retain history and share stock; they are not permission to enter training transactions into the live shop database.

21. Terminology and navigation tips

Term	Meaning
Job ID / automatic reference	Database identity / JOB-formatted reference; separate from unique manual Job Number.
GRN	Goods Received Note: actual received purchase quantity at a location.

Term	Meaning
Draft / issued / finalized	Preparation/posting states; a draft is not proof of payment or receipt of stock.
Allocation	Application of an existing payment to a debt/job/POS target; not another cash receipt.
Refund / reversal	Return of money versus accounting correction; both preserve original records.
Available stock	Sum of posted movements at a location, not quotation/request quantities.
Latest purchase cost	Latest posted purchase cost at location, with known opening fallback; not a rewrite of historical costs.
Duplicate	Marked later document copy with print history.
UAT	Business user acceptance on approved fixtures, including physical hardware checks.

- Use sidebar links, record search and breadcrumbs. Tab/Shift+Tab move through ordinary browser form controls; Ctrl+F searches the displayed document. No special application keyboard shortcuts were verified.
- Wait for saved confirmation; after timeout search before creating another transaction.
- When using browser Print/PDF, inspect paper/scale and signatures; do not repost a sale or receipt to troubleshoot a printer.

22. Current verification and limitations

Fresh automated result: 218 tests; 218 Passed; 0 Failed; 0 skipped/Not Run; 3566 assertions. This is scenario coverage, not complete BRD/owner acceptance.

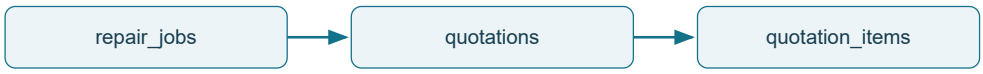
- Not Implemented: public registration/reset/account-settings screens; multi-invoice payment allocation form; partial return of one issued repair quantity.
- Not Verified operationally: private backup destination/Drive upload, host scheduler execution, physical printers/scanners, realistic-volume capacity and owner UAT.
- Pending policy: freight/purchase-wide discount allocation; affected profit remains incomplete.
- Observed UI limitation: unresolved translation keys appear on some quotation labels; English/Sinhala completeness is not certified.
- Screenshots are isolated non-sensitive fixtures. Screen amounts/names are examples from actual fixture execution, not shop transactions.

23. System relationship diagrams

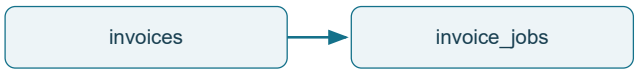
Repair and financial relationships



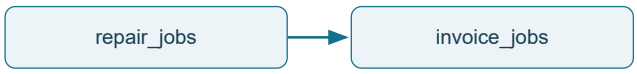
Parent to referencing child; arrows correspond to inspected foreign keys.



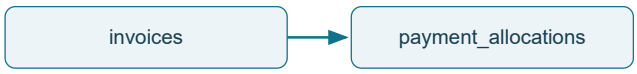
Parent to referencing child; arrows correspond to inspected foreign keys.



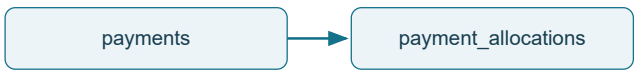
Parent to referencing child; arrows correspond to inspected foreign keys.



Parent to referencing child; arrows correspond to inspected foreign keys.



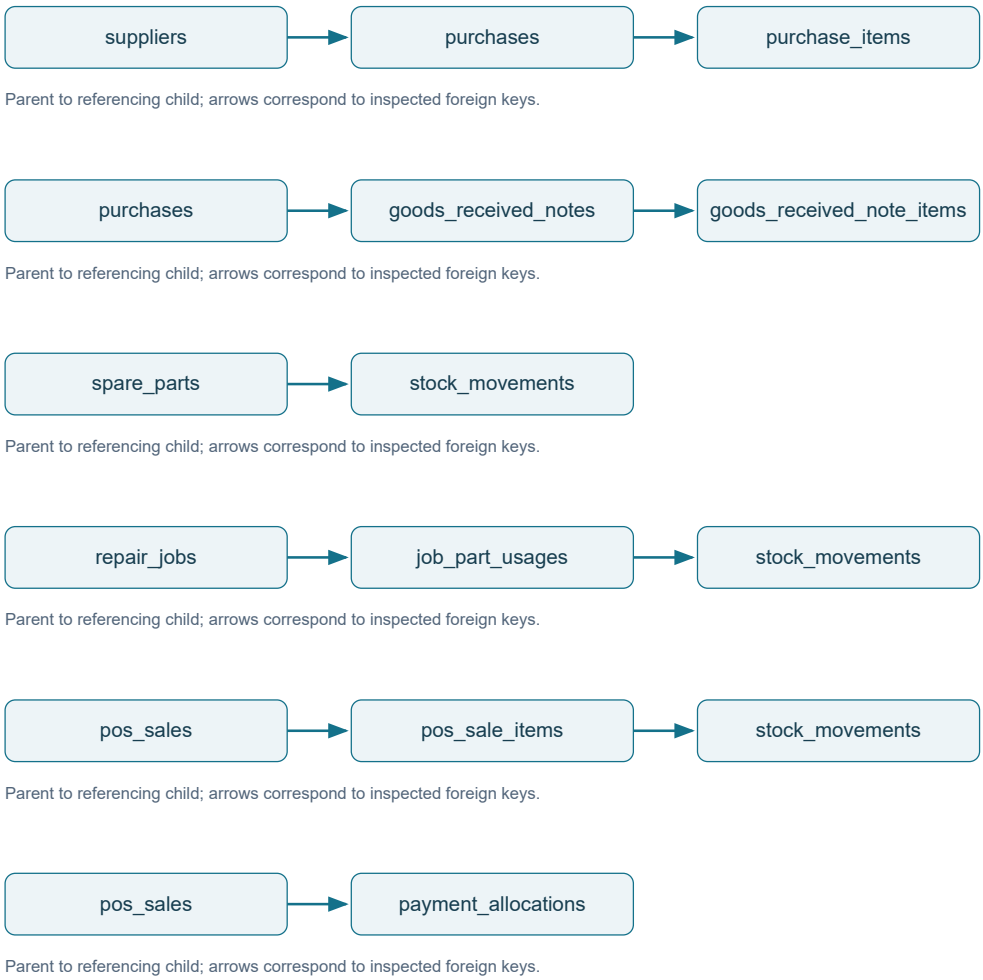
Parent to referencing child; arrows correspond to inspected foreign keys.



Parent to referencing child; arrows correspond to inspected foreign keys.

Verified table relationships, not a sequence of automatic business actions. Join tables connect the separate paths.

Shared stock, purchase and POS relationships



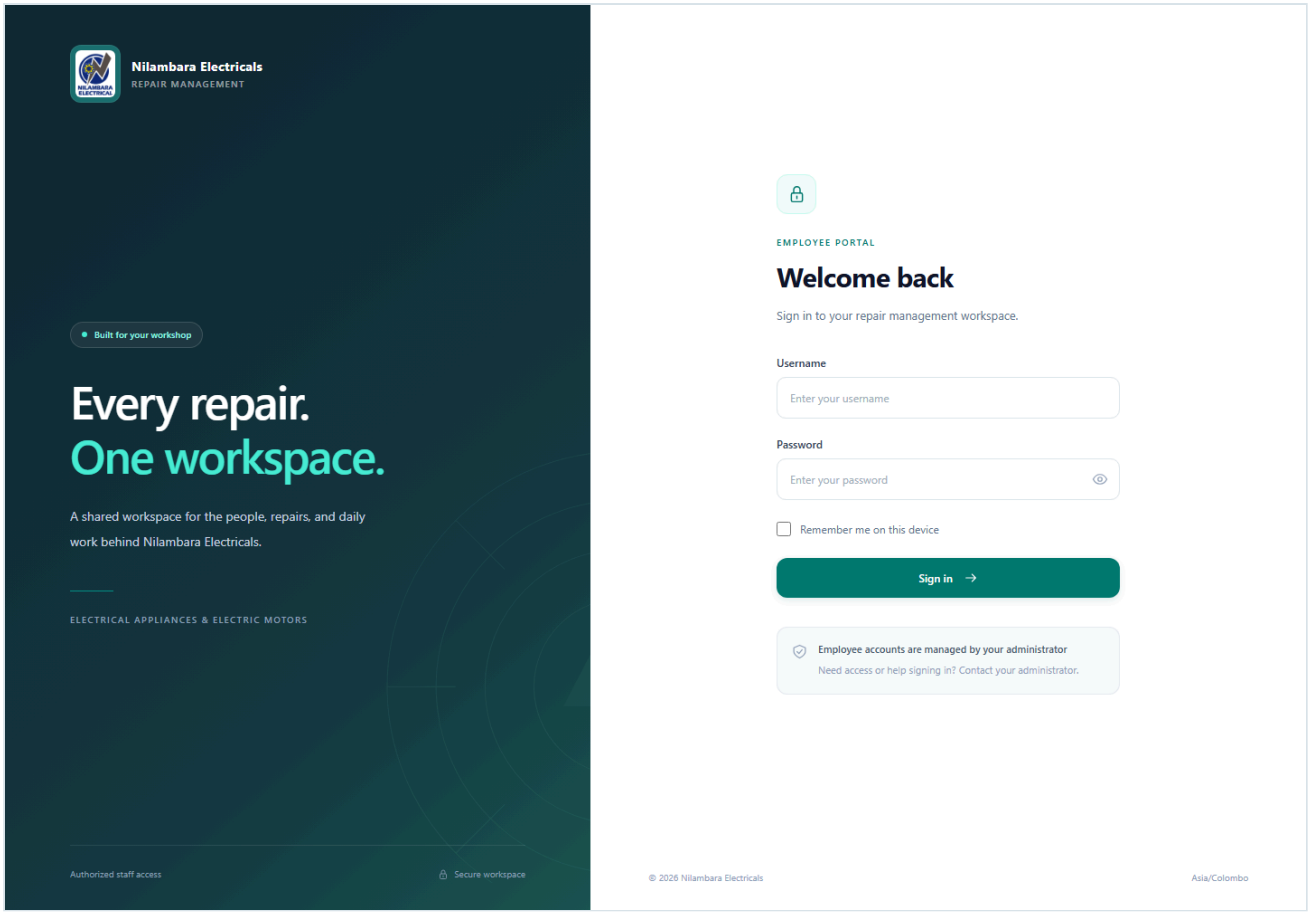
Verified table relationships, not a sequence of automatic business actions. Join tables connect the separate paths.

24. Real application screenshot atlas

The atlas contains genuine captured application pages. Long captures are split into numbered continuation panels with no hidden middle section. Original unmodified captures and the capture manifest are preserved. No application screen was fabricated.

Figure 1.1 - Login page - empty credentials

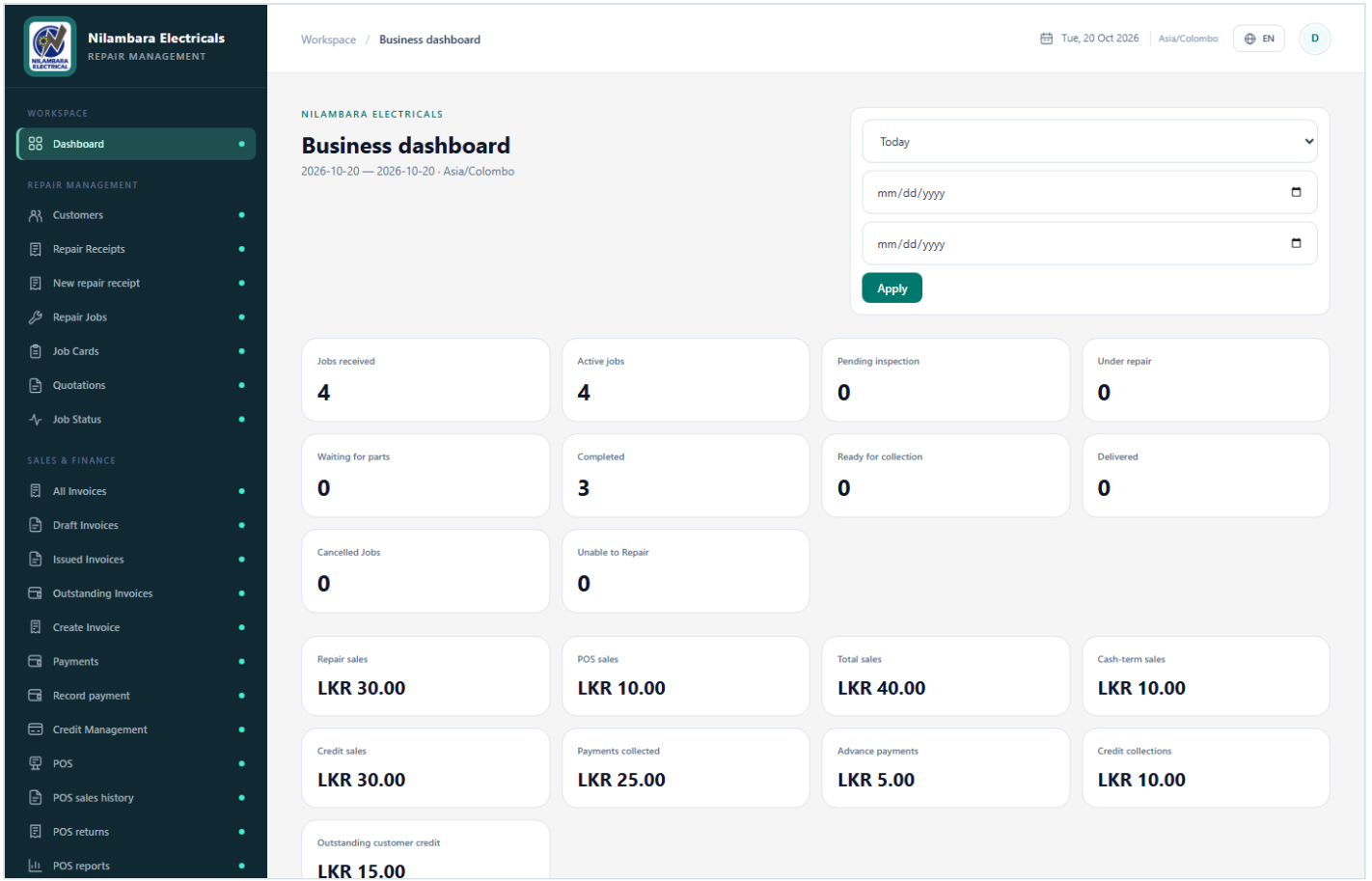
Actual Chrome screenshot | /login | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/login.png. No operational records or passwords are displayed.

Figure 2.1 - Dashboard

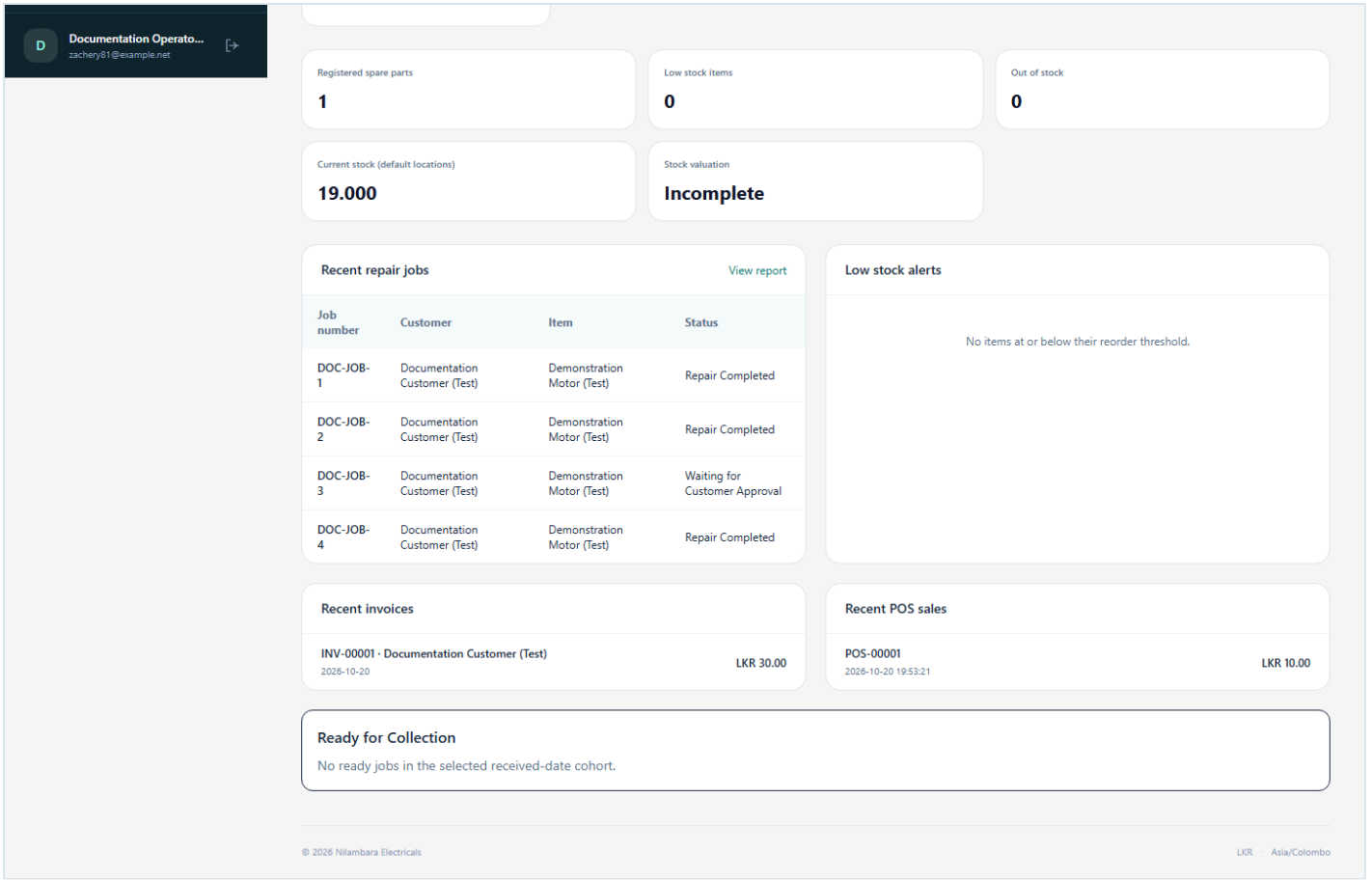
Actual Chrome screenshot | /dashboard | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/dashboard.png. No operational records or passwords are displayed.

Figure 2.2 - Dashboard

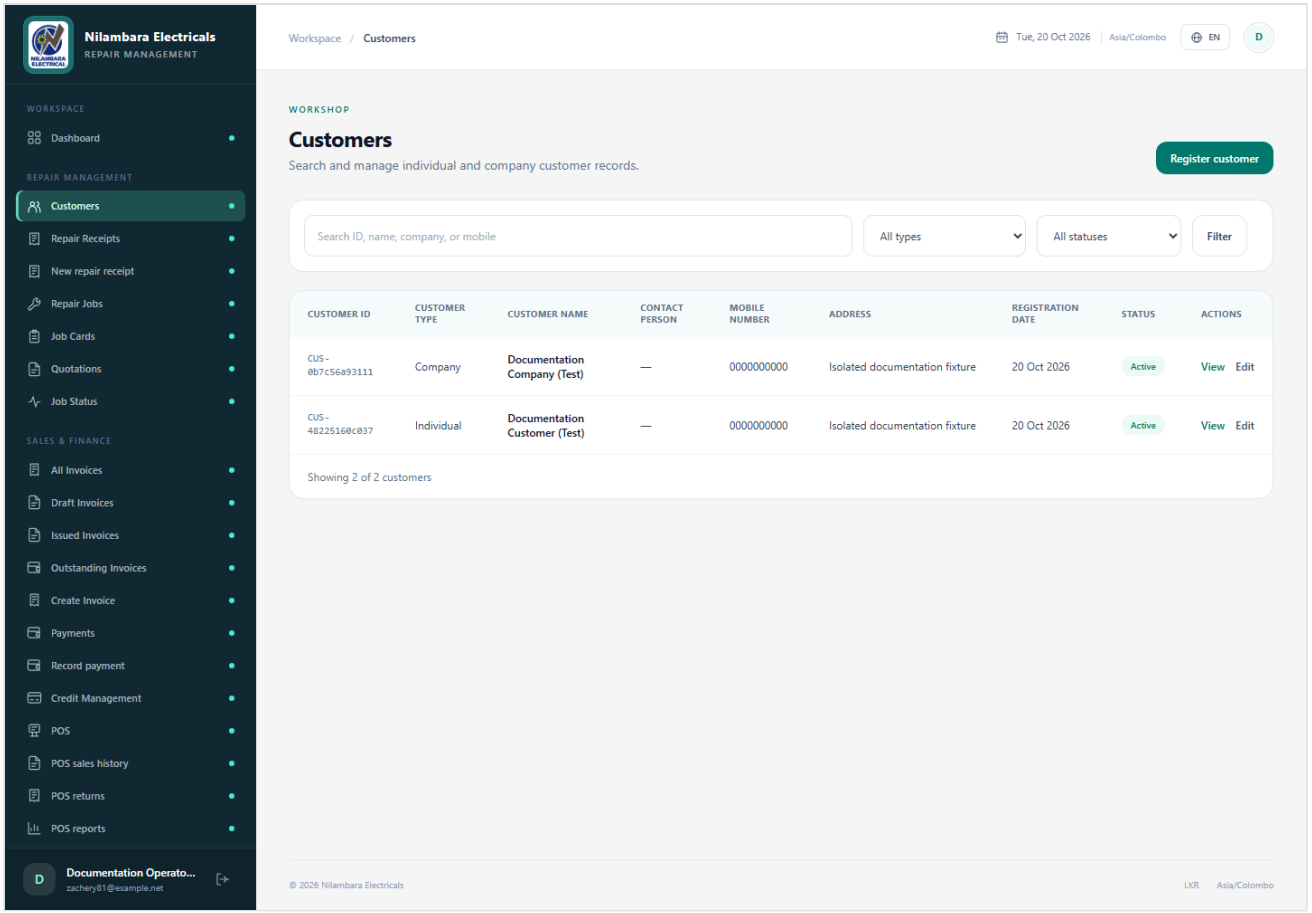
Actual Chrome screenshot | /dashboard | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/dashboard.png. No operational records or passwords are displayed.

Figure 3.1 - Customer list

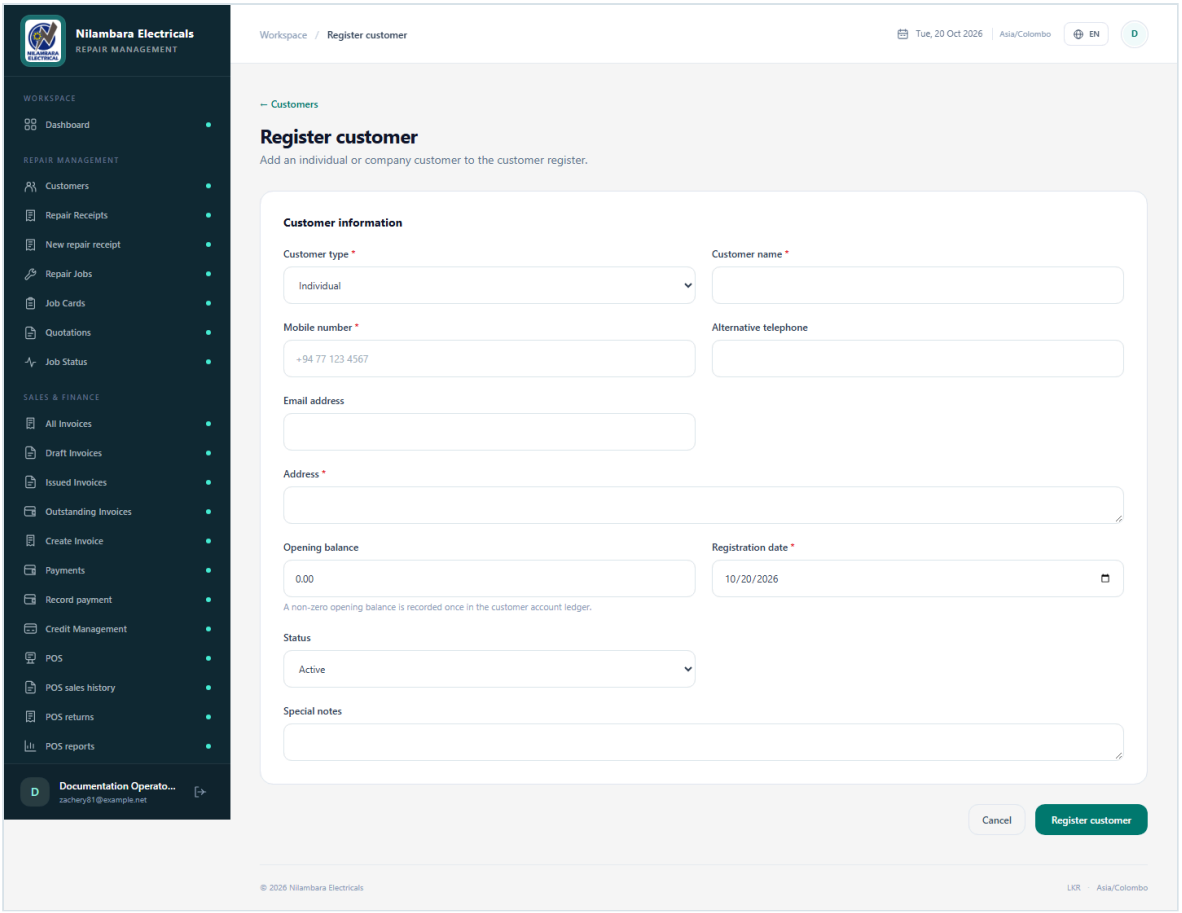
Actual Chrome screenshot | /customers | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/customers.png. No operational records or passwords are displayed.

Figure 4.1 - Register individual customer

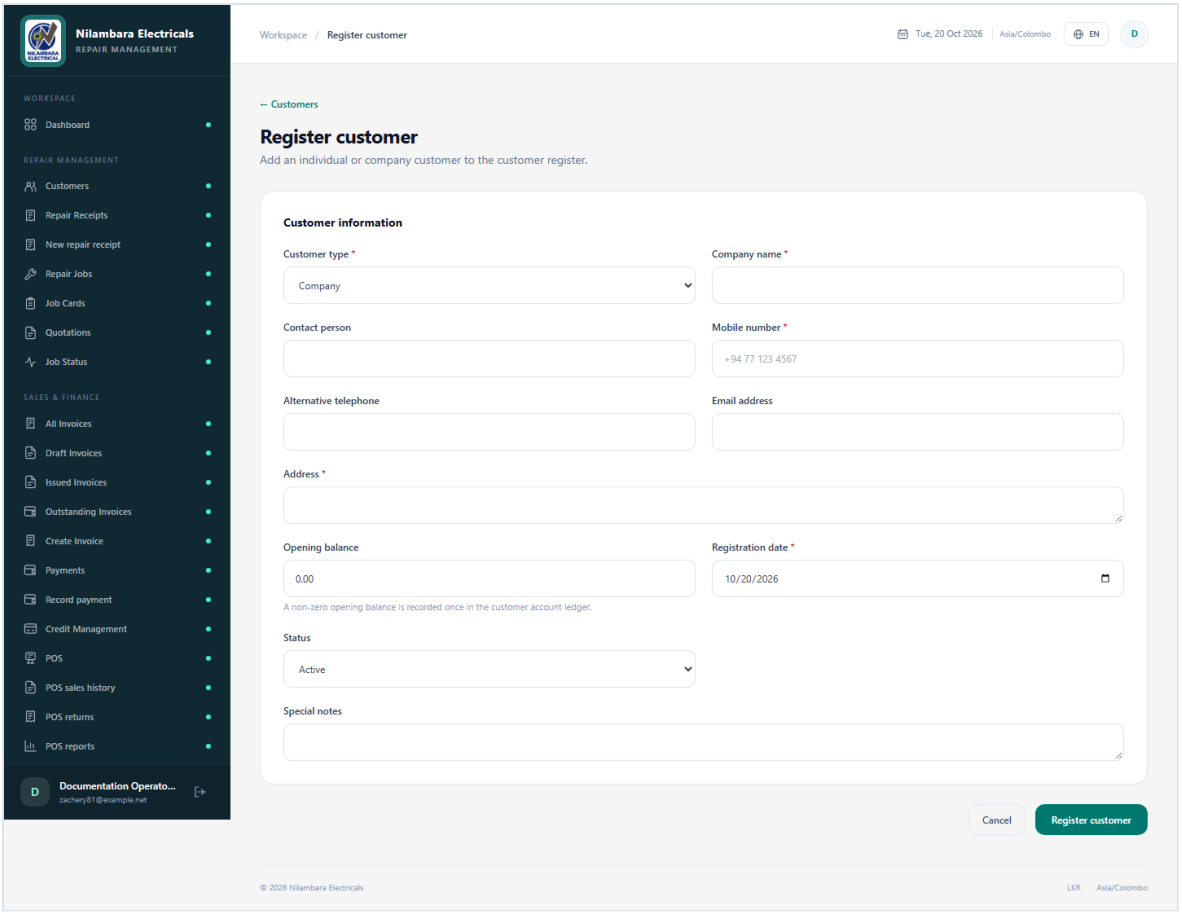
Actual Chrome screenshot | /customers/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/customer-create.png. No operational records or passwords are displayed.

Figure 5.1 - Register company

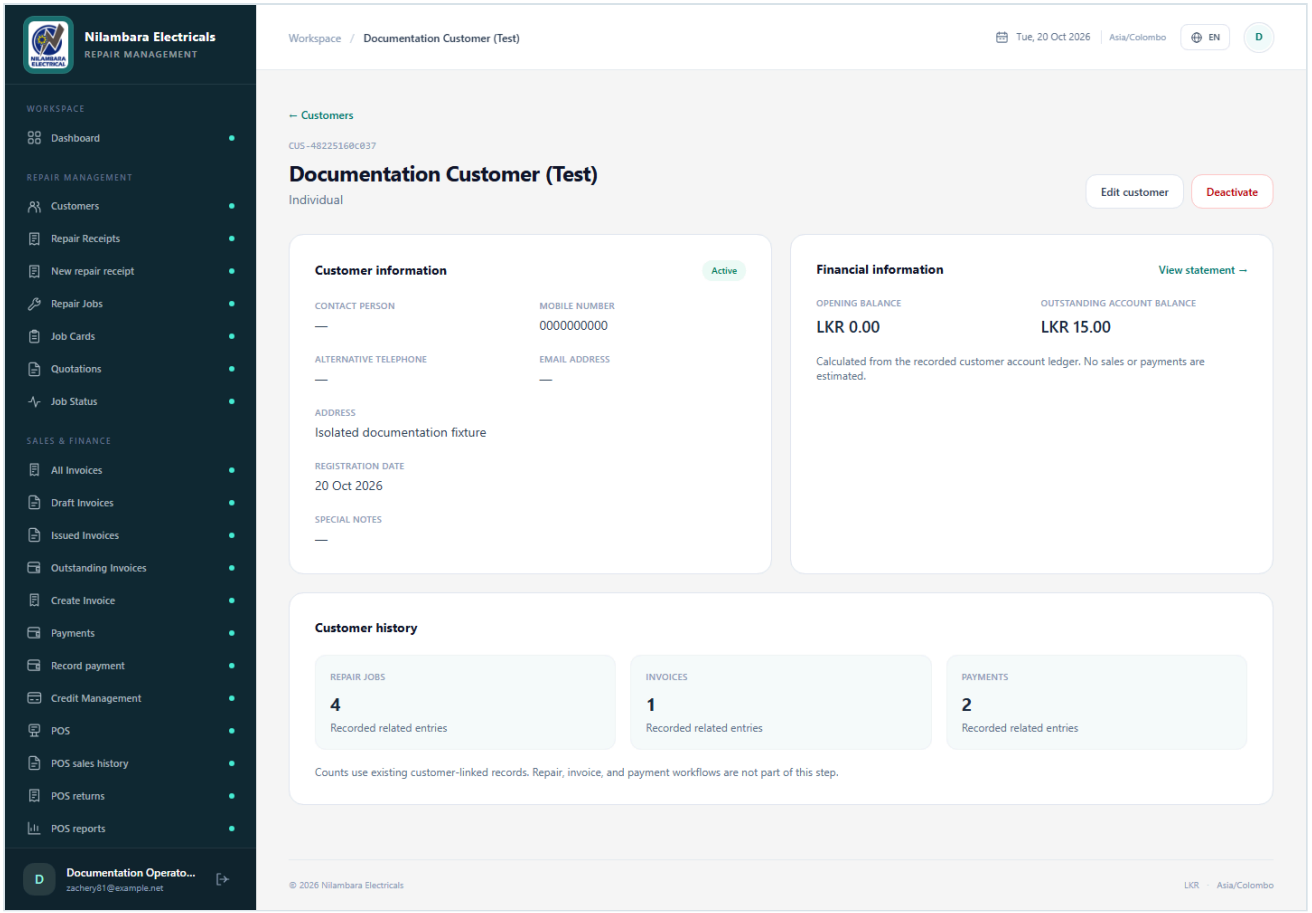
Actual Chrome screenshot | /customers/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/company-create.png. No operational records or passwords are displayed.

Figure 6.1 - Customer profile

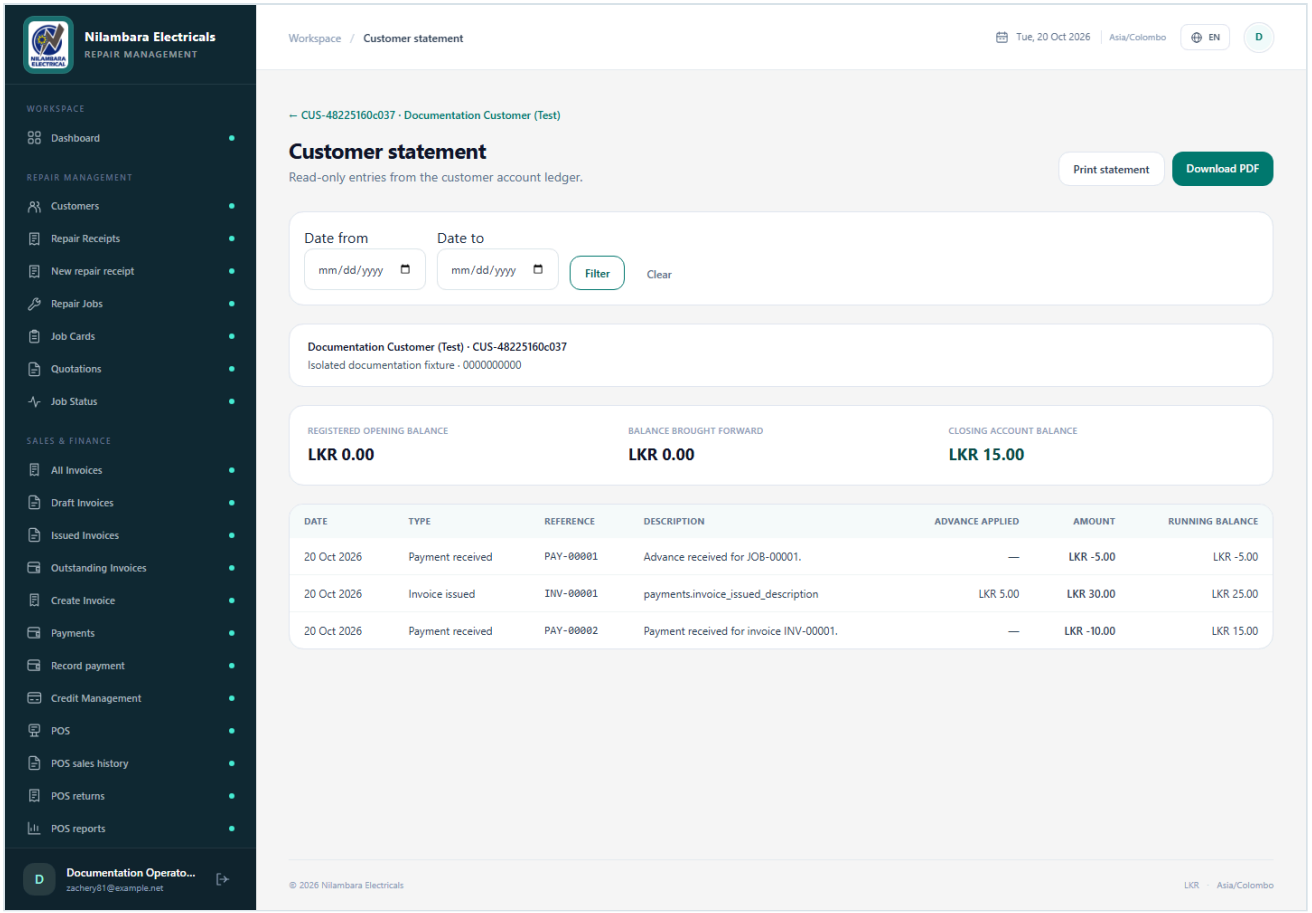
Actual Chrome screenshot | /customers/1 | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/customer-profile.png. No operational records or passwords are displayed.

Figure 7.1 - Customer statement

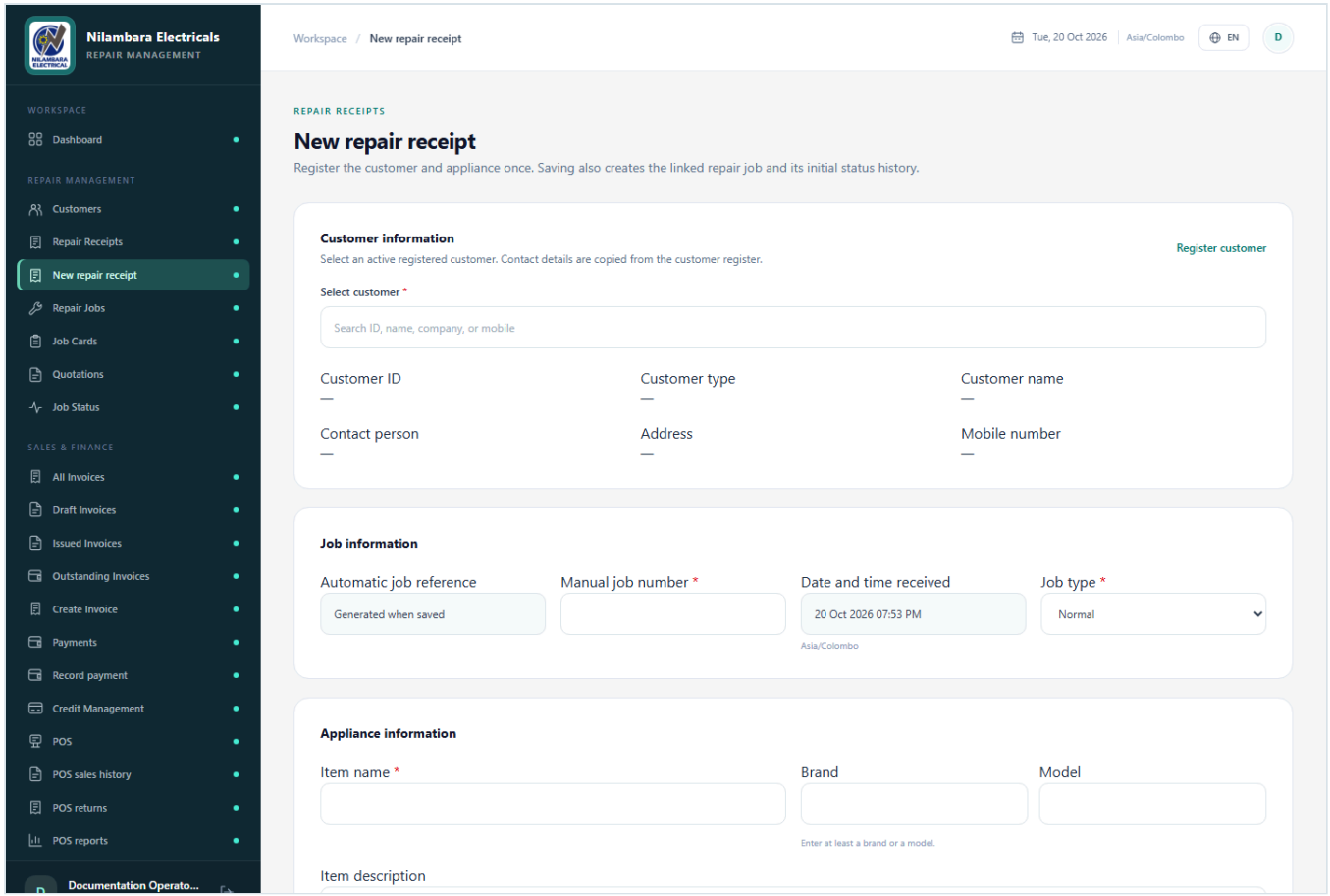
Actual Chrome screenshot | /customers/1/statement | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/customer-statement.png. No operational records or passwords are displayed.

Figure 8.1 - Repair receipt registration

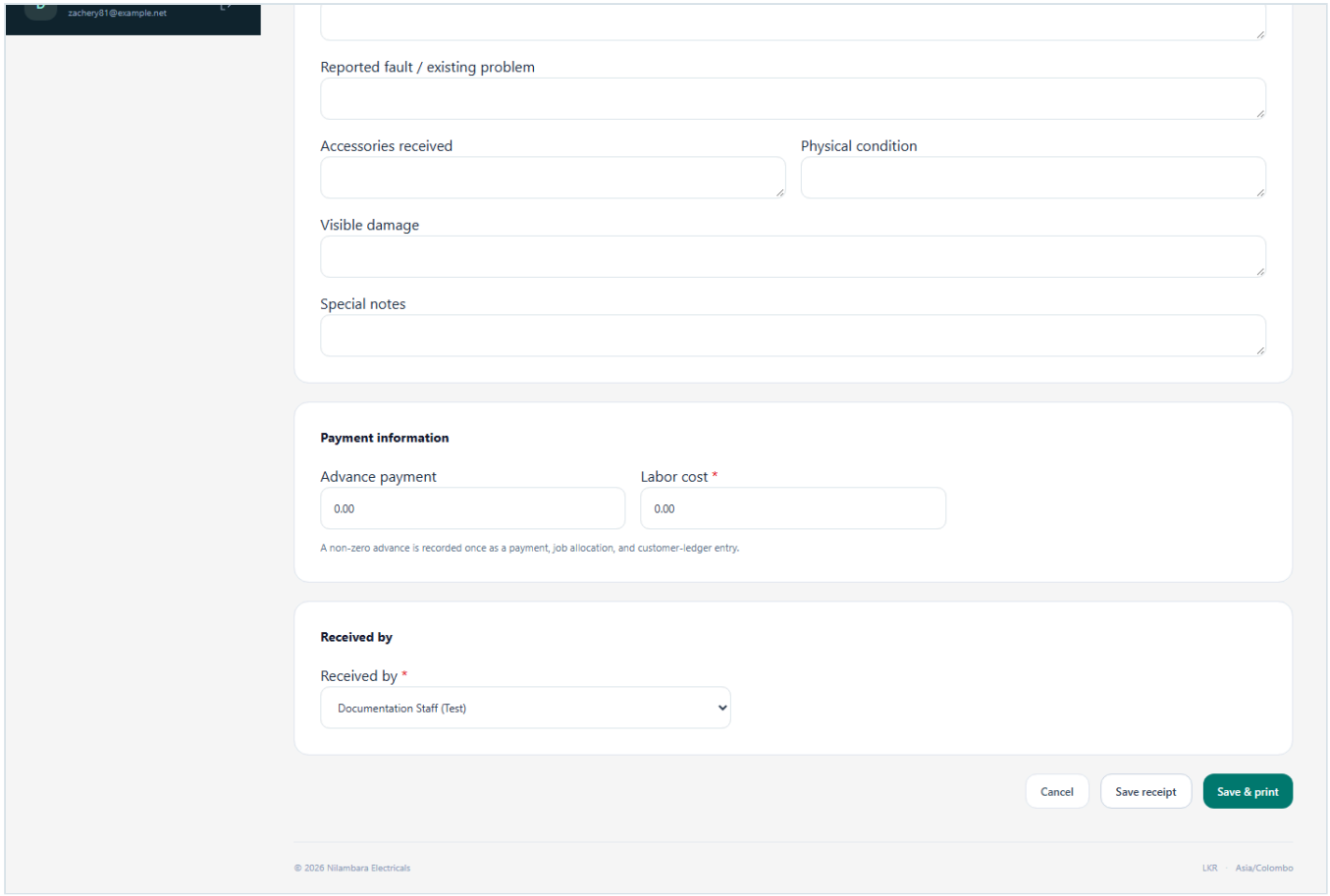
Actual Chrome screenshot | /repair-receipts/create | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/receipt-create.png. No operational records or passwords are displayed.

Figure 8.2 - Repair receipt registration

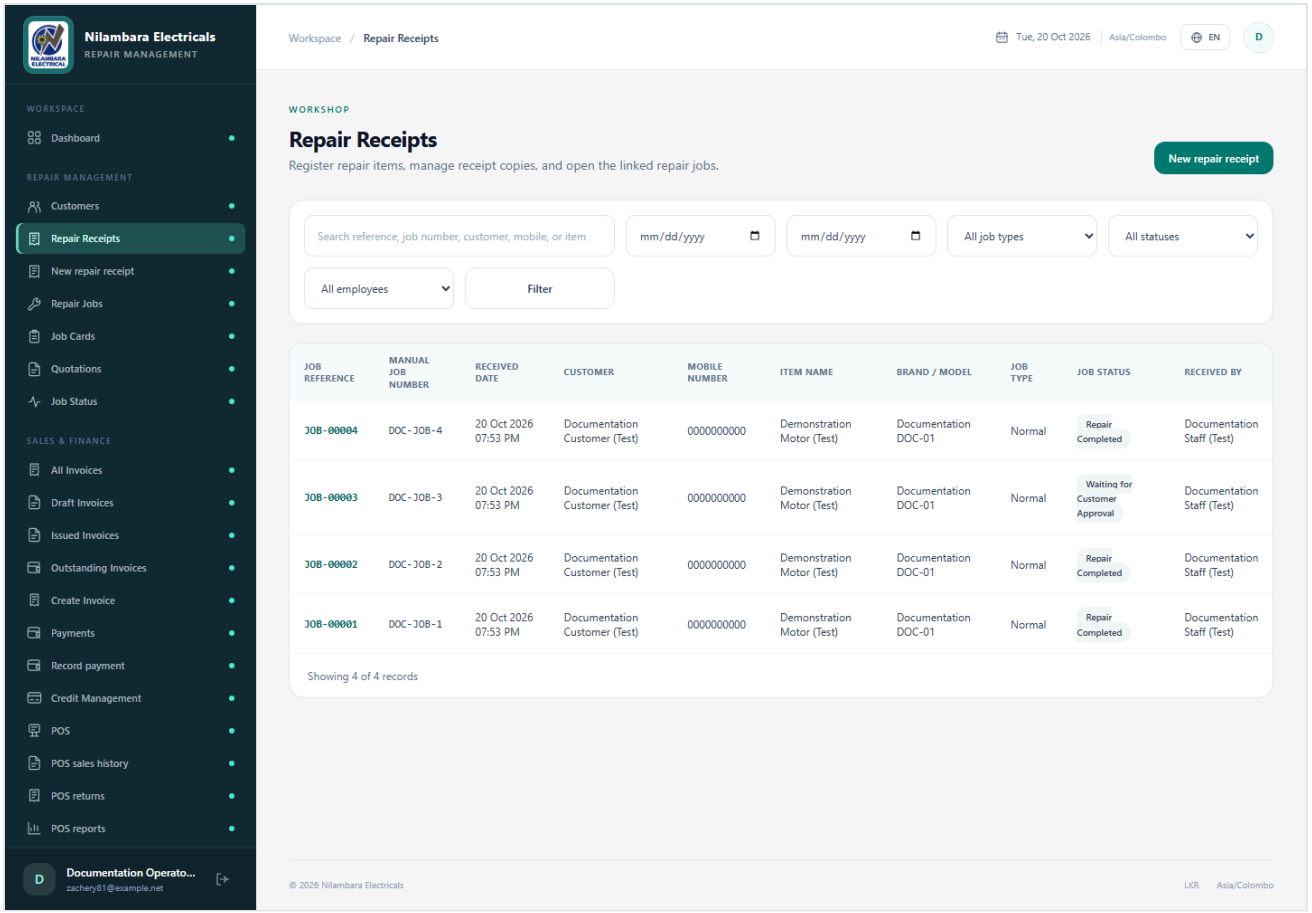
Actual Chrome screenshot | /repair-receipts/create | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/receipt-create.png. No operational records or passwords are displayed.

Figure 9.1 - Repair receipt list

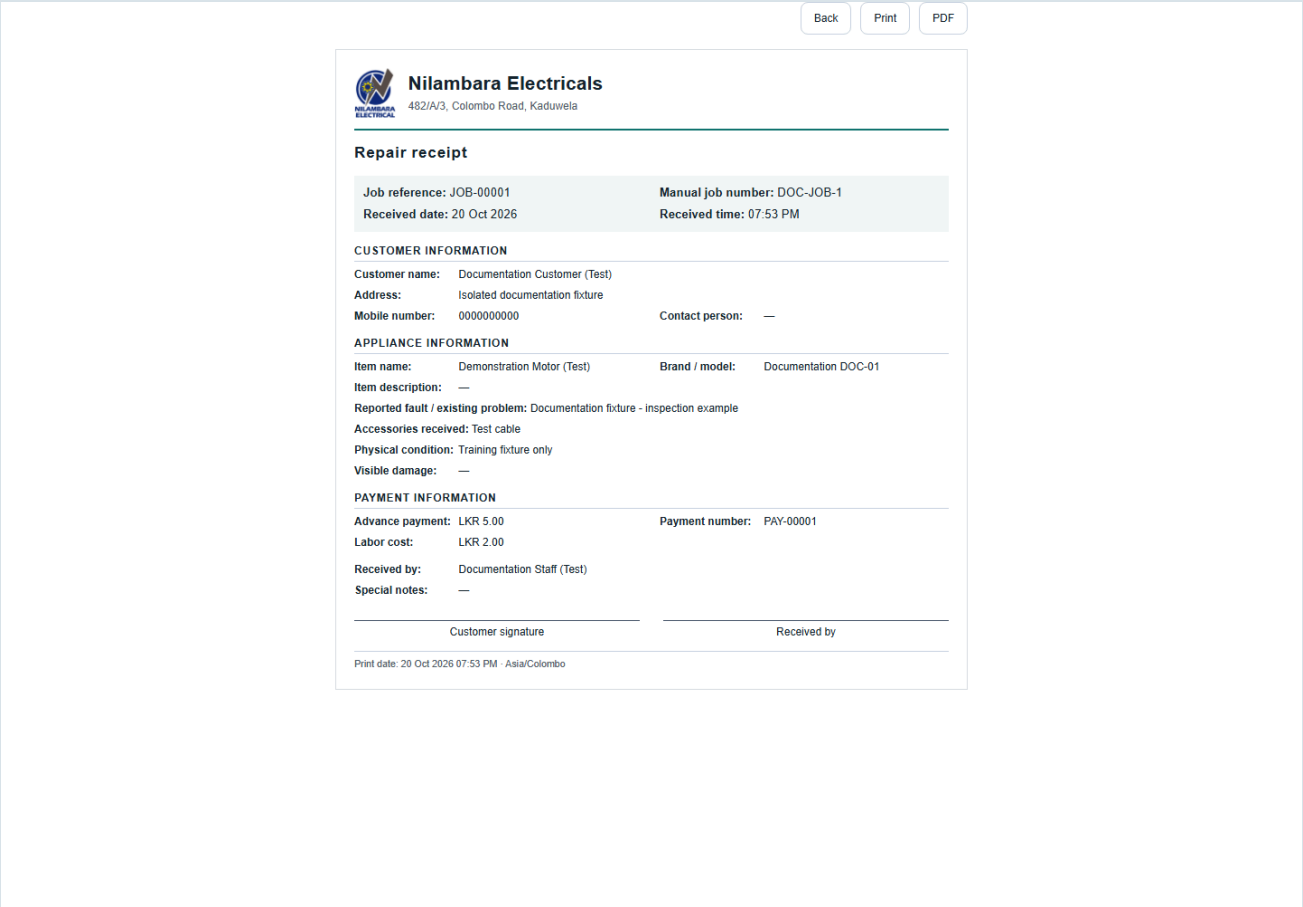
Actual Chrome screenshot | /repair-receipts | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/receipts.png. No operational records or passwords are displayed.

Figure 10.1 - A5 receipt preview

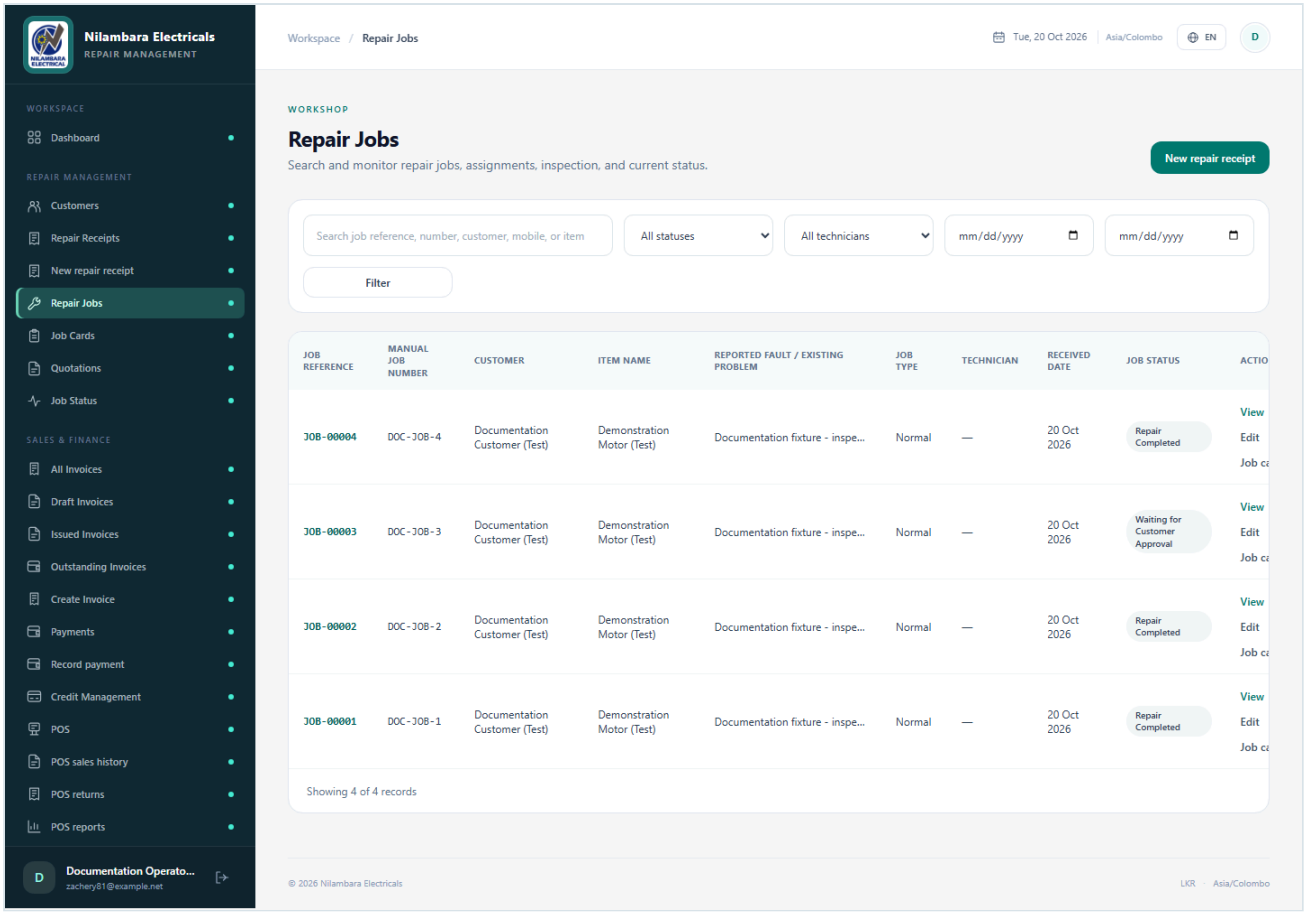
Actual Chrome screenshot | /repair-receipts/1/print | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/receipt-preview.png. No operational records or passwords are displayed.

Figure 11.1 - Repair jobs

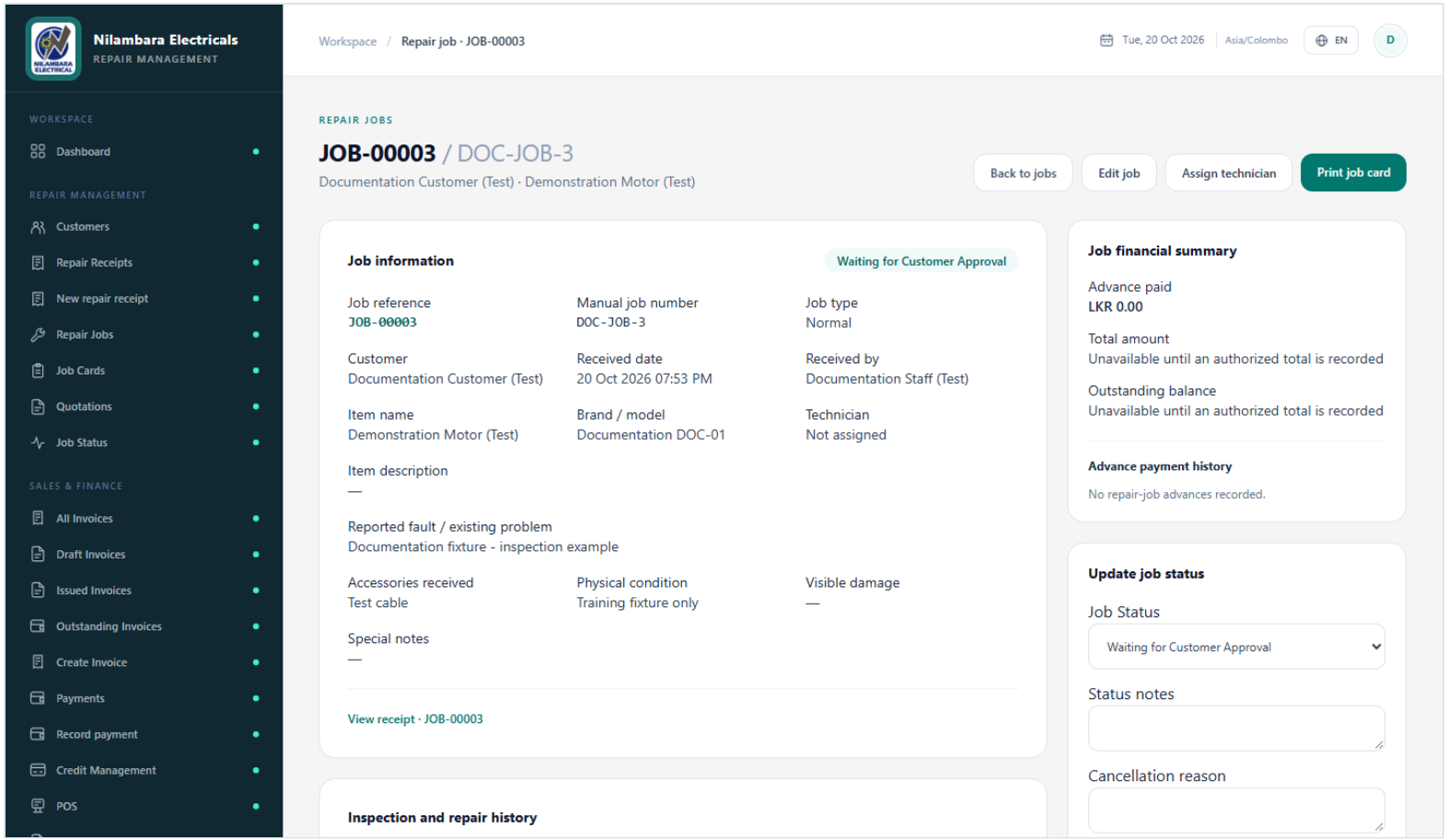
Actual Chrome screenshot | /repair-jobs | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/jobs.png. No operational records or passwords are displayed.

Figure 12.1 - Job details and status history

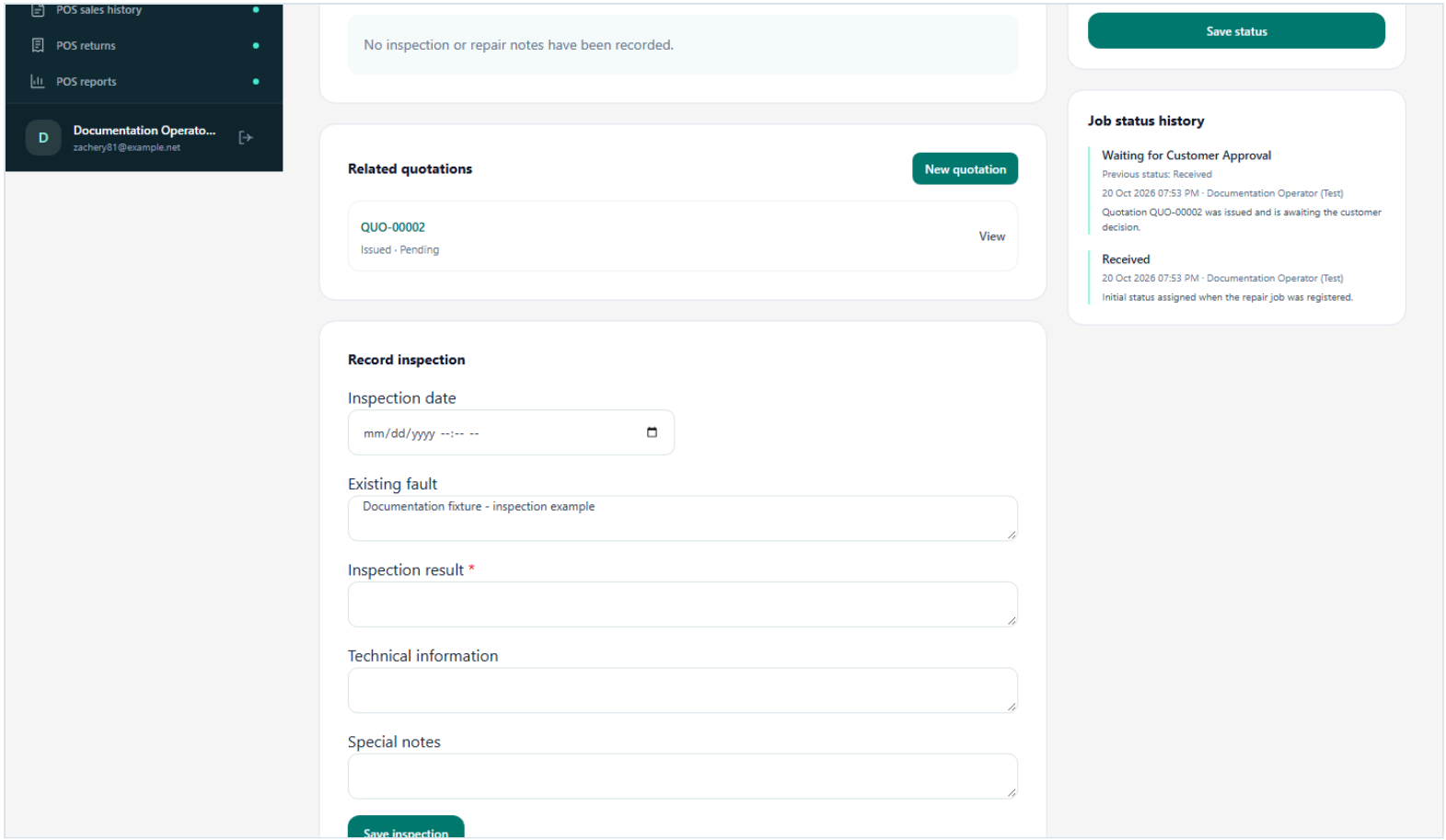
Actual Chrome screenshot | /repair-jobs/3 | isolated test fixtures | panel 1/3



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-detail.png. No operational records or passwords are displayed.

Figure 12.2 - Job details and status history

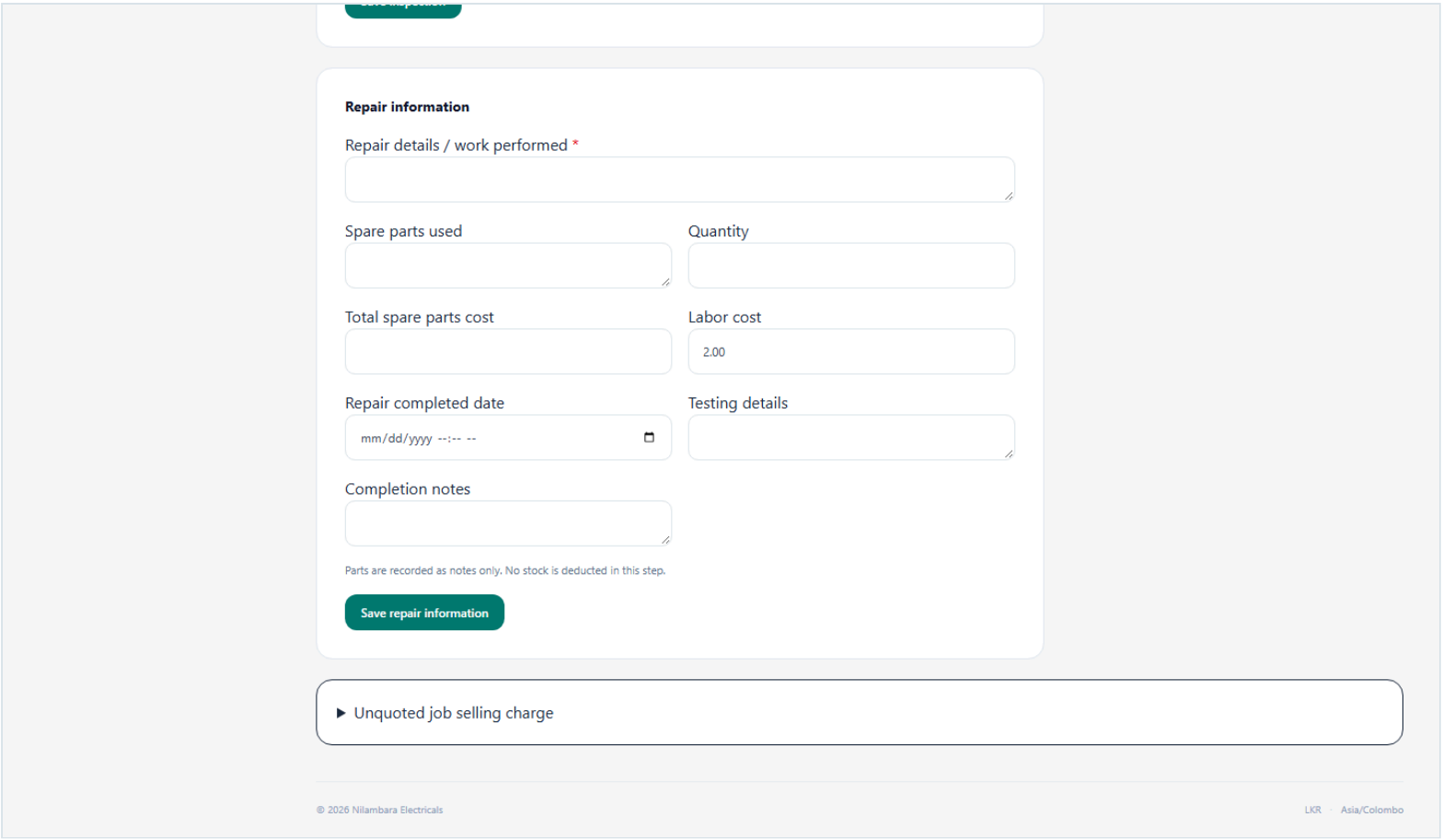
Actual Chrome screenshot | /repair-jobs/3 | isolated test fixtures | panel 2/3



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-detail.png. No operational records or passwords are displayed.

Figure 12.3 - Job details and status history

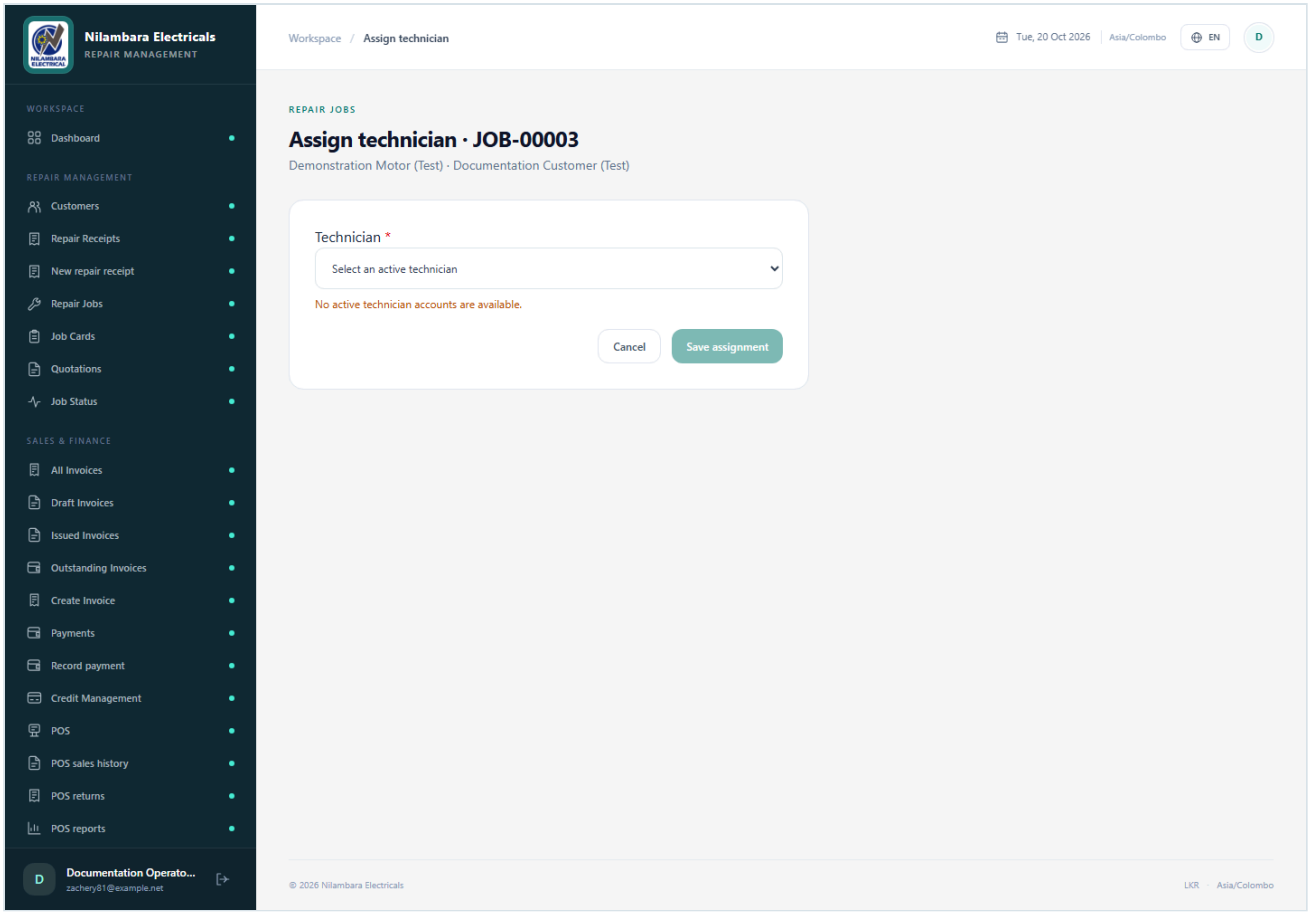
Actual Chrome screenshot | /repair-jobs/3 | isolated test fixtures | panel 3/3



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-detail.png. No operational records or passwords are displayed.

Figure 13.1 - Technician assignment

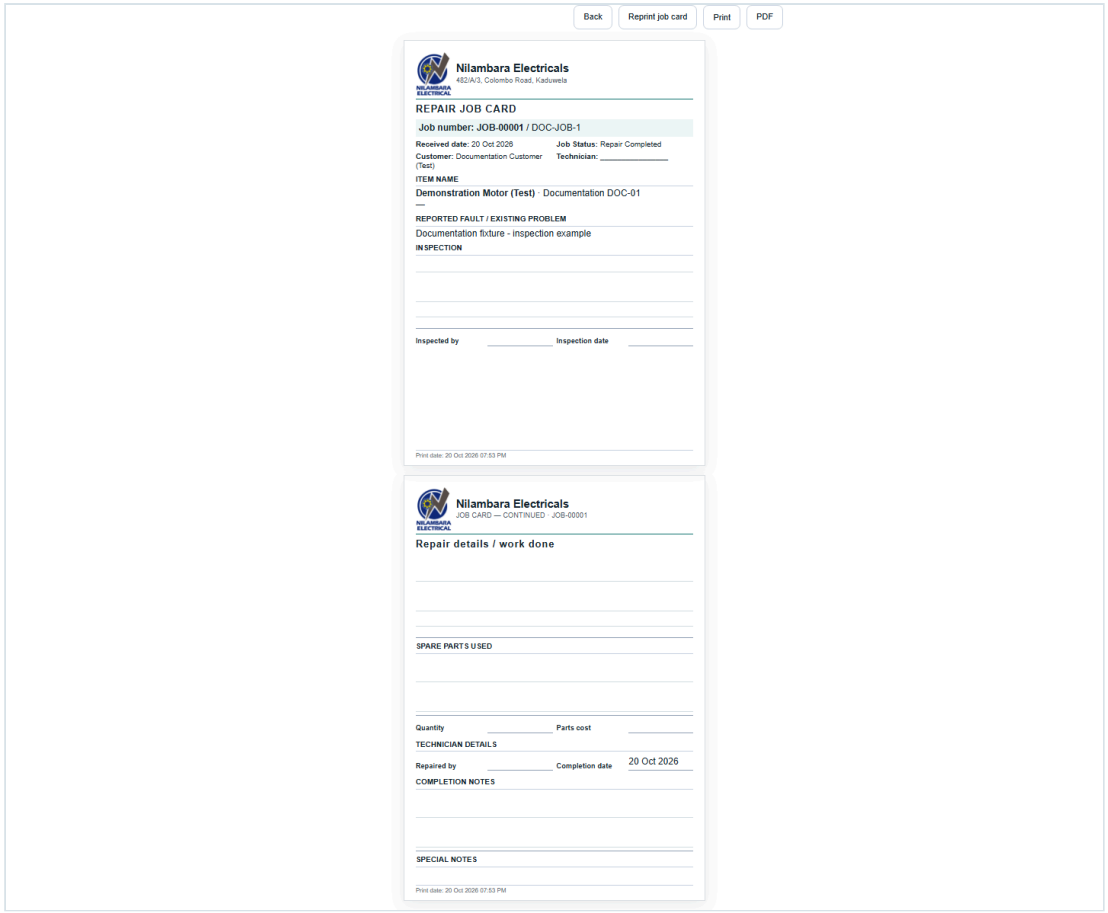
Actual Chrome screenshot | /repair-jobs/3/assign | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-assign.png. No operational records or passwords are displayed.

Figure 14.1 - A6 job card preview

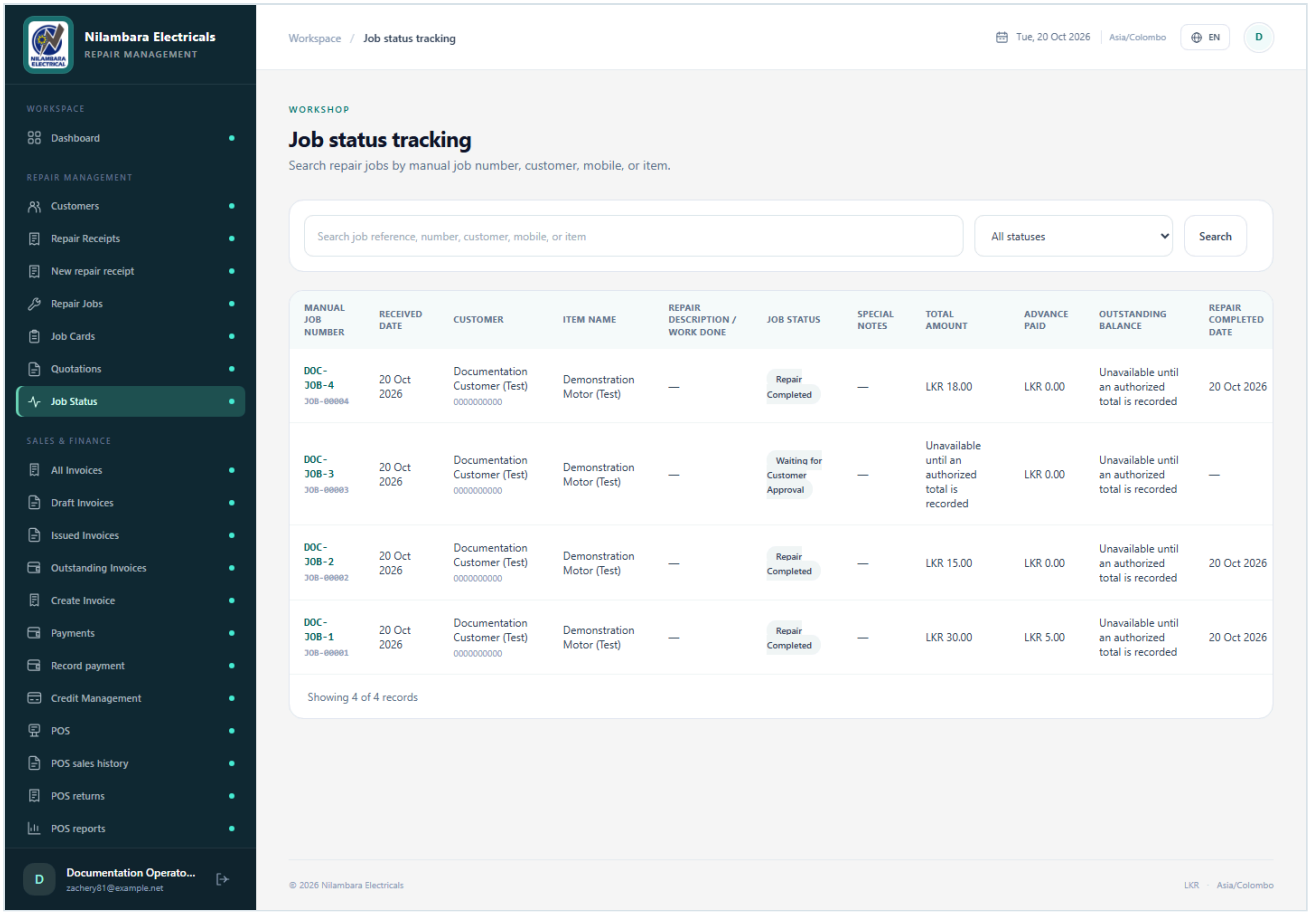
Actual Chrome screenshot | /repair-jobs/1/job-card | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-card.png. No operational records or passwords are displayed.

Figure 15.1 - Job status tracking

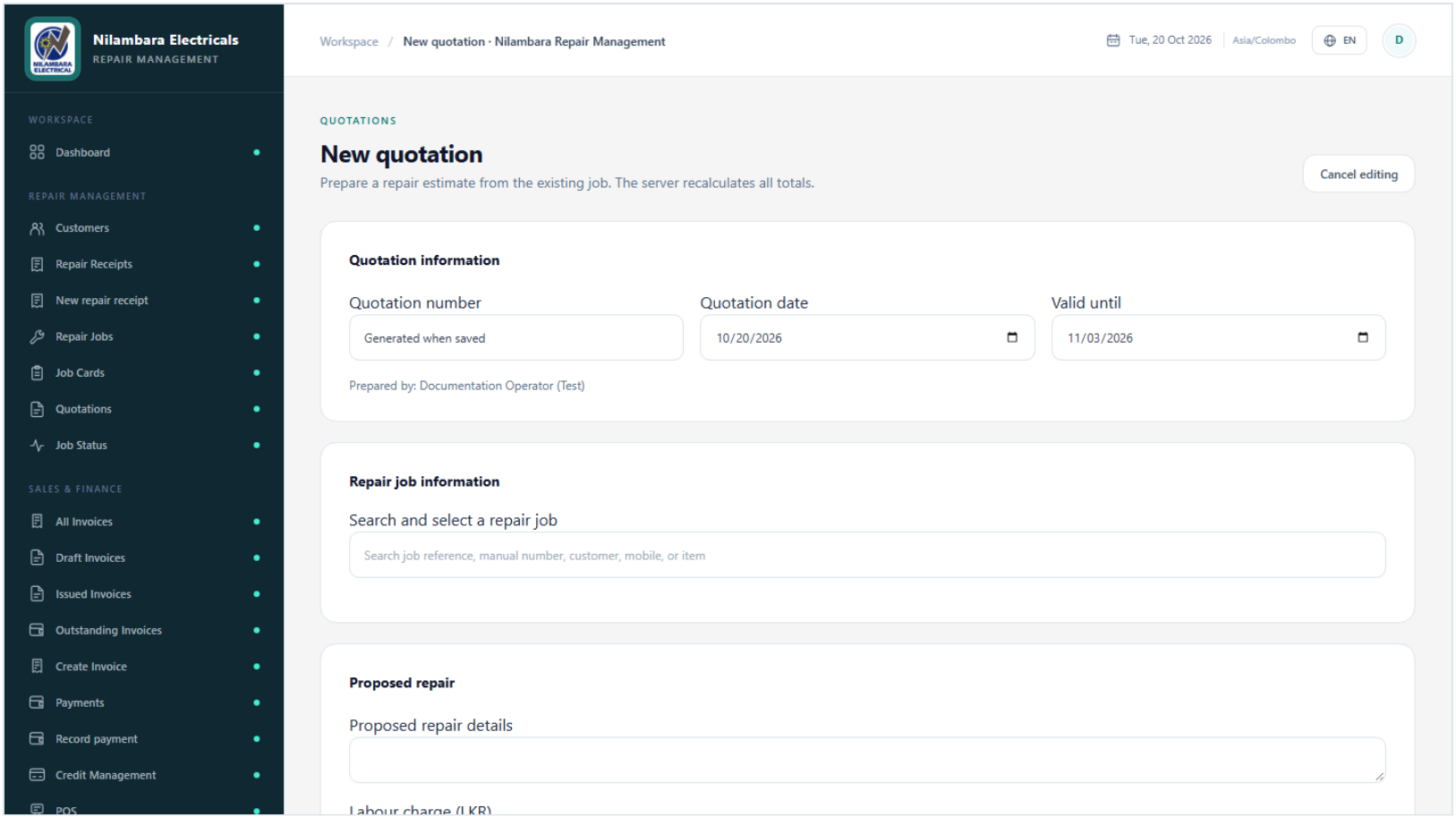
Actual Chrome screenshot | /repair-jobs/status | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/job-status.png. No operational records or passwords are displayed.

Figure 16.1 - Quotation creation

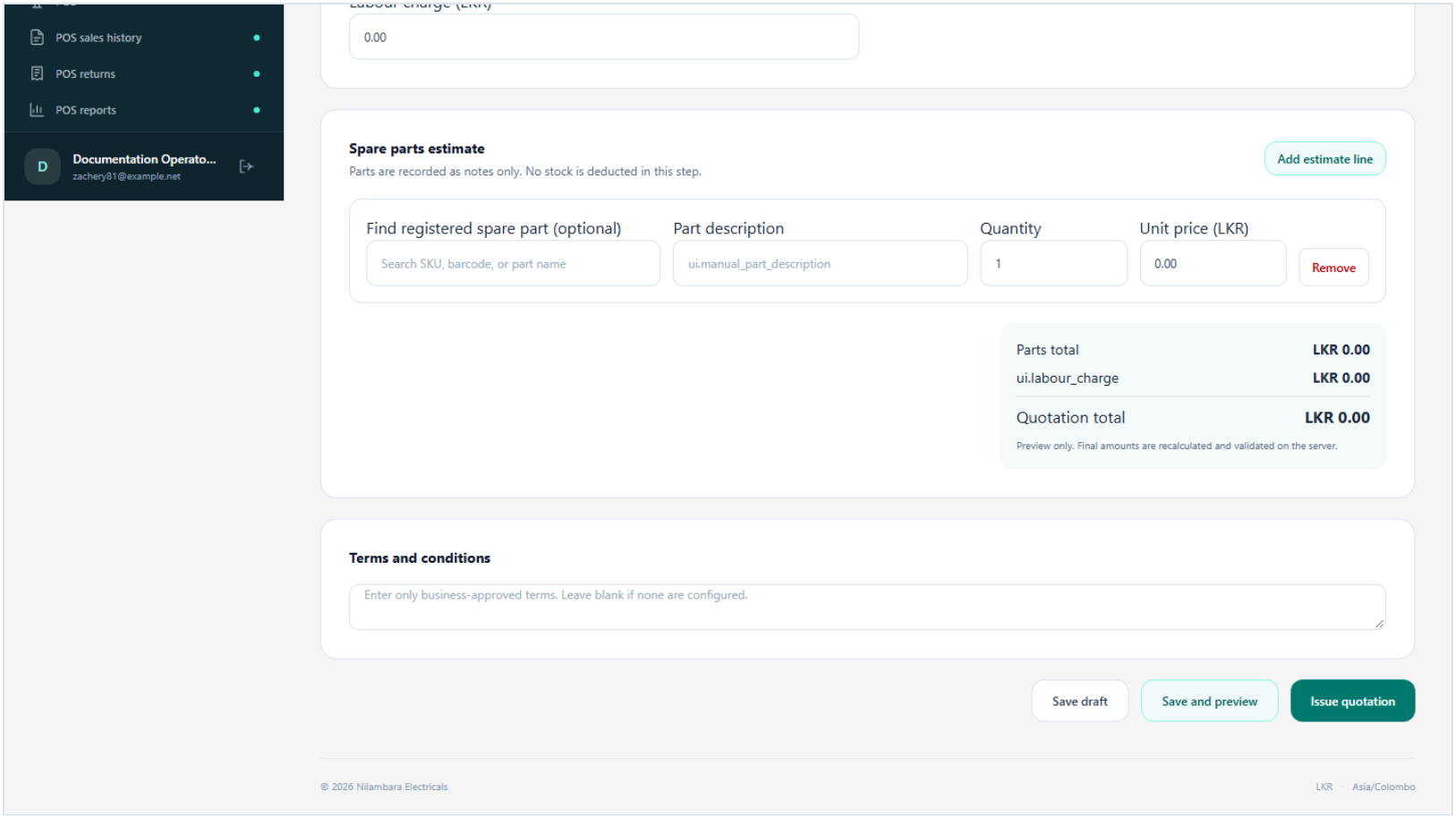
Actual Chrome screenshot | /quotations/create | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-create.png. No operational records or passwords are displayed.

Figure 16.2 - Quotation creation

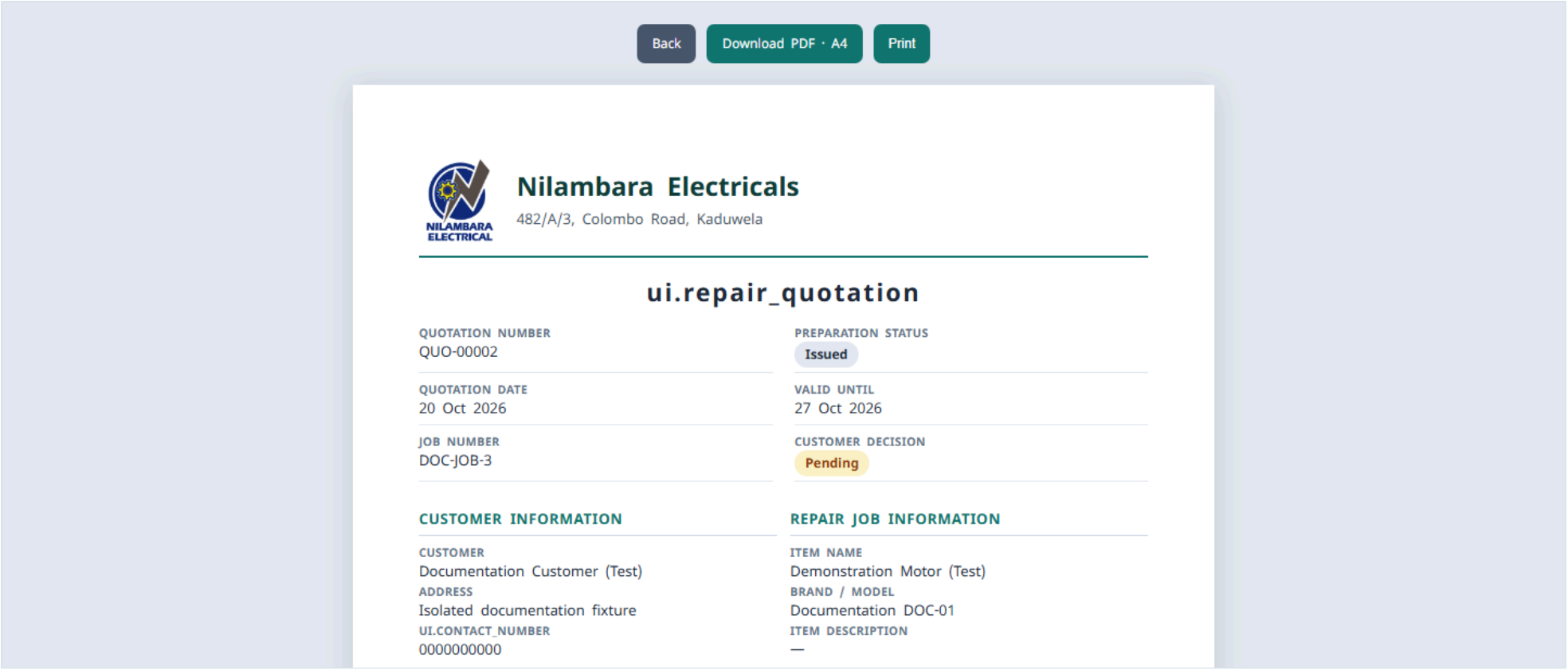
Actual Chrome screenshot | /quotations/create | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-create.png. No operational records or passwords are displayed.

Figure 17.1 - Quotation preview

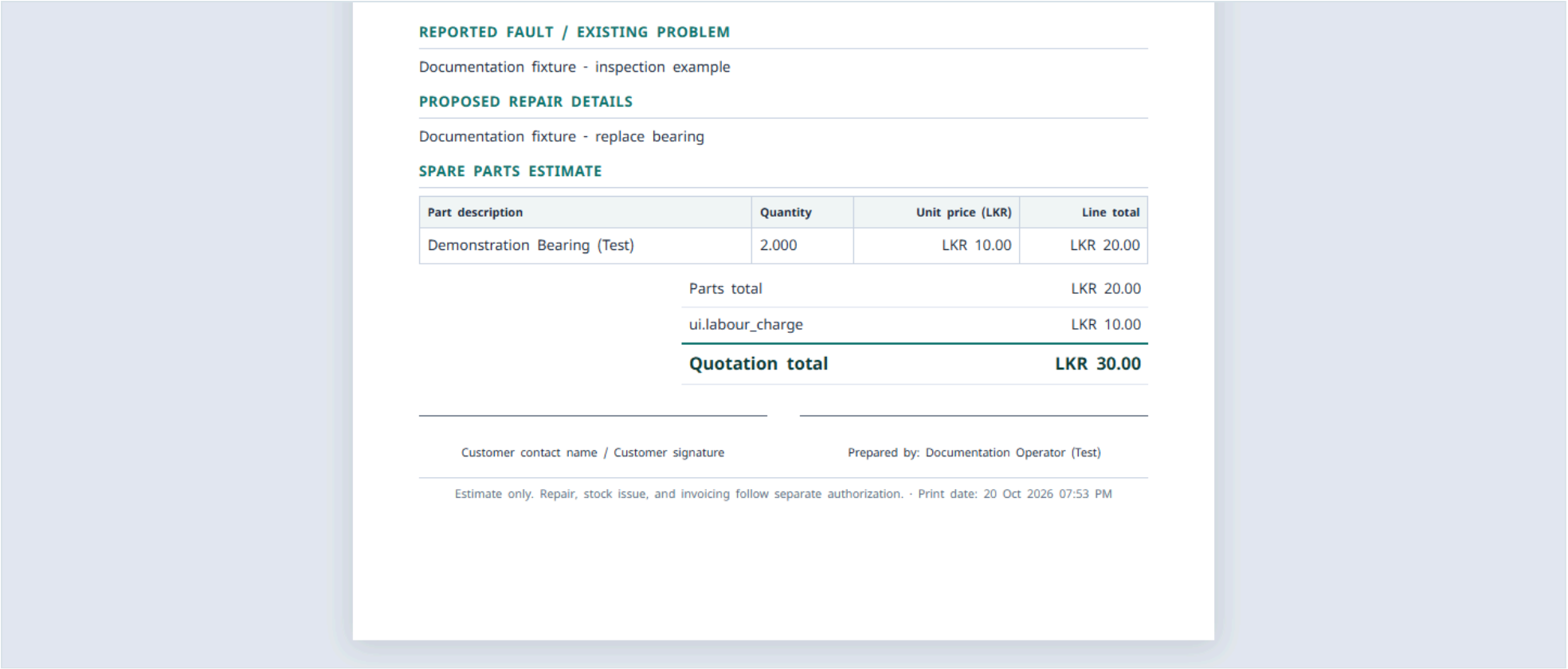
Actual Chrome screenshot | /quotations/2/print | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-preview.png. No operational records or passwords are displayed.

Figure 17.2 - Quotation preview

Actual Chrome screenshot | /quotations/2/print | isolated test fixtures | panel 2/2



REPORTED FAULT / EXISTING PROBLEM

Documentation fixture - inspection example

PROPOSED REPAIR DETAILS

Documentation fixture - replace bearing

SPARE PARTS ESTIMATE

Part description	Quantity	Unit price (LKR)	Line total
Demonstration Bearing (Test)	2.000	LKR 10.00	LKR 20.00

Parts total	LKR 20.00
ui.labour_charge	LKR 10.00
Quotation total	**LKR 30.00**

Customer contact name / Customer signature

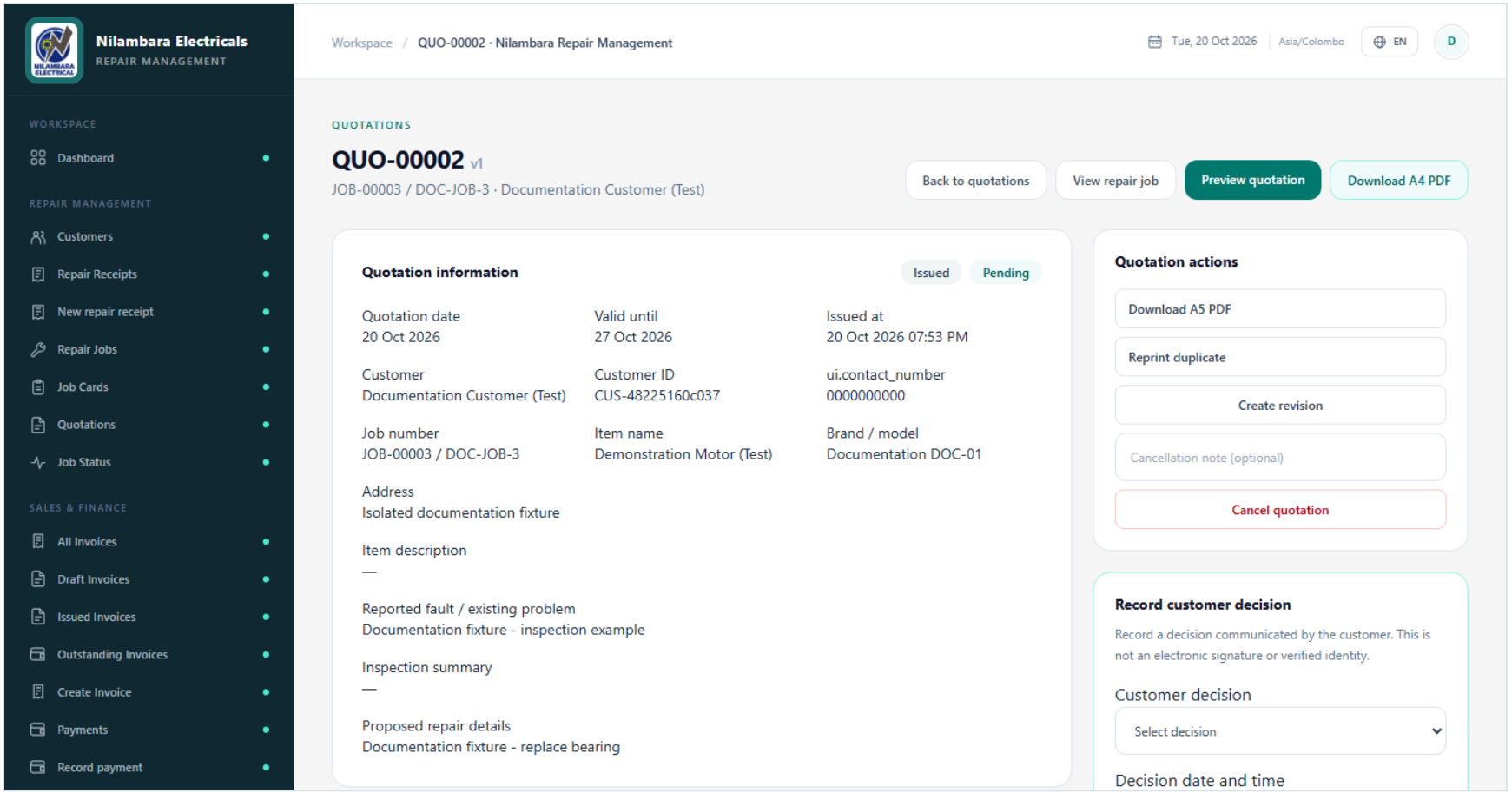
Prepared by: Documentation Operator (Test)

Estimate only. Repair, stock issue, and invoicing follow separate authorization. · Print date: 20 Oct 2026 07:53 PM

Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-preview.png. No operational records or passwords are displayed.

Figure 18.1 - Quotation details and customer decision

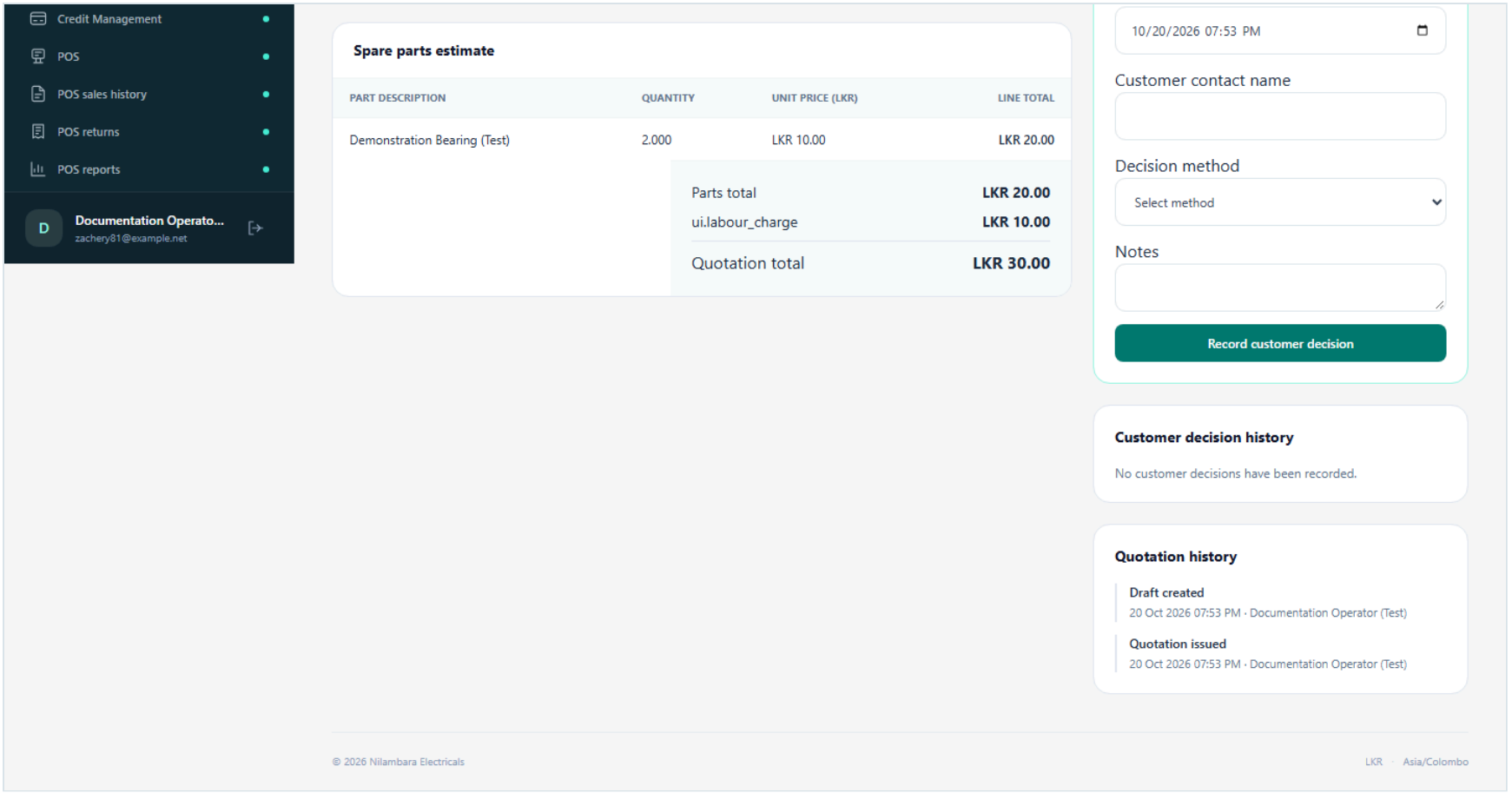
Actual Chrome screenshot | /quotations/2 | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-approval.png. No operational records or passwords are displayed.

Figure 18.2 - Quotation details and customer decision

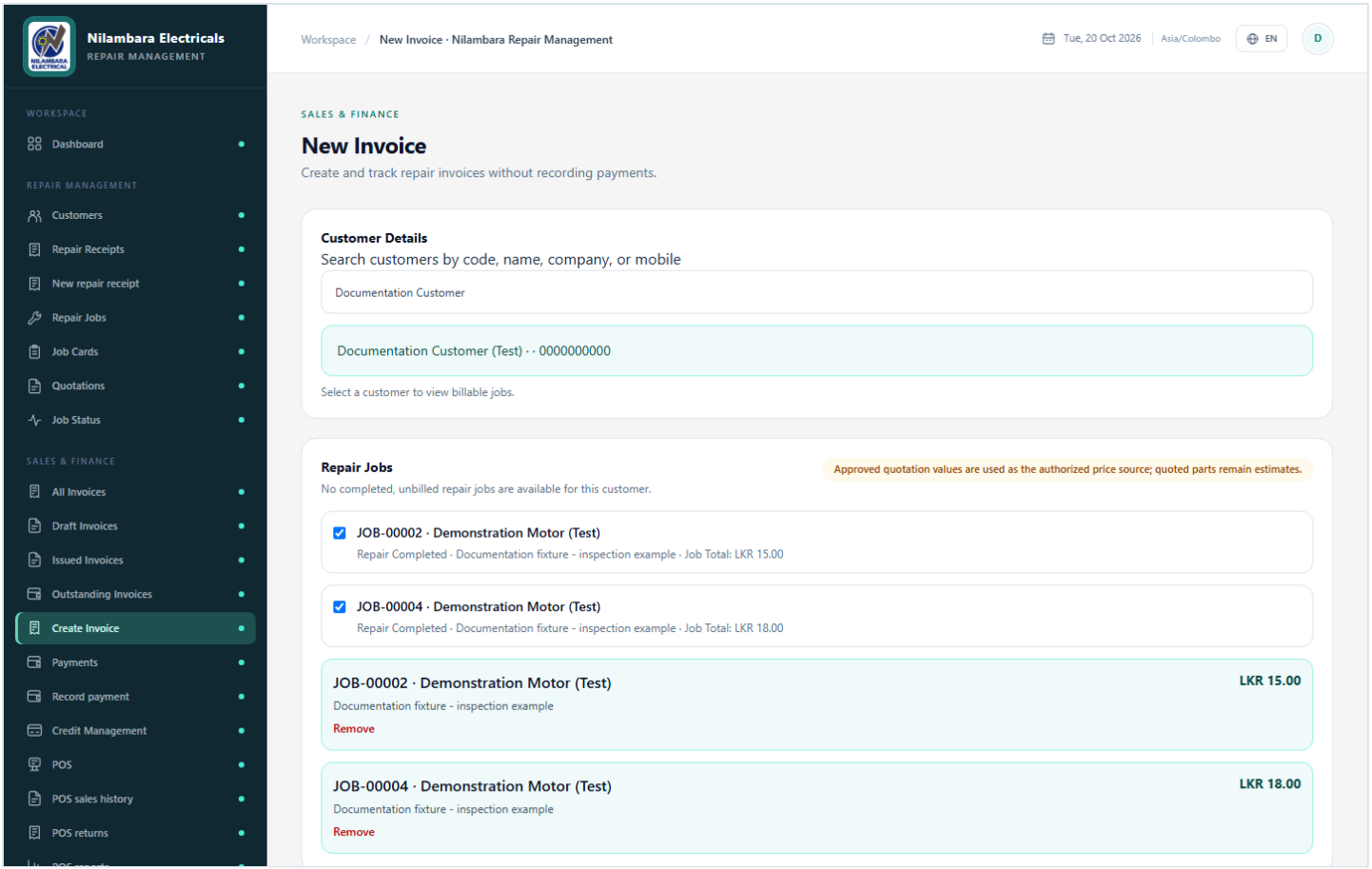
Actual Chrome screenshot | /quotations/2 | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/quotation-approval.png. No operational records or passwords are displayed.

Figure 19.1 - Invoice creation - two eligible same-customer jobs selected (not submitted)

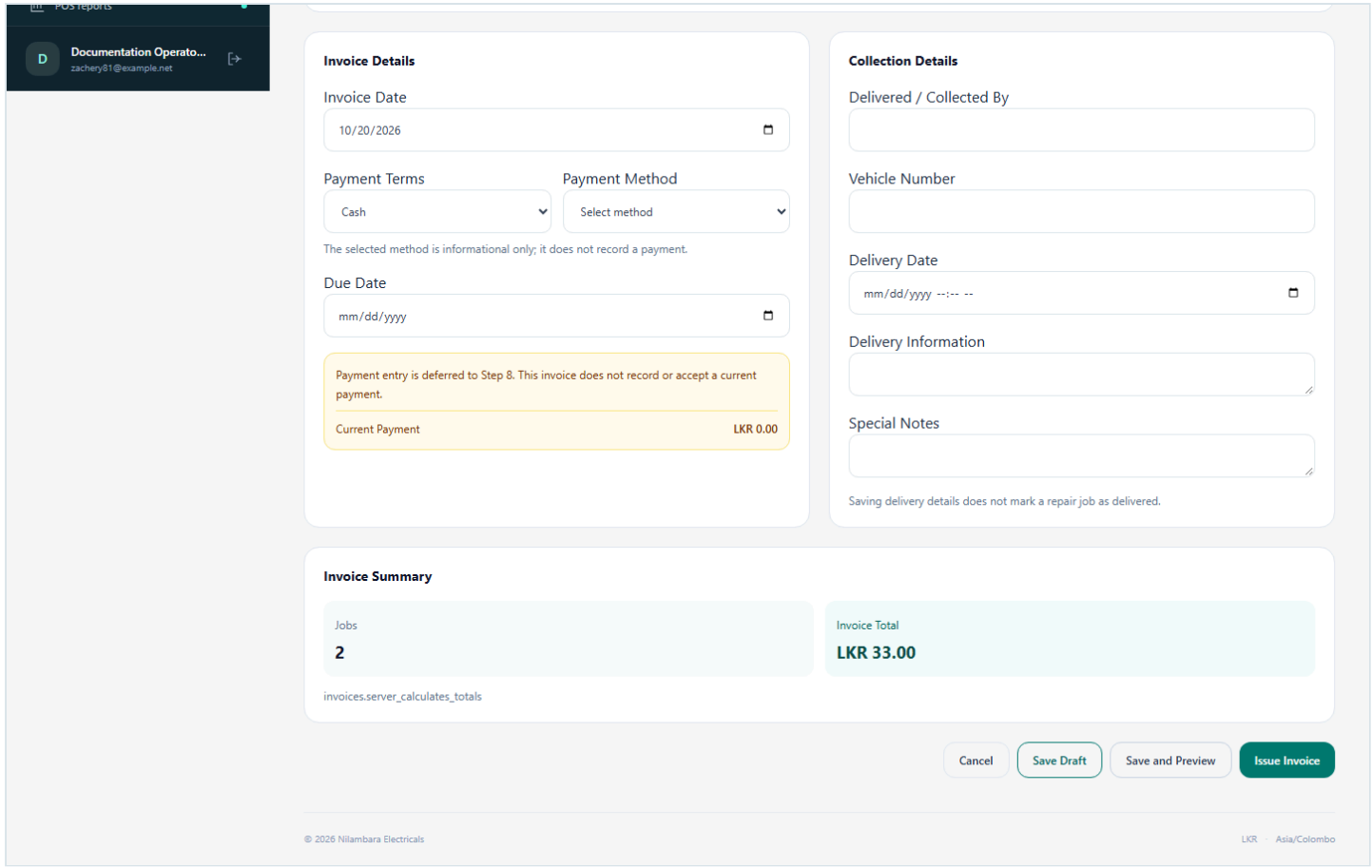
Actual Chrome screenshot | /invoices/create?customer_id=1 | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/invoice-create.png. No operational records or passwords are displayed.

Figure 19.2 - Invoice creation - two eligible same-customer jobs selected (not submitted)

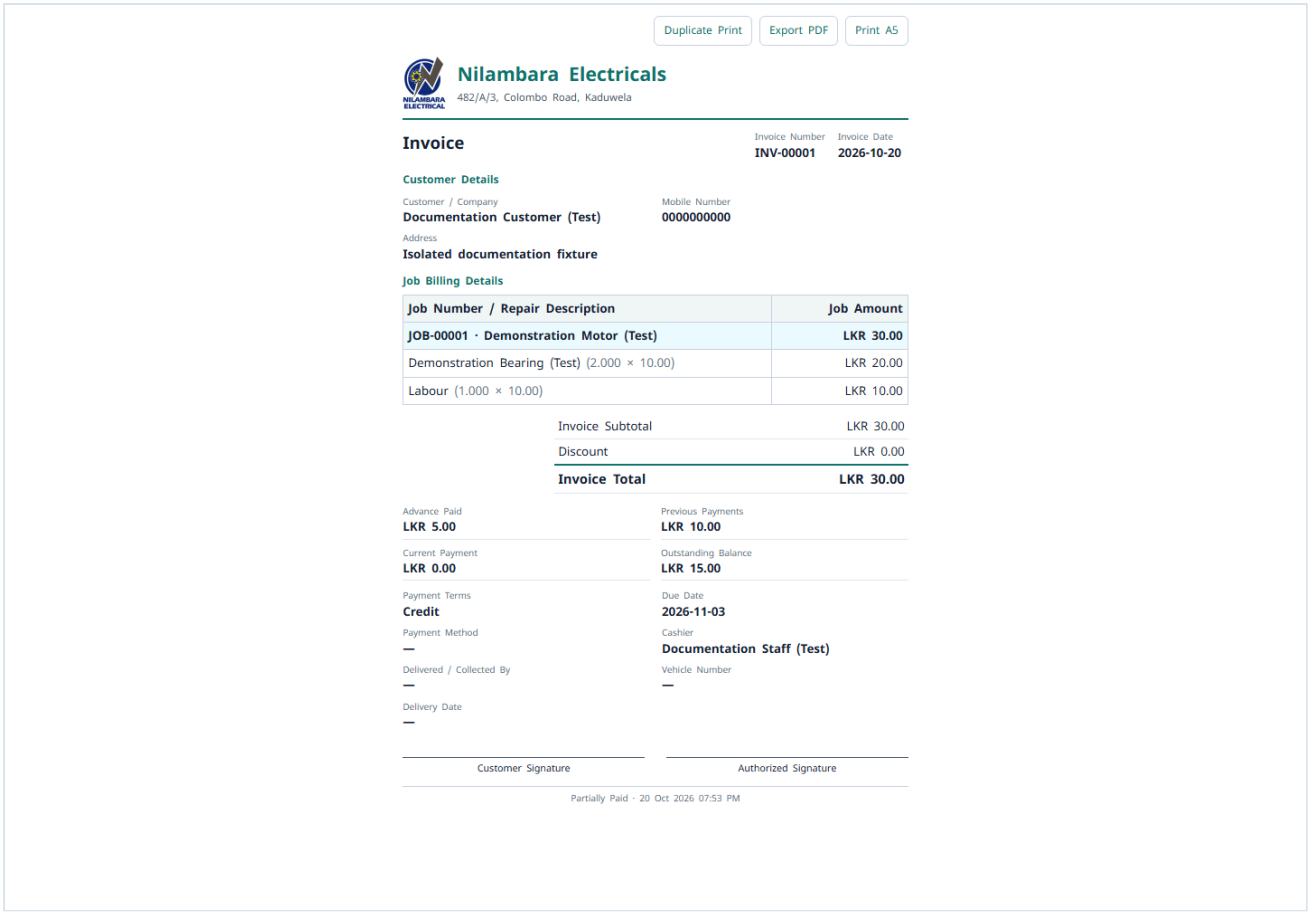
Actual Chrome screenshot | /invoices/create?customer_id=1 | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/invoice-create.png. No operational records or passwords are displayed.

Figure 20.1 - A5 invoice preview

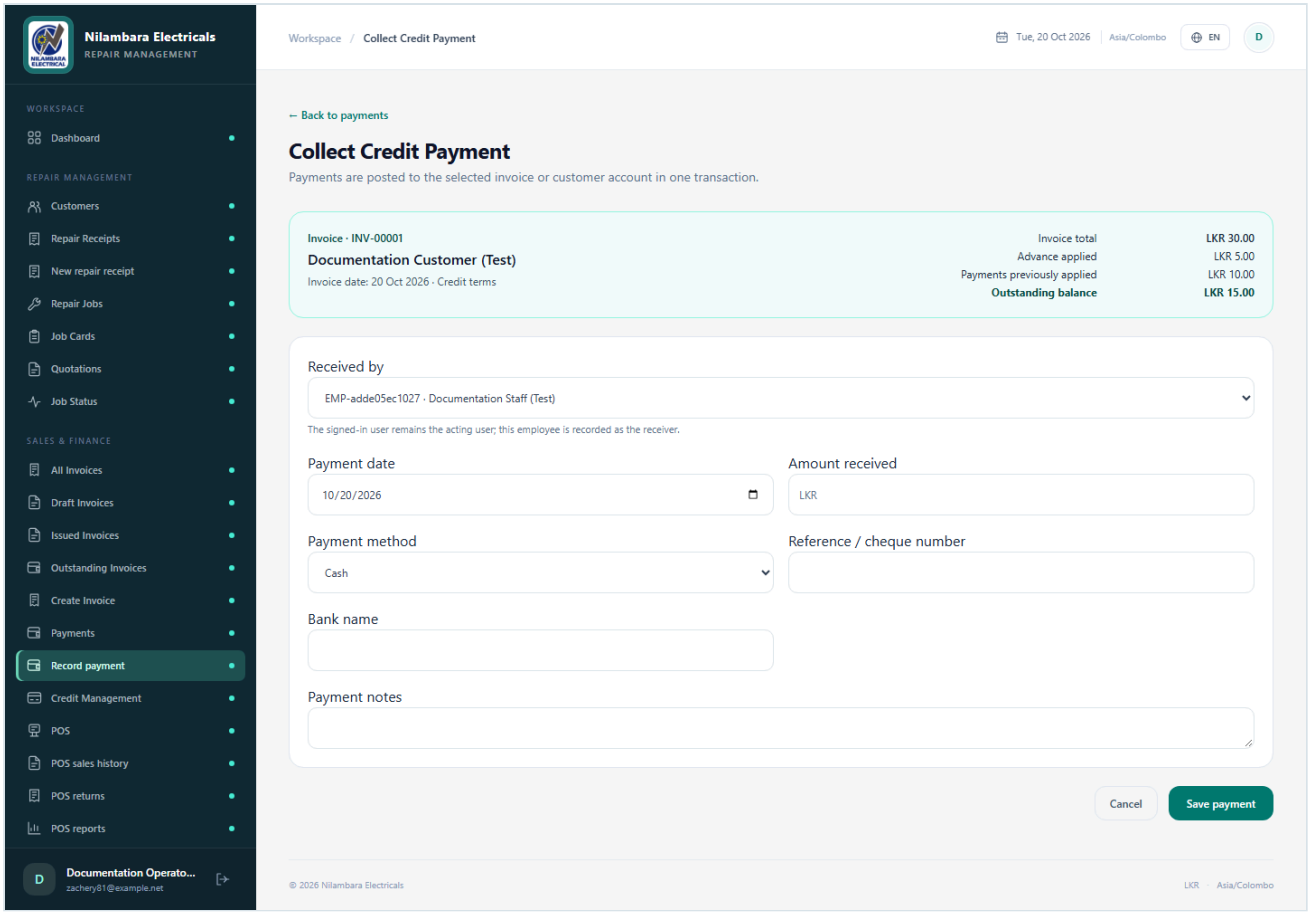
Actual Chrome screenshot | /invoices/1/print | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/invoice-preview.png. No operational records or passwords are displayed.

Figure 21.1 - Payment entry

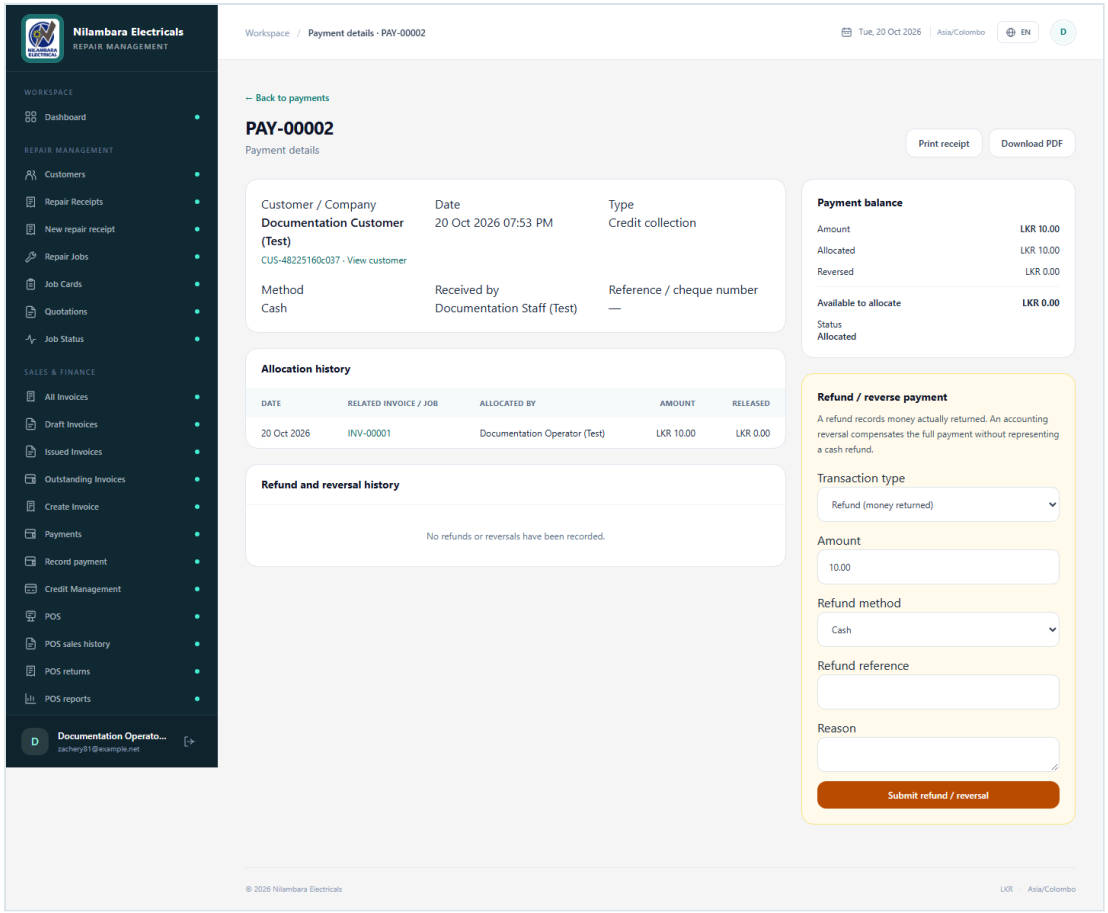
Actual Chrome screenshot | /payments/create?invoice_id=1 | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/payment-create.png. No operational records or passwords are displayed.

Figure 22.1 - Payment details

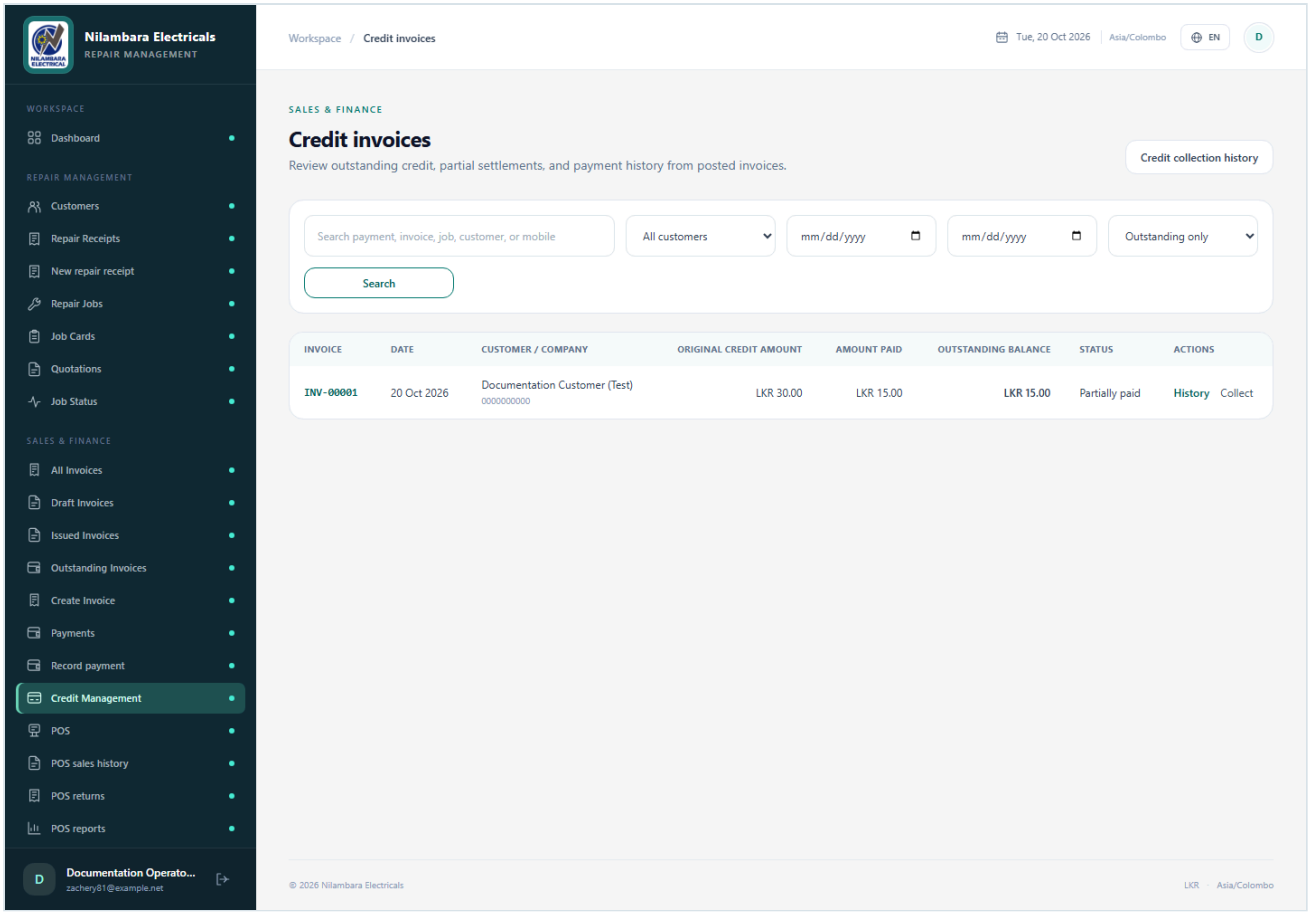
Actual Chrome screenshot | /payments/2 | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/payment-detail.png. No operational records or passwords are displayed.

Figure 23.1 - Credit invoice management

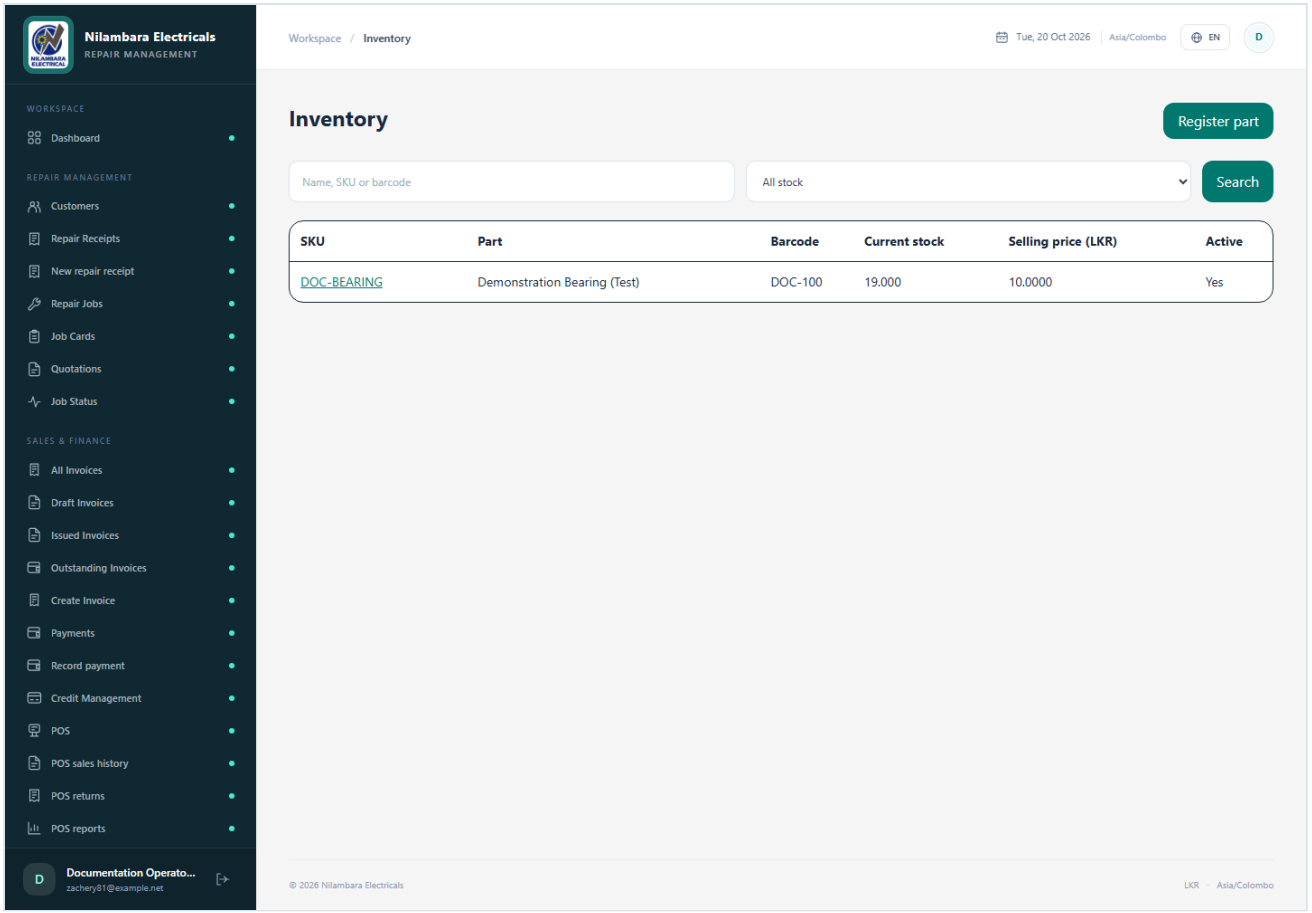
Actual Chrome screenshot | /credit-invoices | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/credit.png. No operational records or passwords are displayed.

Figure 24.1 - Inventory list

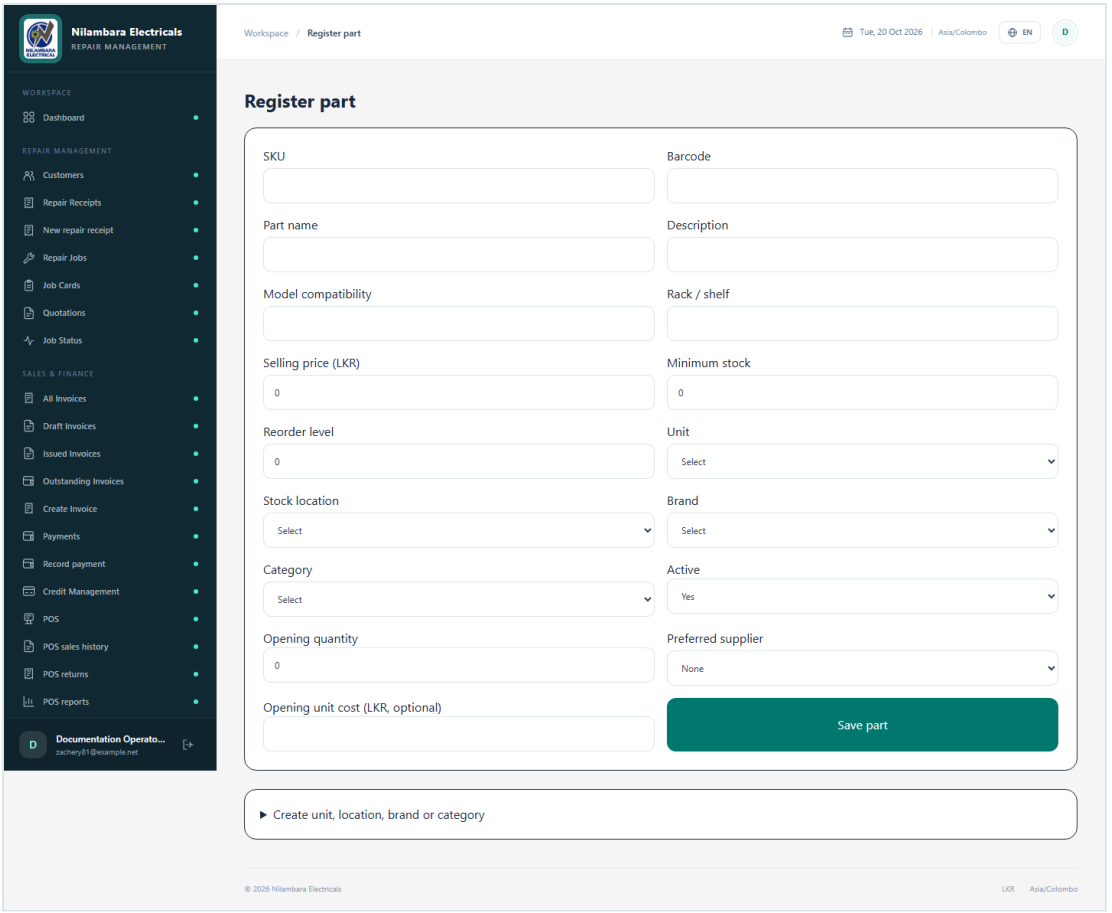
Actual Chrome screenshot | /inventory | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/inventory.png. No operational records or passwords are displayed.

Figure 25.1 - Spare part registration

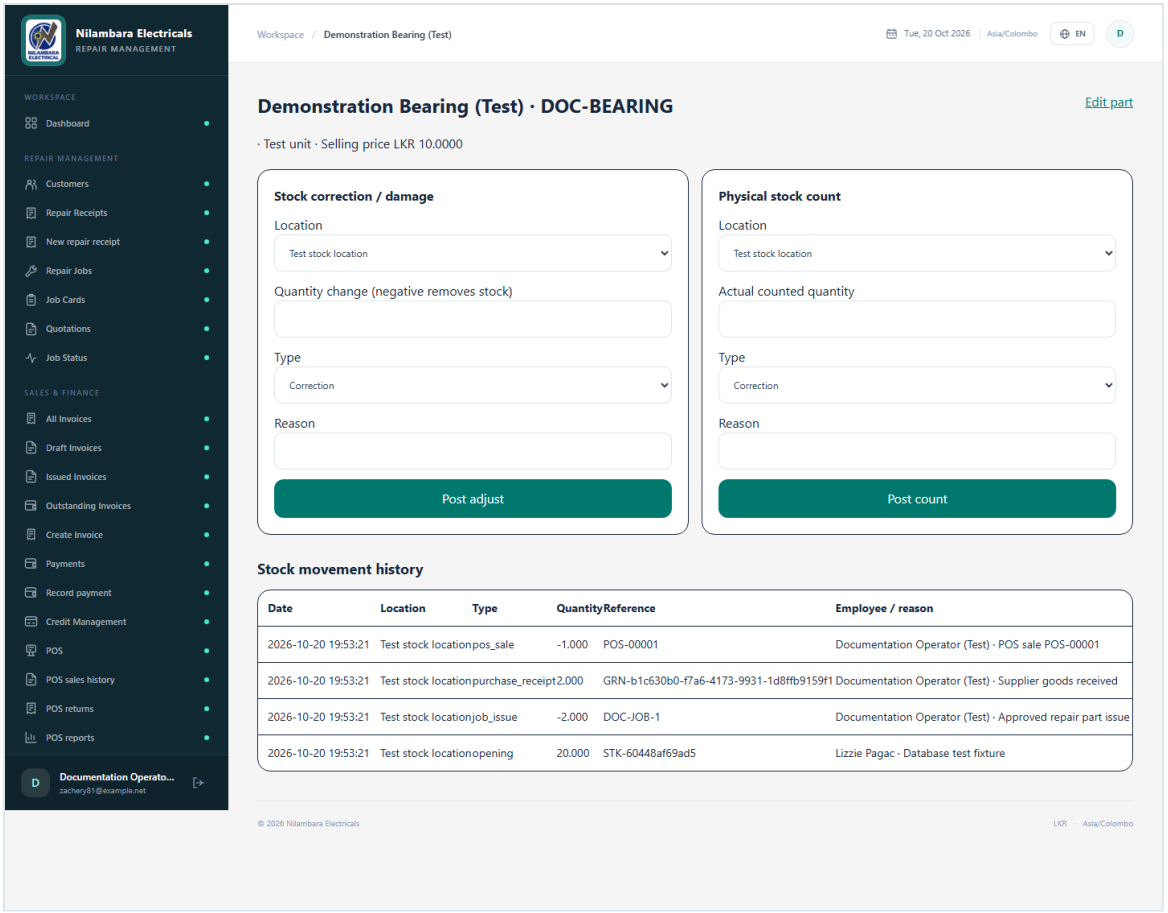
Actual Chrome screenshot | /inventory/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/part-create.png. No operational records or passwords are displayed.

Figure 26.1 - Part details and movements

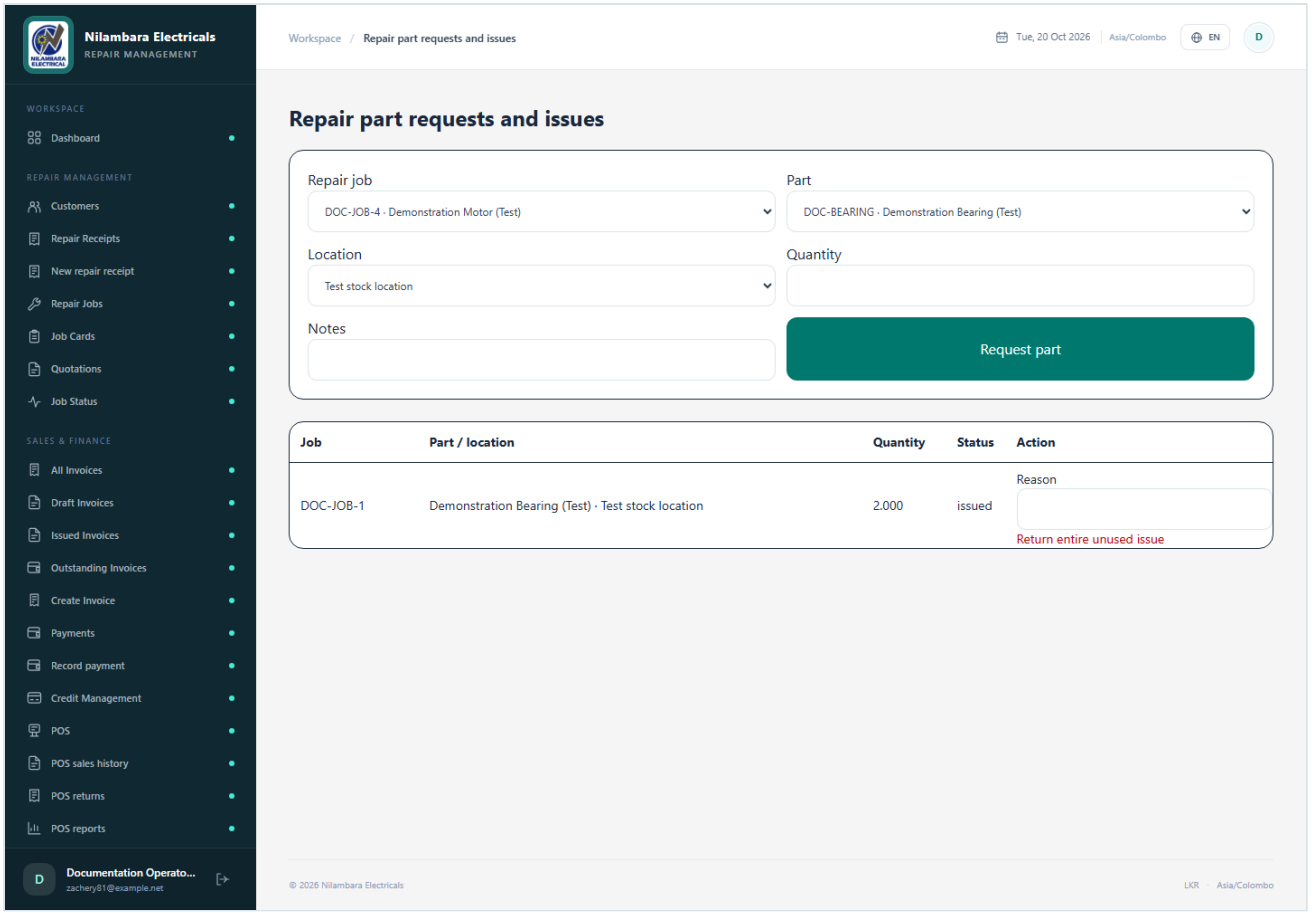
Actual Chrome screenshot | /inventory/1 | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/stock-movements.png. No operational records or passwords are displayed.

Figure 27.1 - Repair part requests and issue

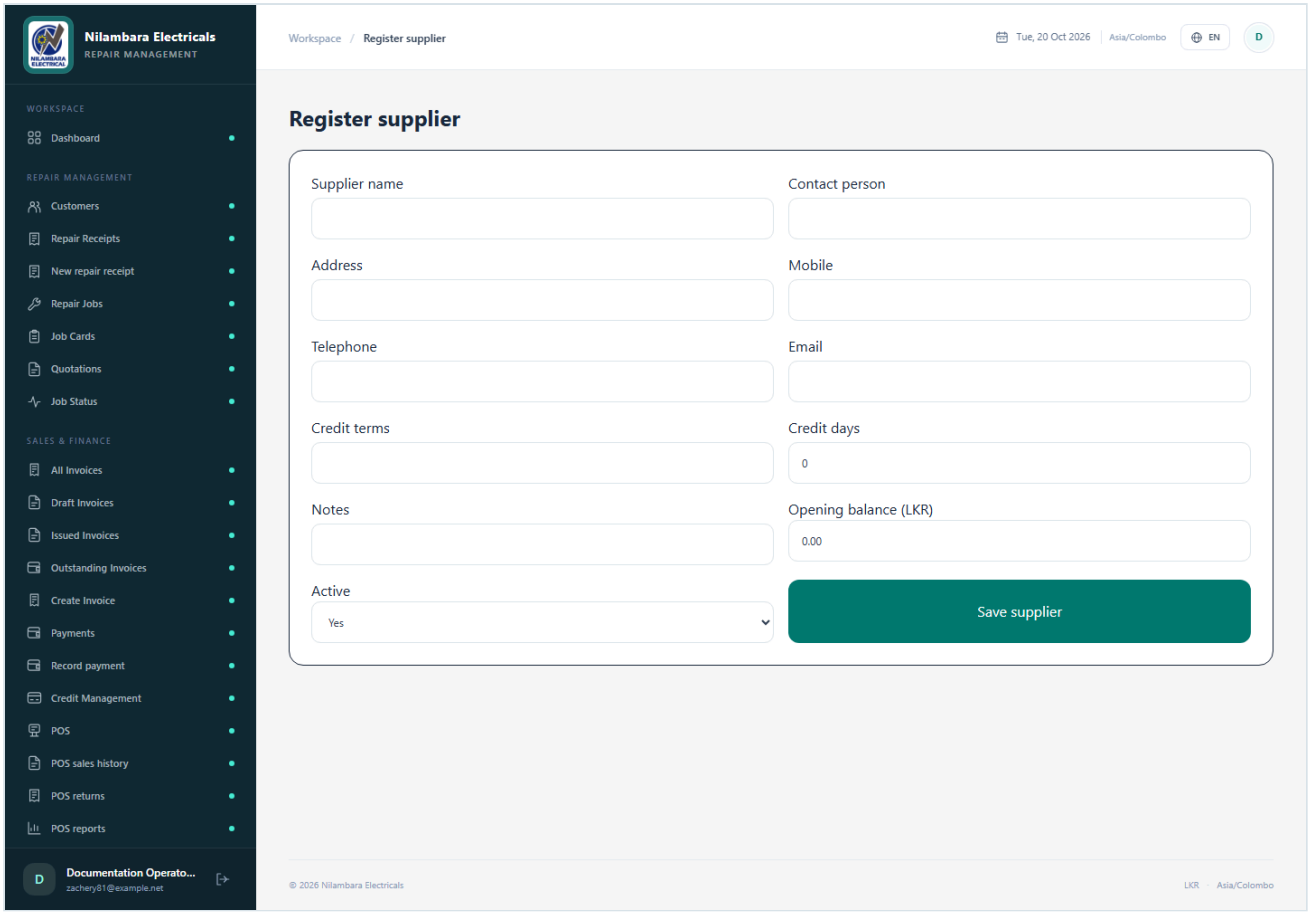
Actual Chrome screenshot | /stock-issues | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/stock-issues.png. No operational records or passwords are displayed.

Figure 28.1 - Supplier registration

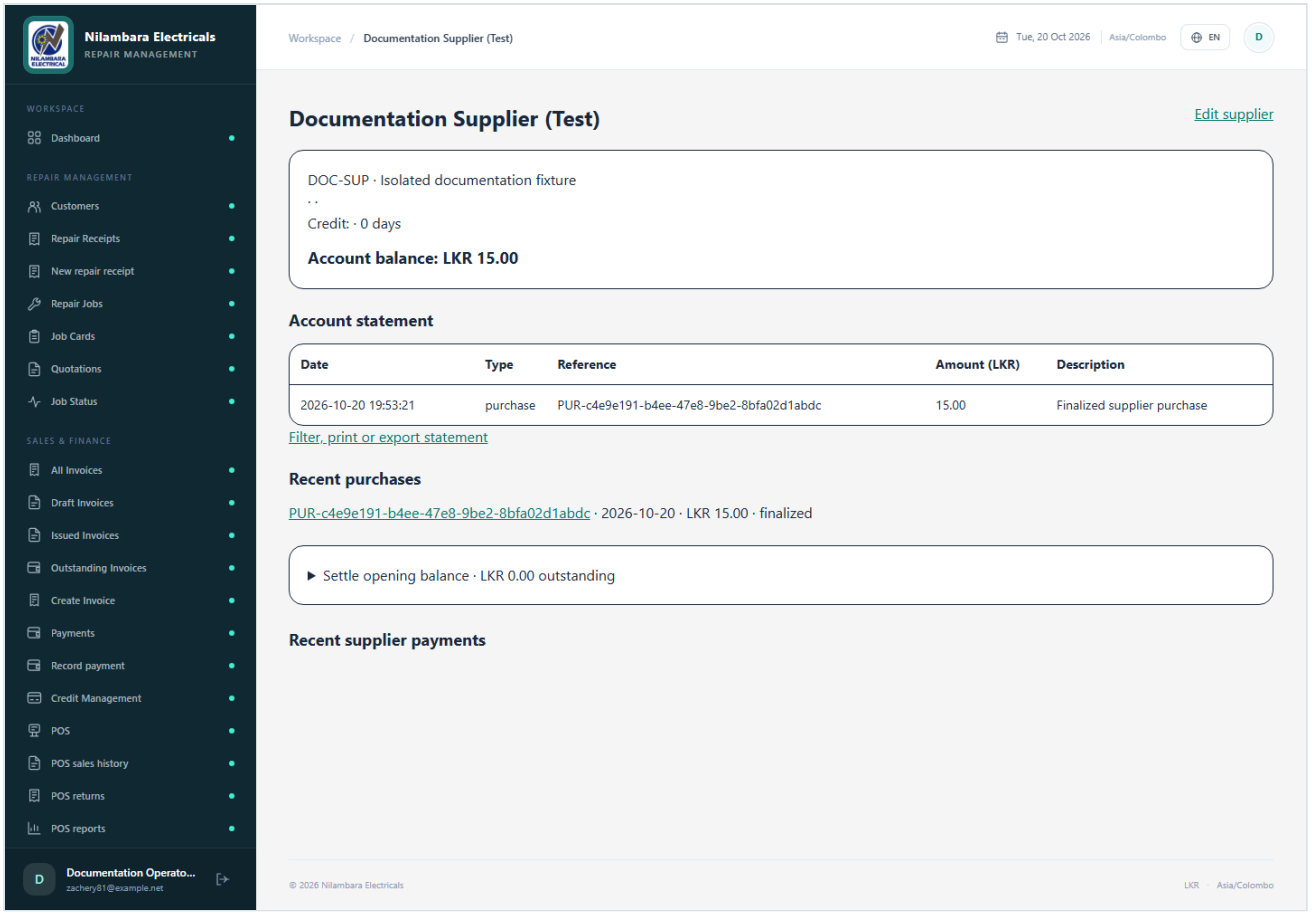
Actual Chrome screenshot | /suppliers/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/supplier-create.png. No operational records or passwords are displayed.

Figure 29.1 - Supplier account

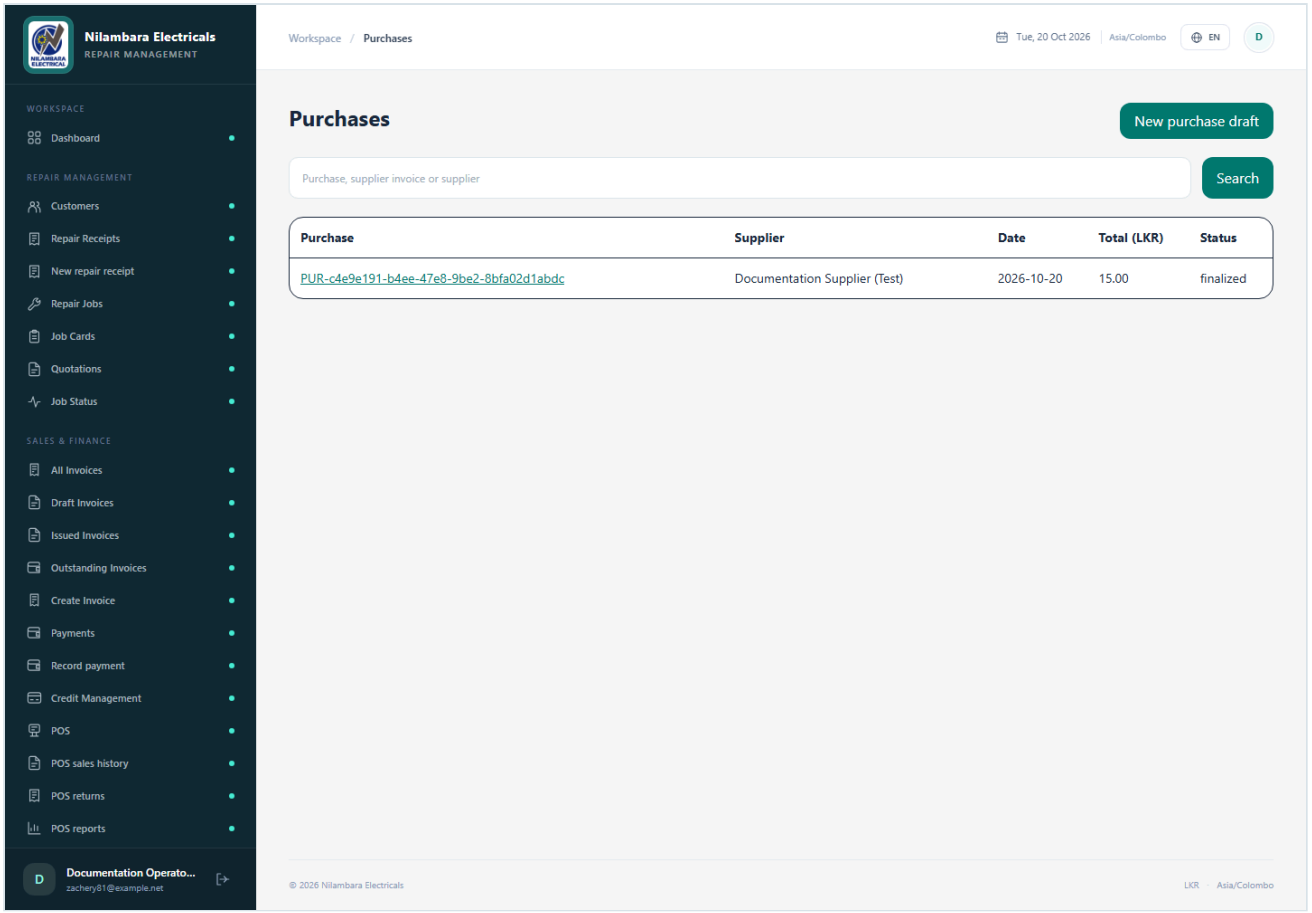
Actual Chrome screenshot | /suppliers/1 | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/supplier-detail.png. No operational records or passwords are displayed.

Figure 30.1 - Purchase management

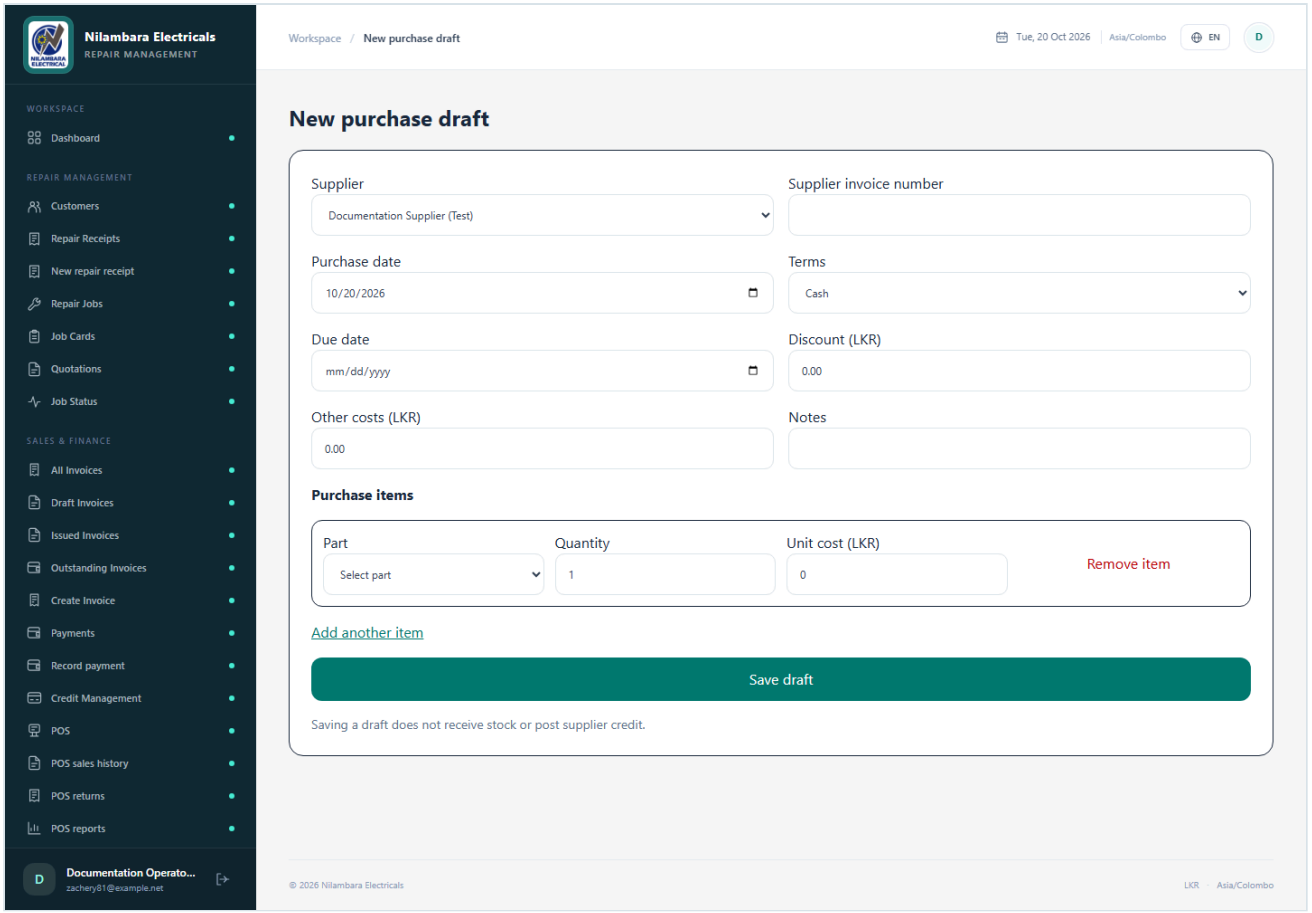
Actual Chrome screenshot | /purchases | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/purchases.png. No operational records or passwords are displayed.

Figure 31.1 - Purchase draft entry

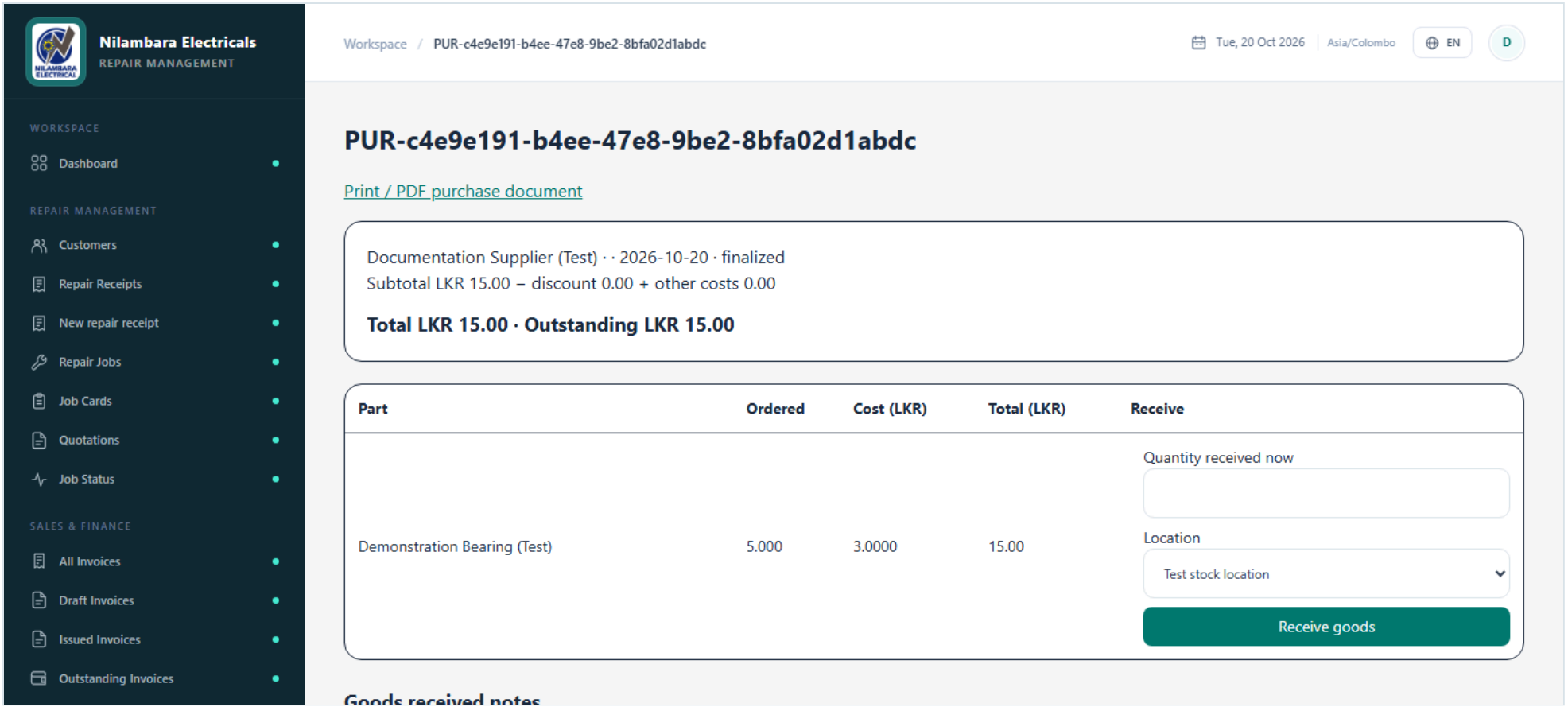
Actual Chrome screenshot | /purchases/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/purchase-create.png. No operational records or passwords are displayed.

Figure 32.1 - Purchase details and GRN receiving

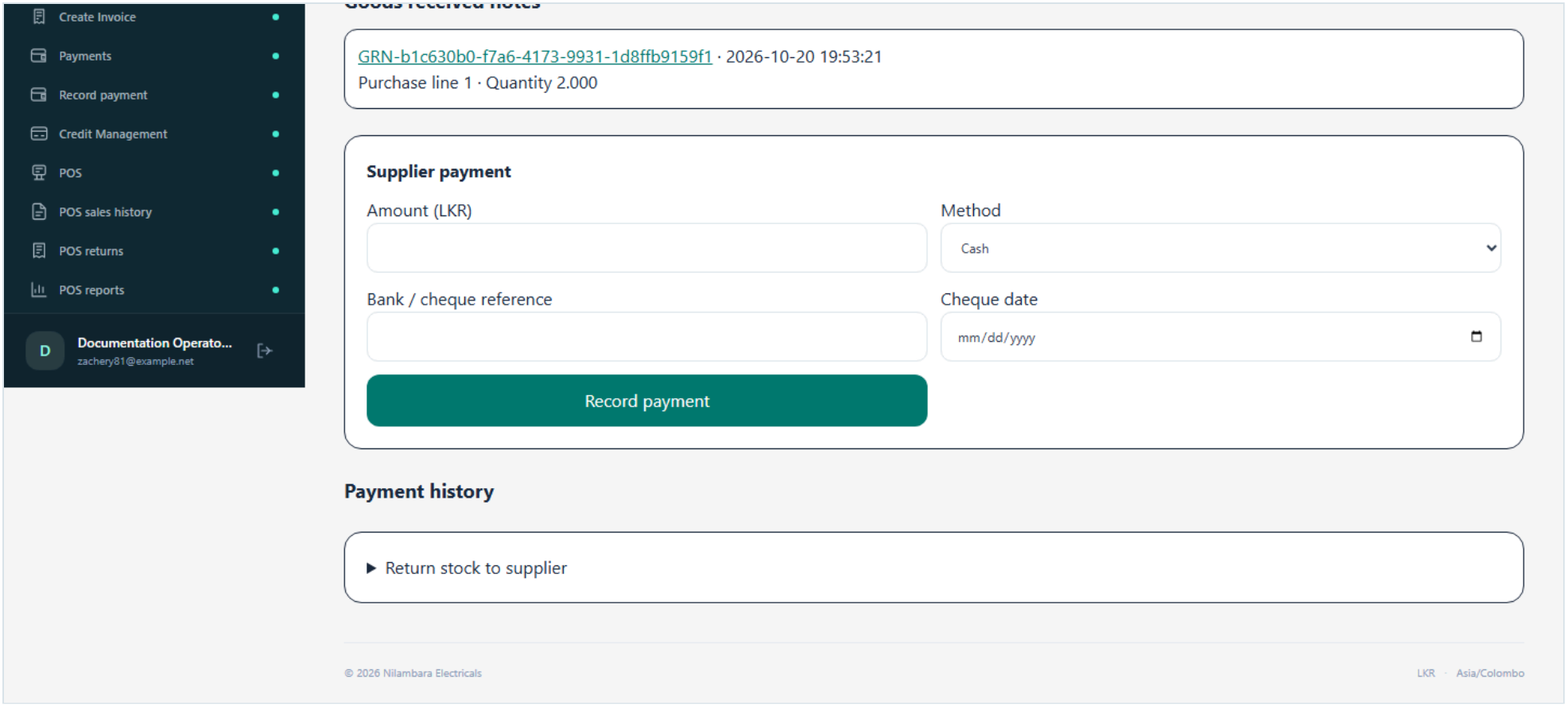
Actual Chrome screenshot | /purchases/1 | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/grn.png. No operational records or passwords are displayed.

Figure 32.2 - Purchase details and GRN receiving

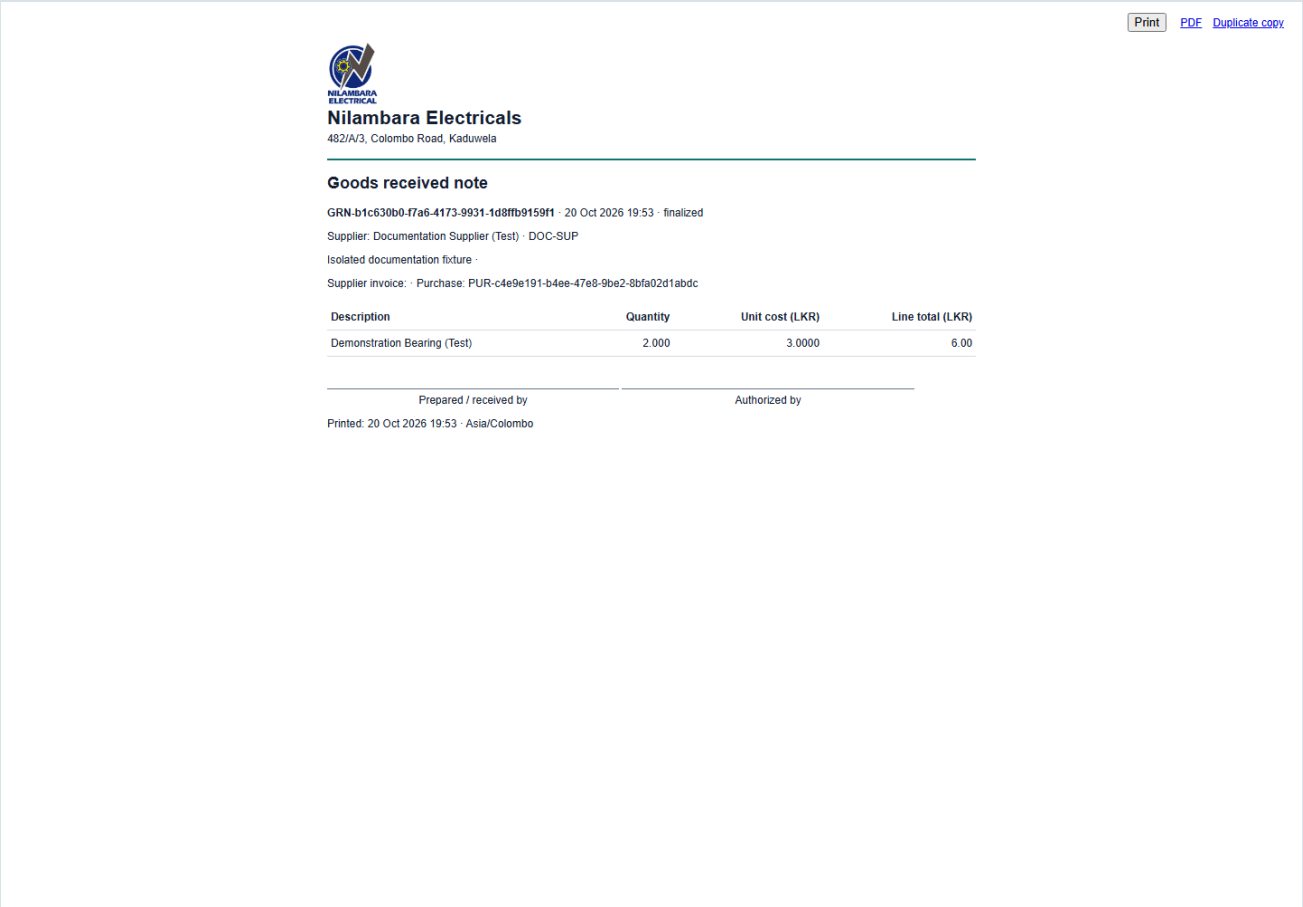
Actual Chrome screenshot | /purchases/1 | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/grn.png. No operational records or passwords are displayed.

Figure 33.1 - GRN document preview

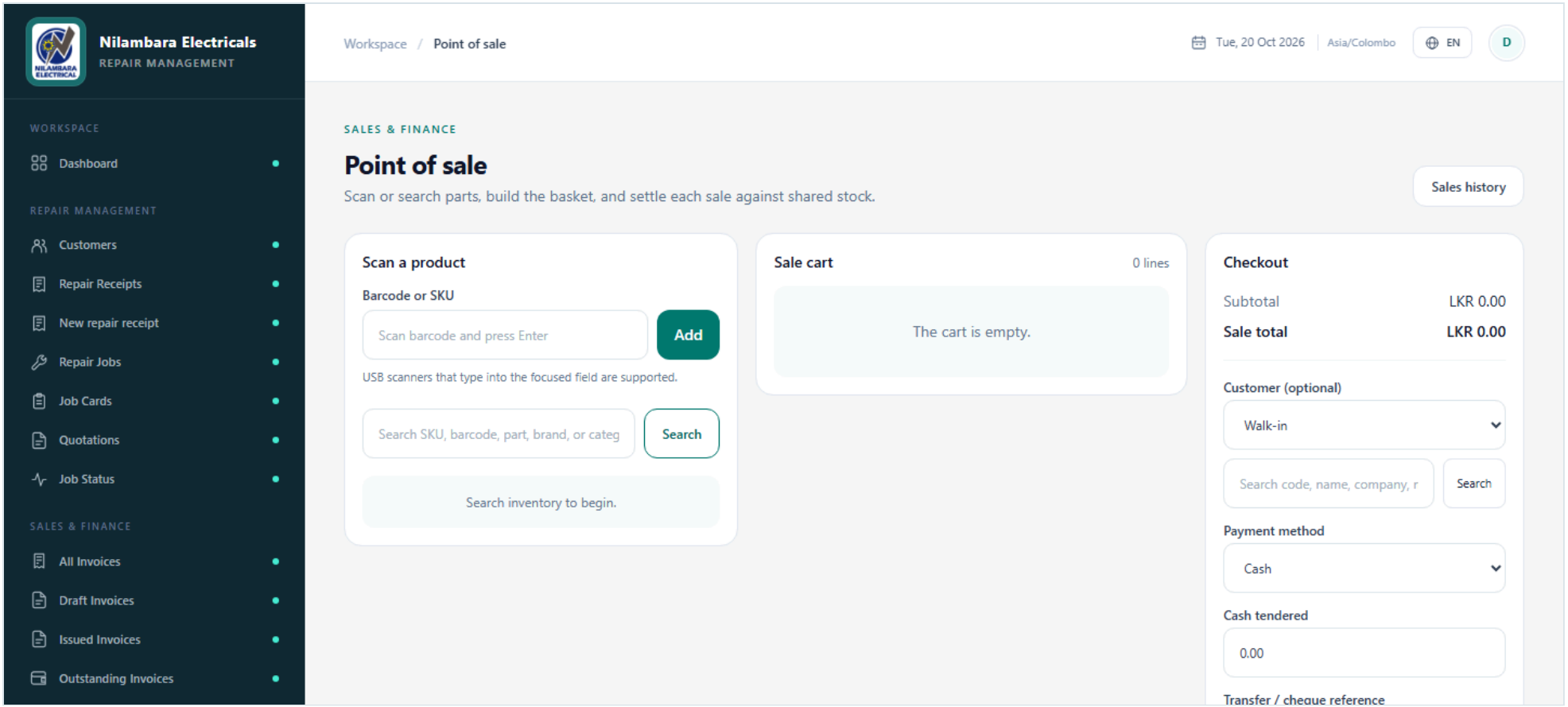
Actual Chrome screenshot | /goods-received-notes/1/document/html | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/grn-preview.png. No operational records or passwords are displayed.

Figure 34.1 - POS checkout

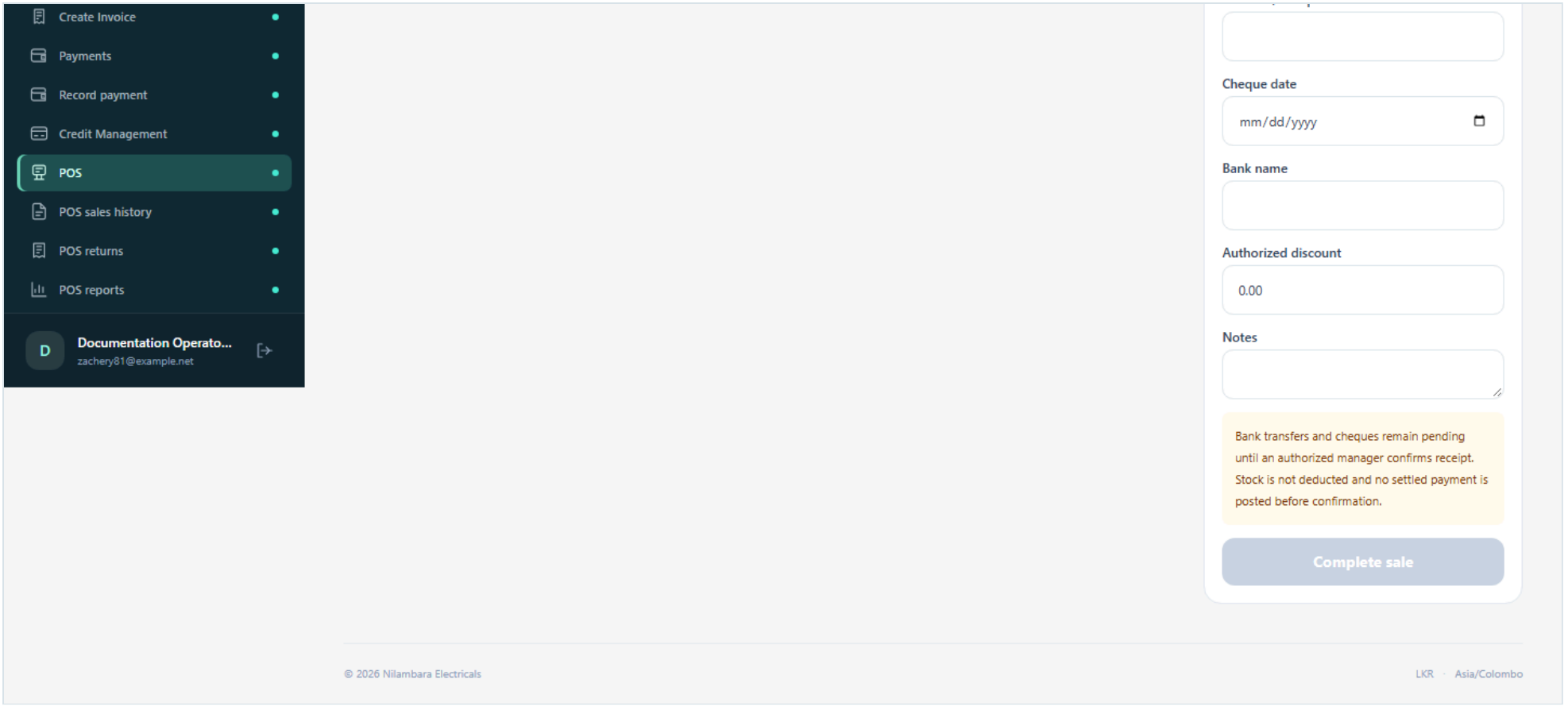
Actual Chrome screenshot | /pos | isolated test fixtures | panel 1/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/pos.png. No operational records or passwords are displayed.

Figure 34.2 - POS checkout

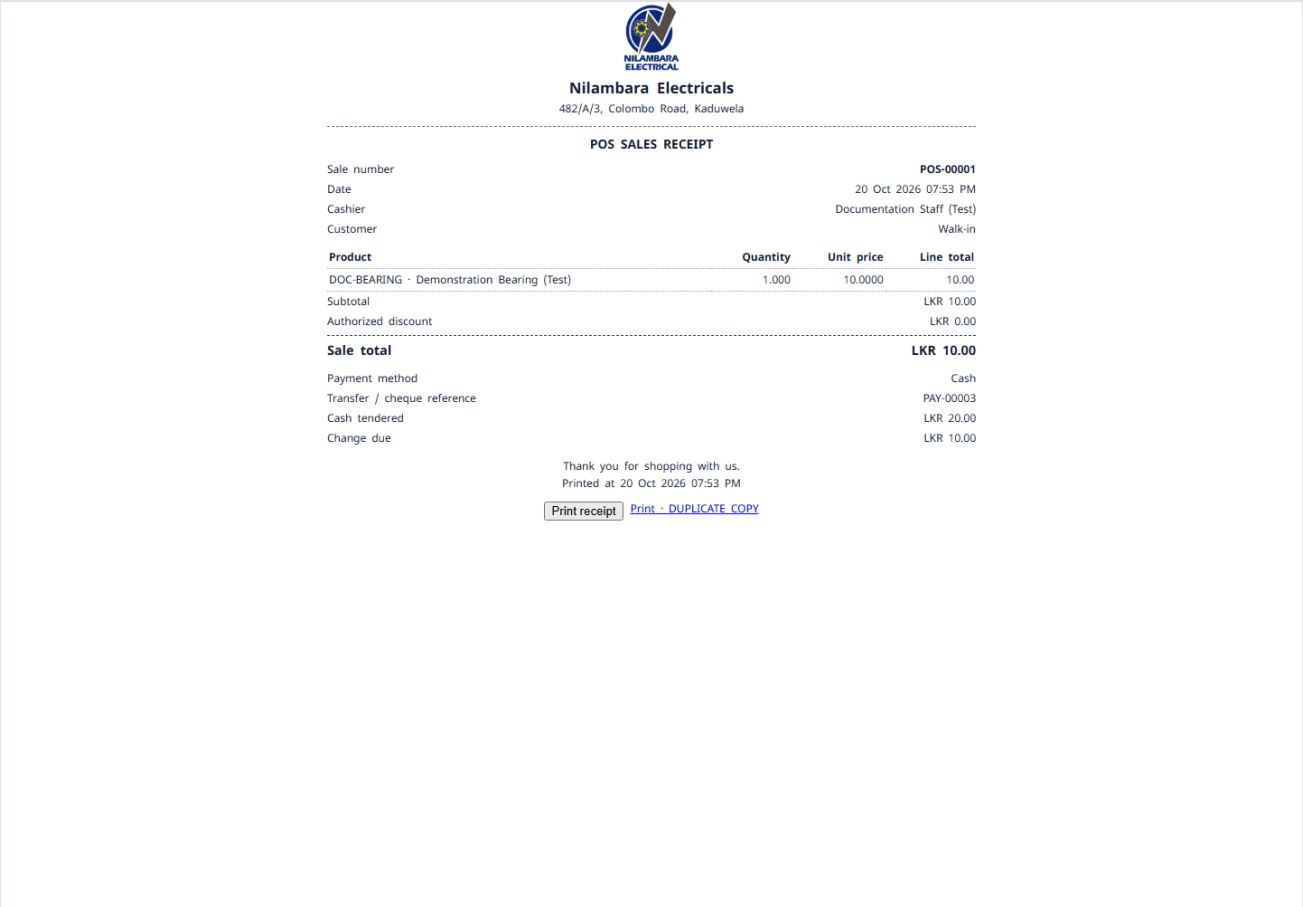
Actual Chrome screenshot | /pos | isolated test fixtures | panel 2/2



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/pos.png. No operational records or passwords are displayed.

Figure 35.1 - POS receipt preview

Actual Chrome screenshot | /pos/sales/1/print | isolated test fixtures | panel 1/1



Nilambara Electricals

482/A/3, Colombo Road, Kaduwela

POS SALES RECEIPT

Sale number	POS-00001
Date	20 Oct 2026 07:53 PM
Cashier	Documentation Staff (Test)
Customer	Walk-in

Product	Quantity	Unit price	Line total
DOC-BEARING · Demonstration Bearing (Test)	1.000	10.0000	10.00

Subtotal	LKR 10.00
Authorized discount	LKR 0.00
Sale total	**LKR 10.00**
Payment method	Cash
Transfer / cheque reference	PAY-00003
Cash tendered	LKR 20.00
Change due	LKR 10.00

Thank you for shopping with us.

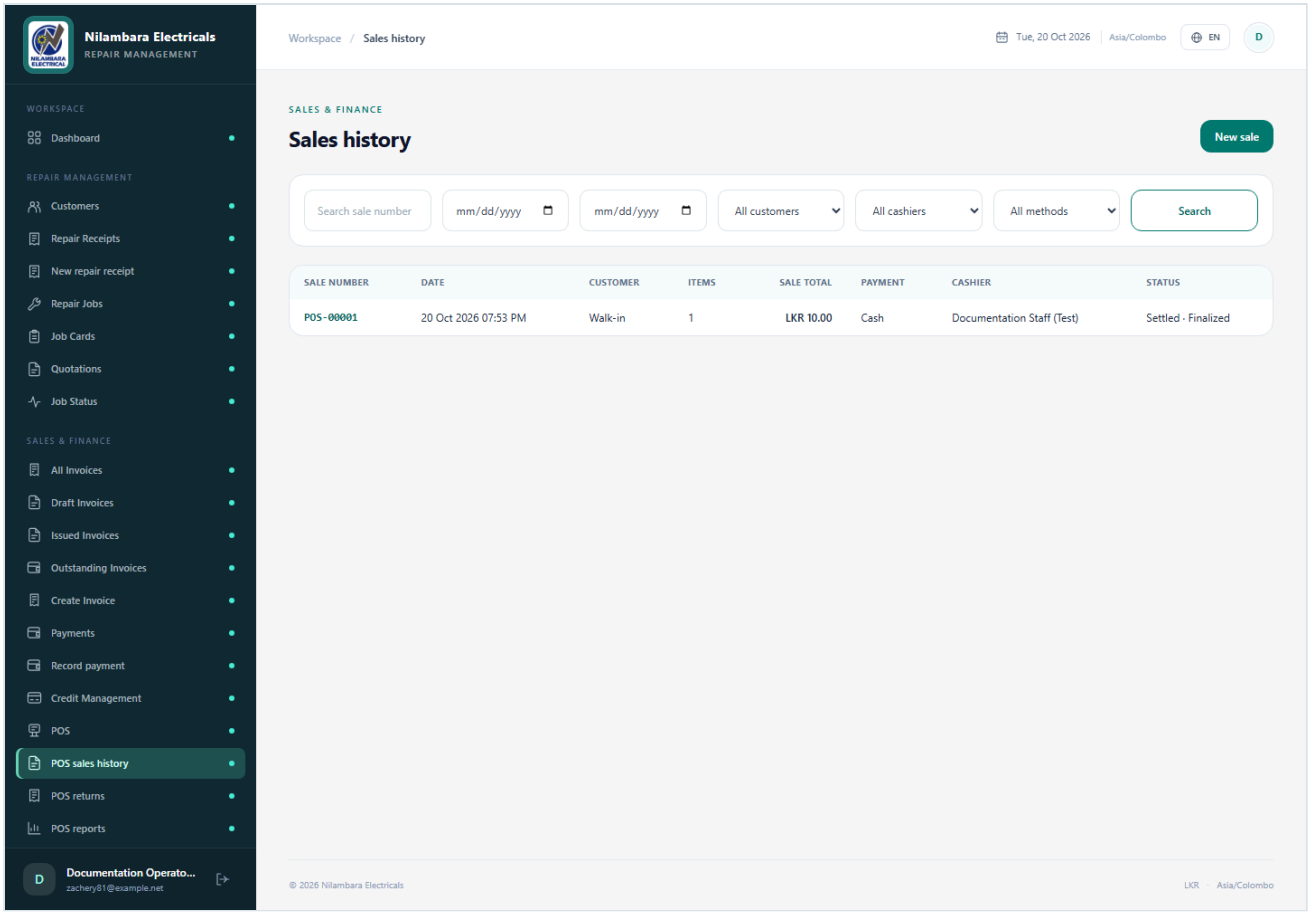
Printed at 20 Oct 2026 07:53 PM

Print receipt Print · DUPLICATE COPY

Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/pos-receipt.png. No operational records or passwords are displayed.

Figure 36.1 - POS sales history

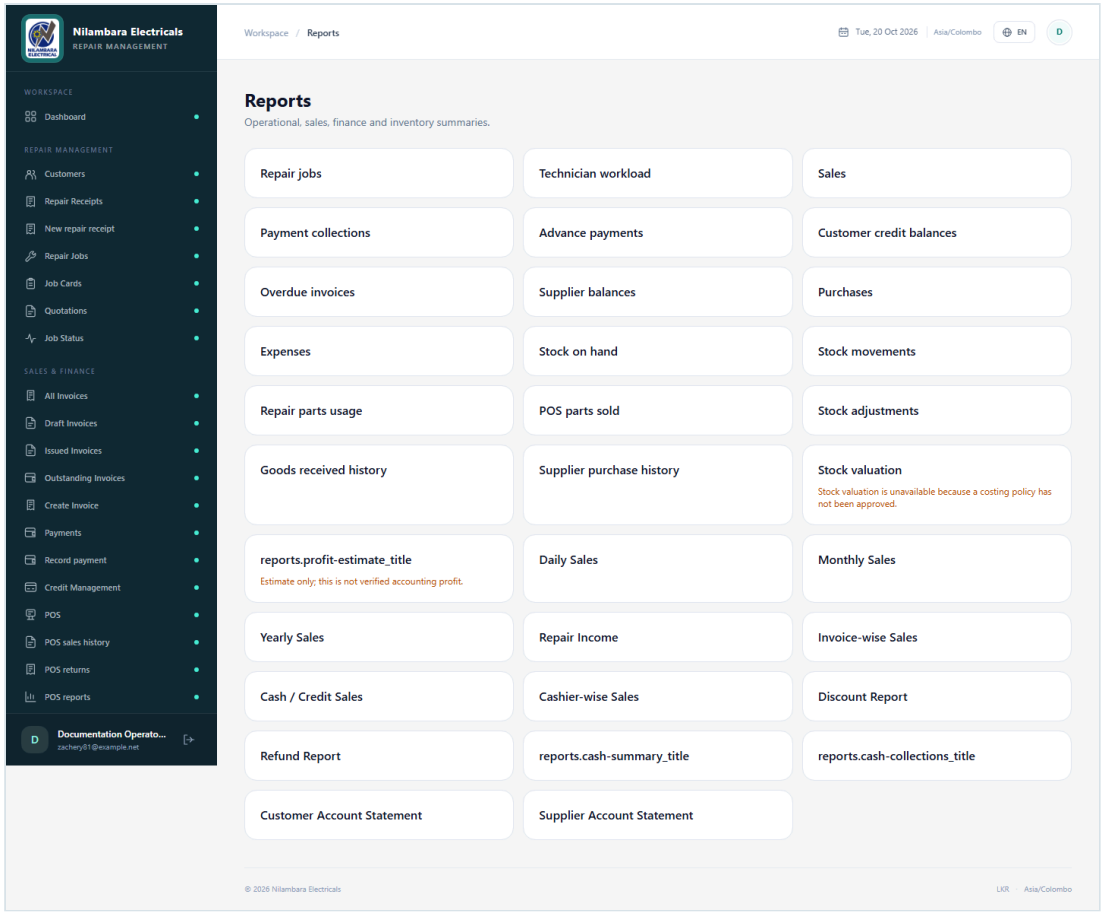
Actual Chrome screenshot | /pos/sales | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/pos-sales.png. No operational records or passwords are displayed.

Figure 37.1 - Report catalogue

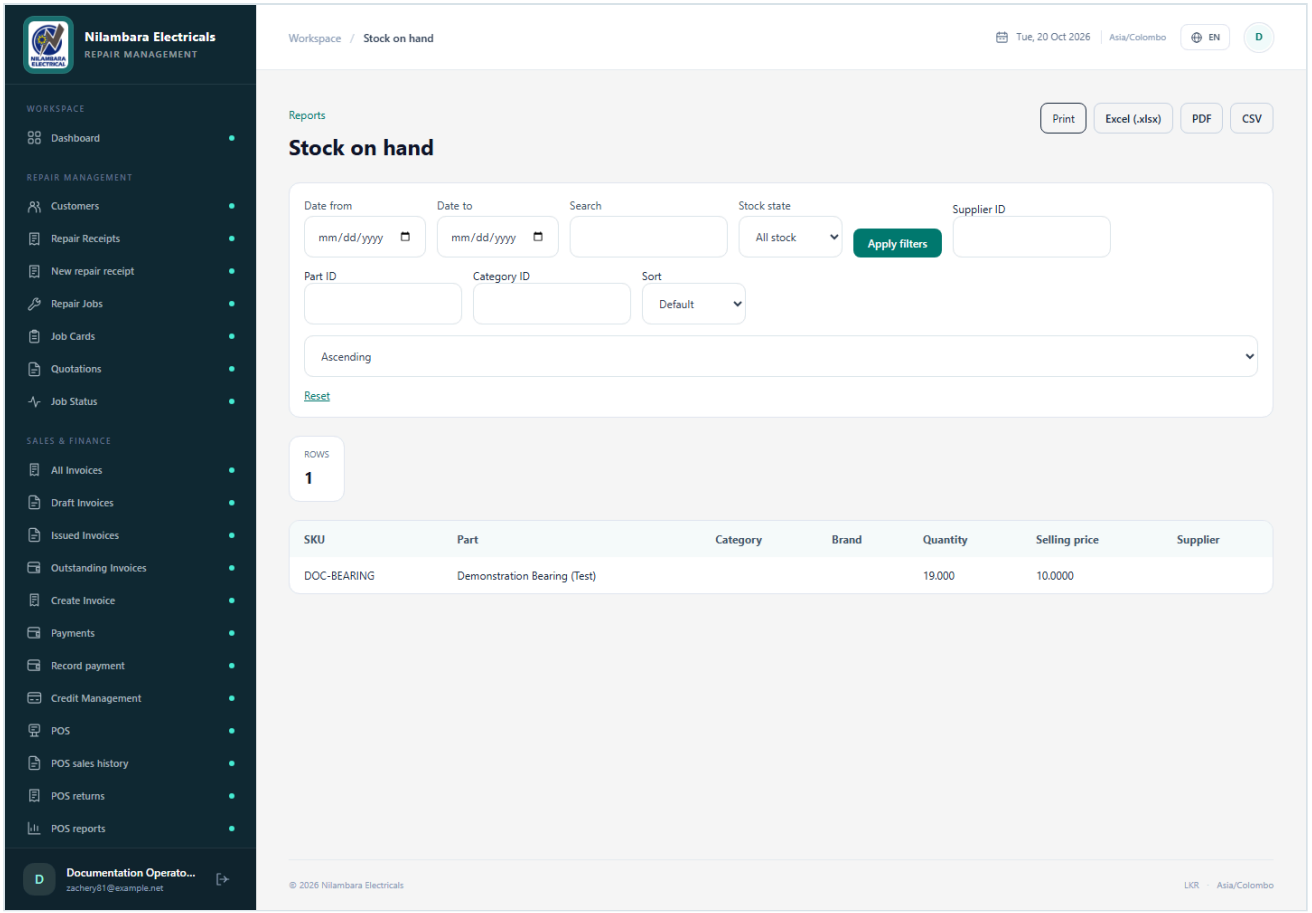
Actual Chrome screenshot | /reports | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/reports.png. No operational records or passwords are displayed.

Figure 38.1 - Stock report and export actions

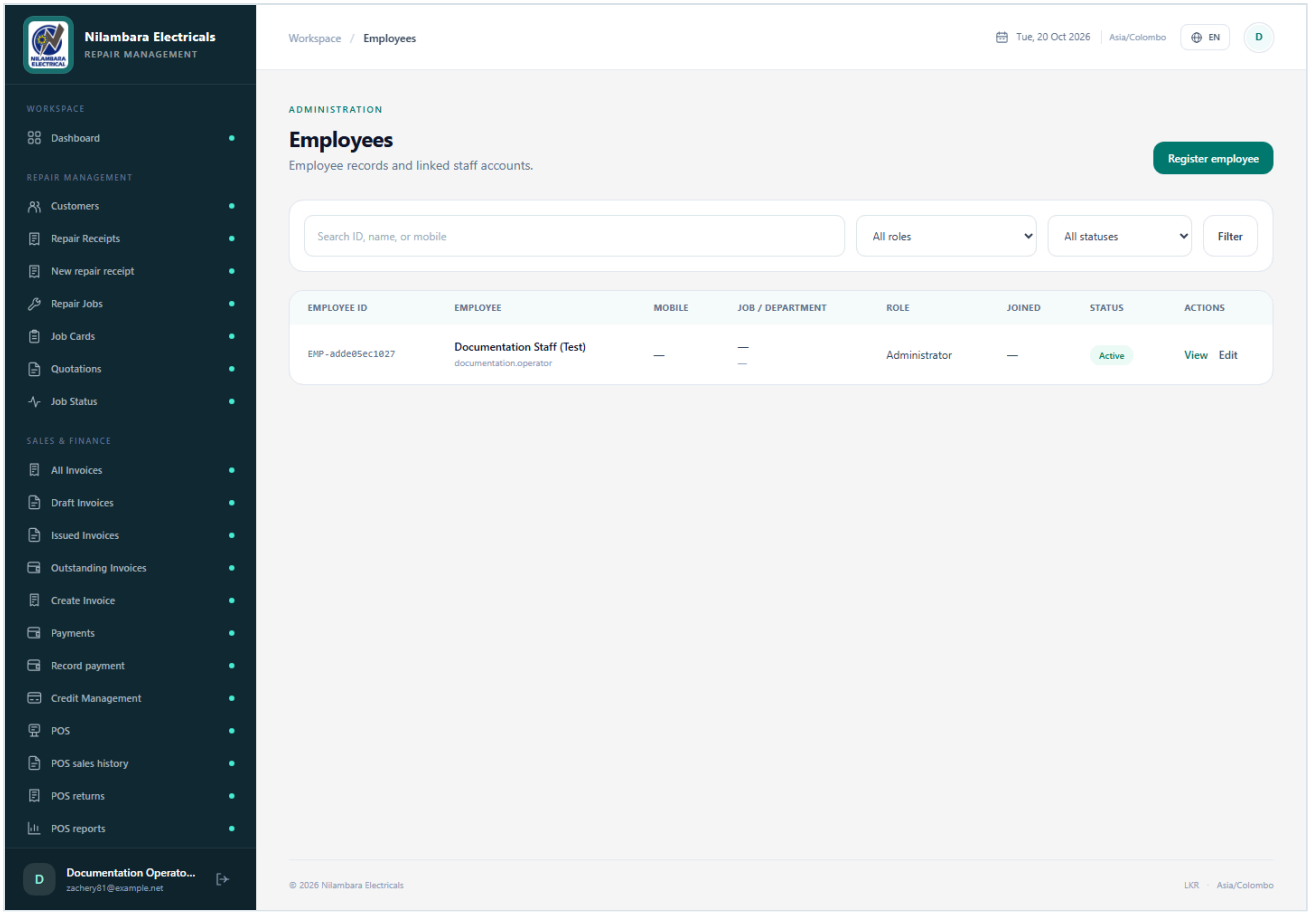
Actual Chrome screenshot | /reports/stock | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/report-detail.png. No operational records or passwords are displayed.

Figure 39.1 - Employee management

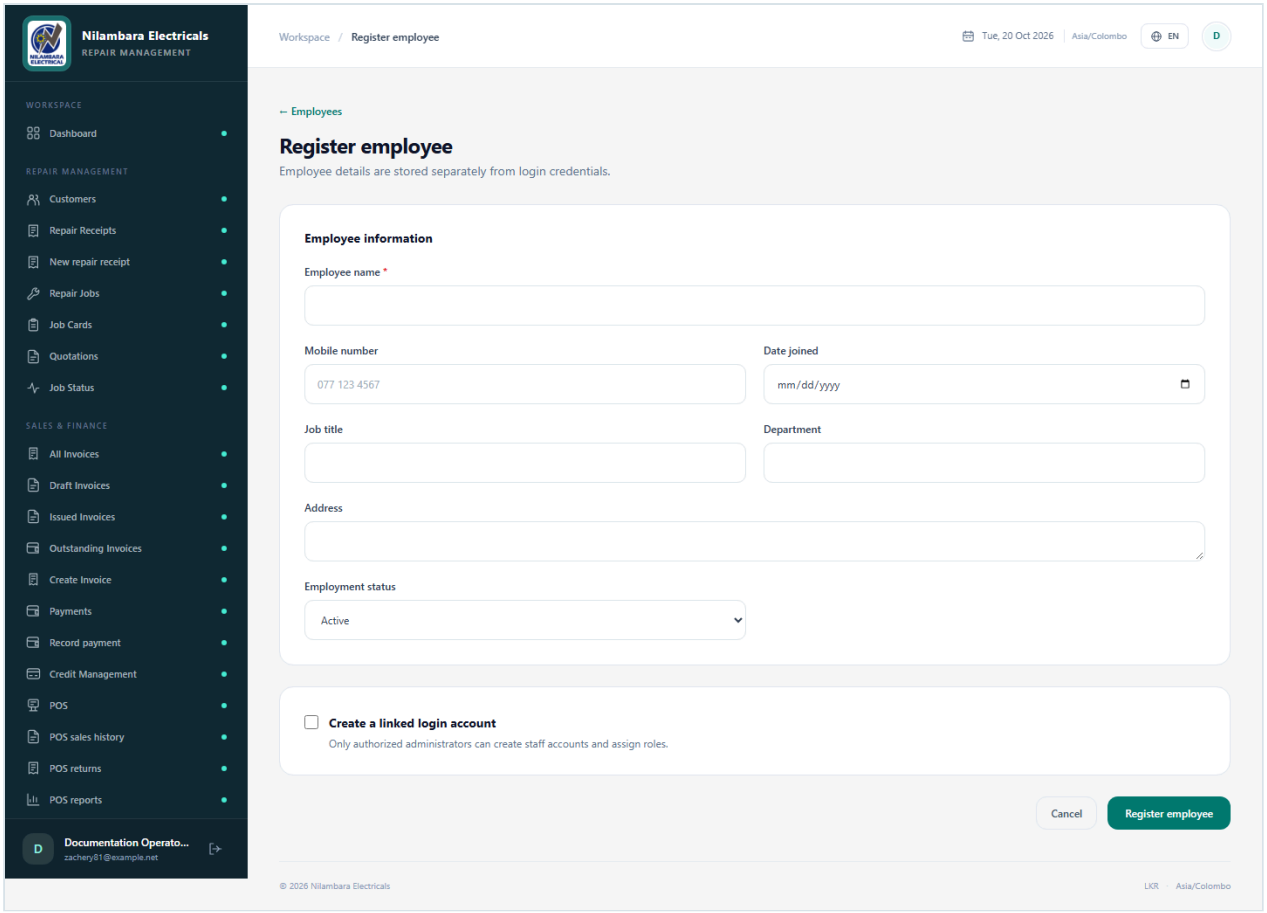
Actual Chrome screenshot | /employees | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/employees.png. No operational records or passwords are displayed.

Figure 40.1 - Employee registration

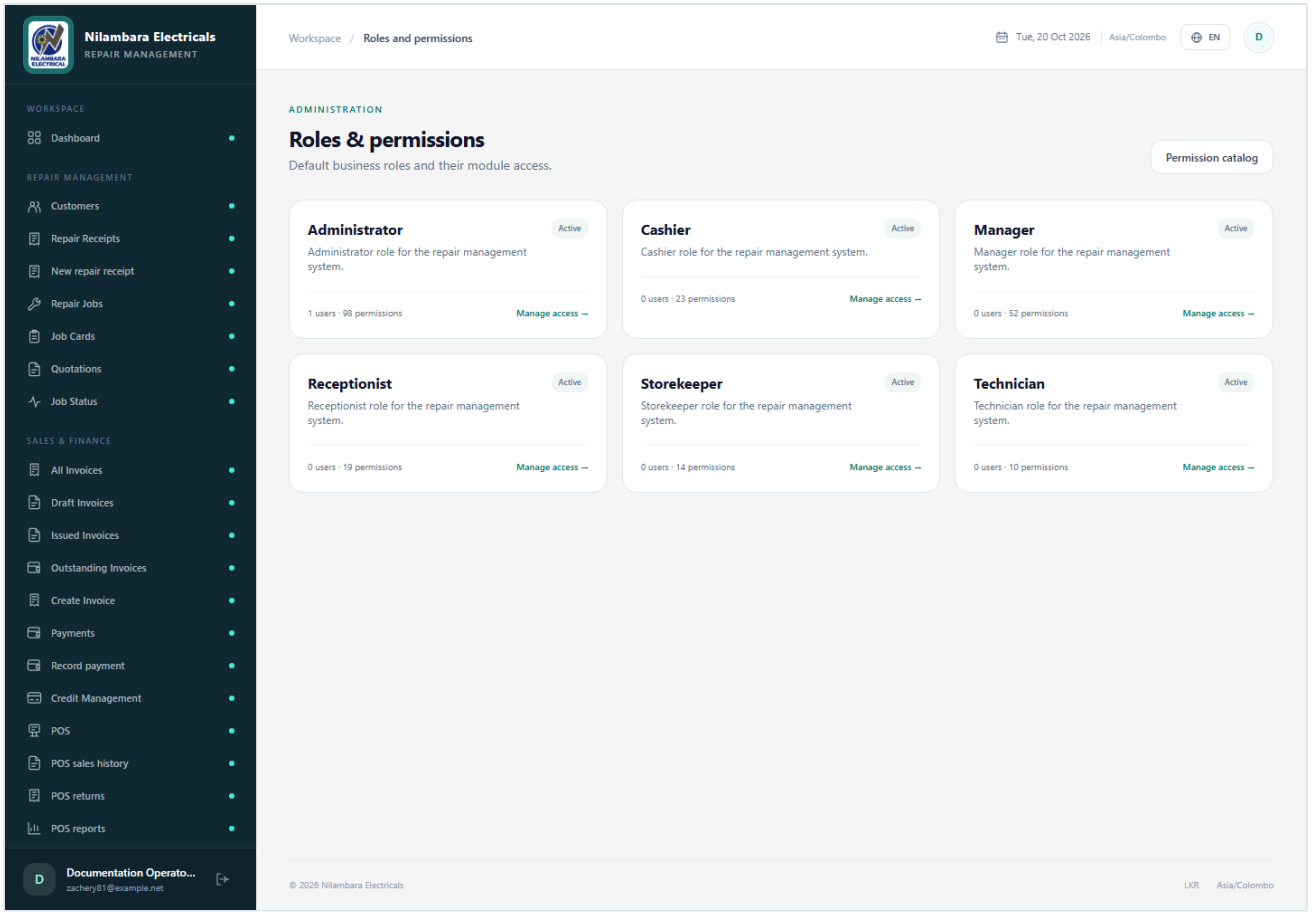
Actual Chrome screenshot | /employees/create | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/employee-create.png. No operational records or passwords are displayed.

Figure 41.1 - Roles and permissions

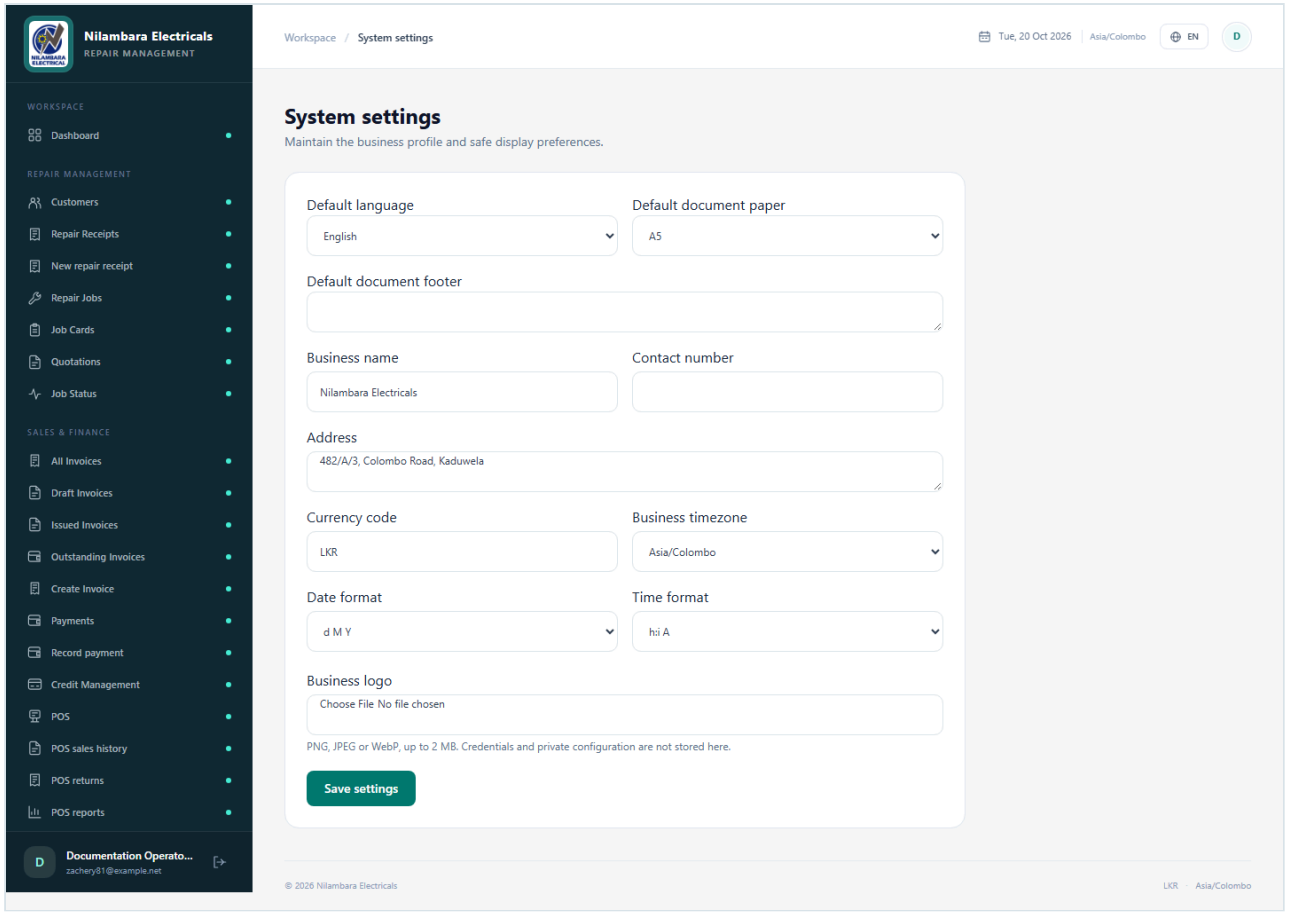
Actual Chrome screenshot | /admin/roles | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/roles.png. No operational records or passwords are displayed.

Figure 42.1 - System settings

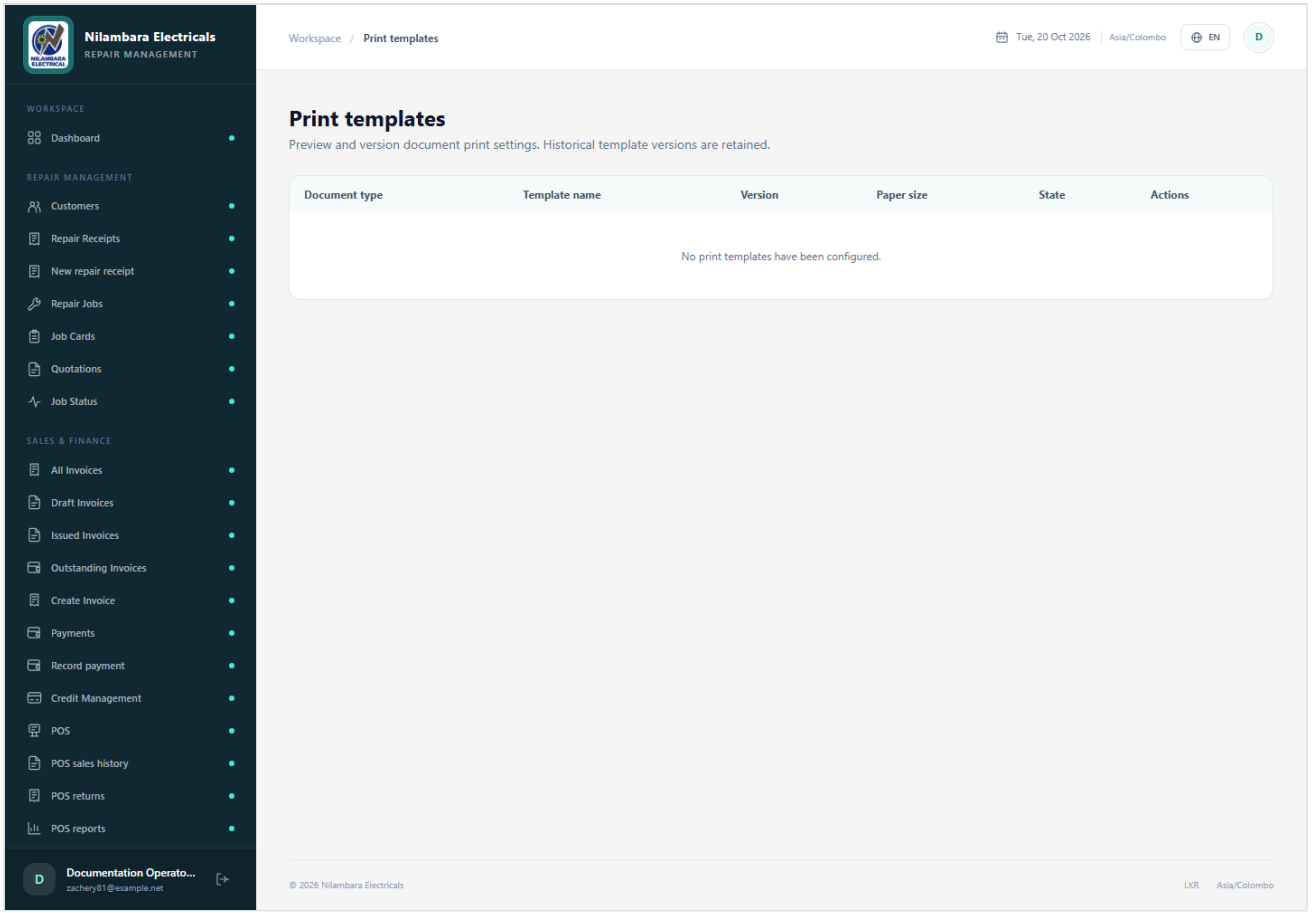
Actual Chrome screenshot | /settings | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/settings.png. No operational records or passwords are displayed.

Figure 43.1 - Print template management

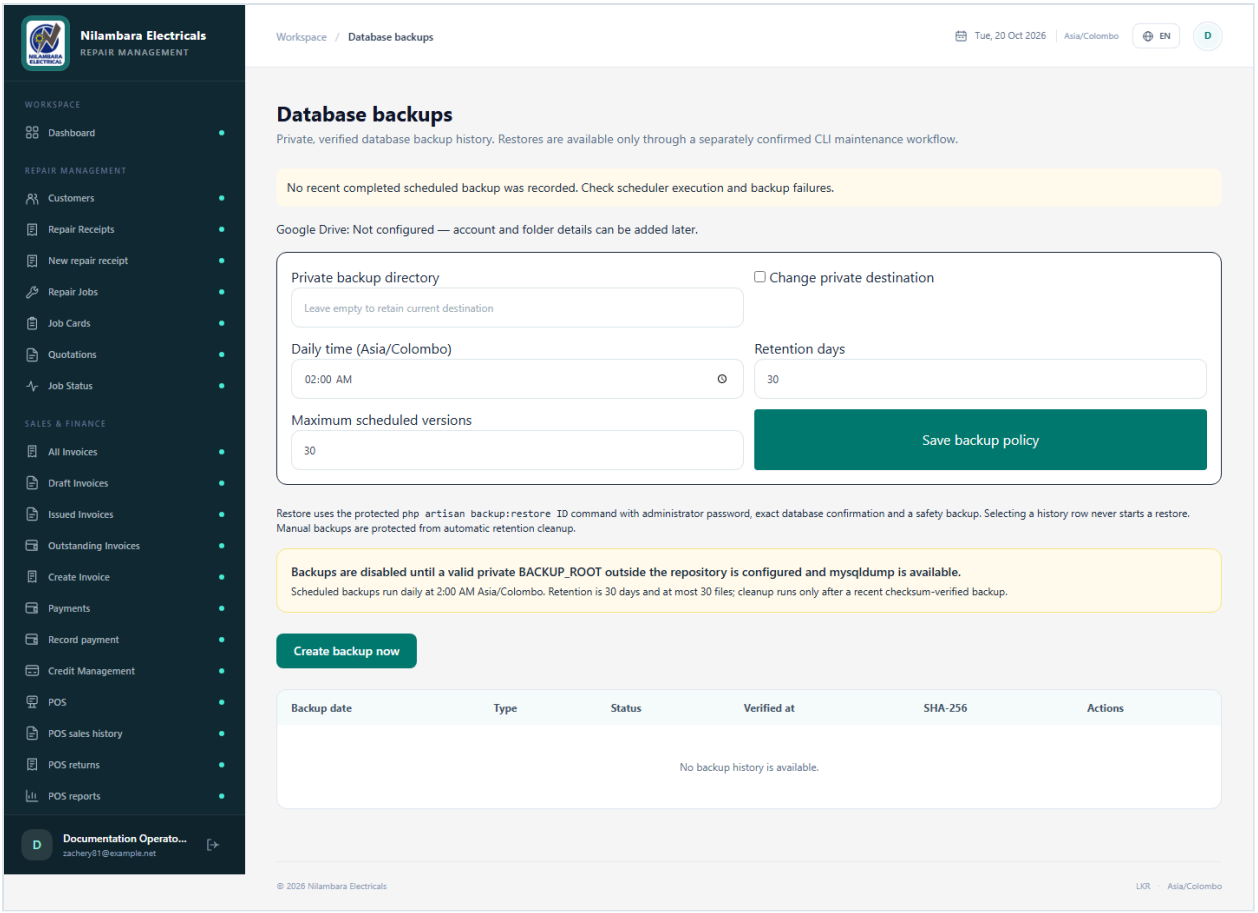
Actual Chrome screenshot | /print-templates | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/print-templates.png. No operational records or passwords are displayed.

Figure 44.1 - Backup management - unconfigured destination

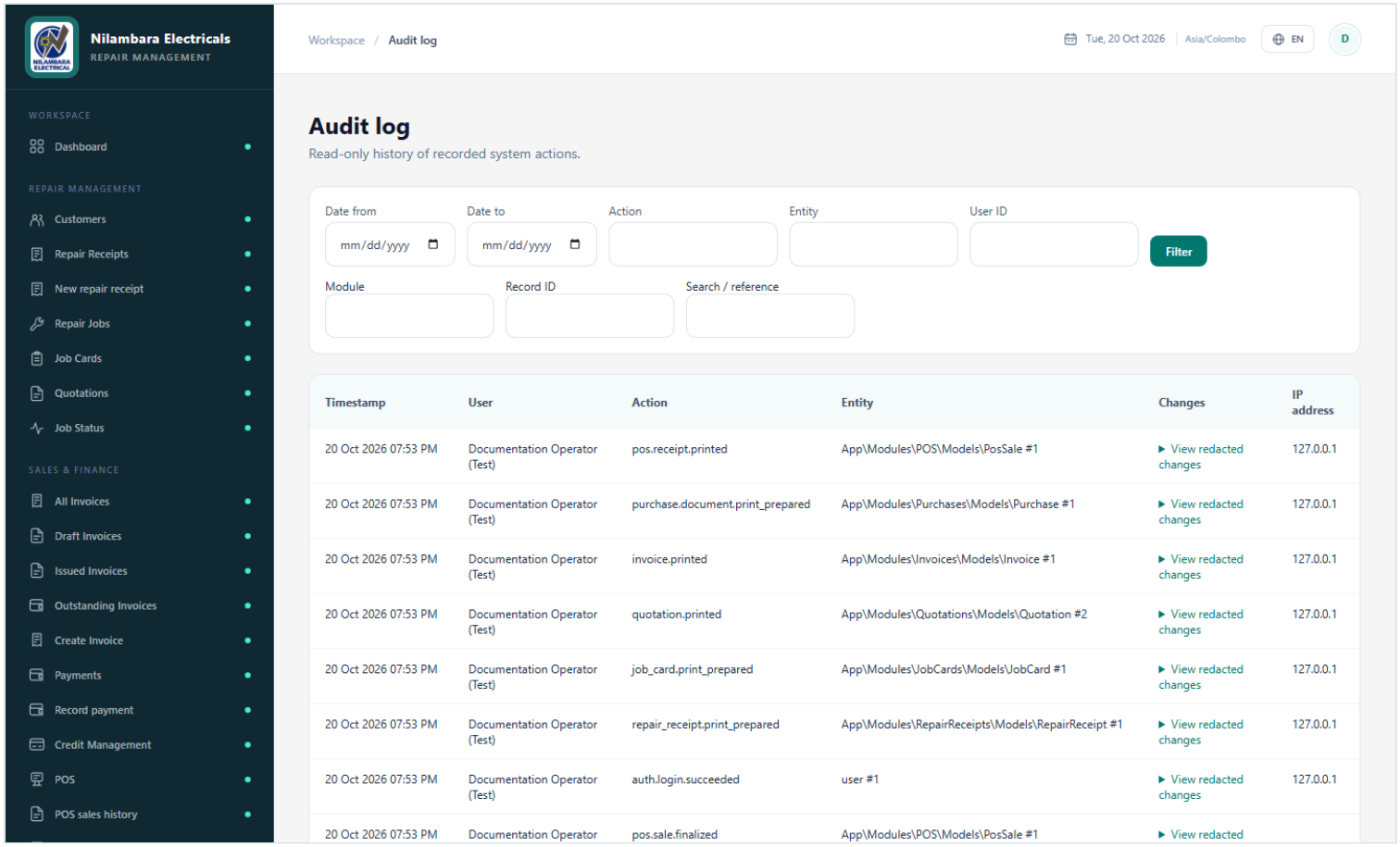
Actual Chrome screenshot | /admin/backups | isolated test fixtures | panel 1/1



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/backups.png. No operational records or passwords are displayed.

Figure 45.1 - Audit log

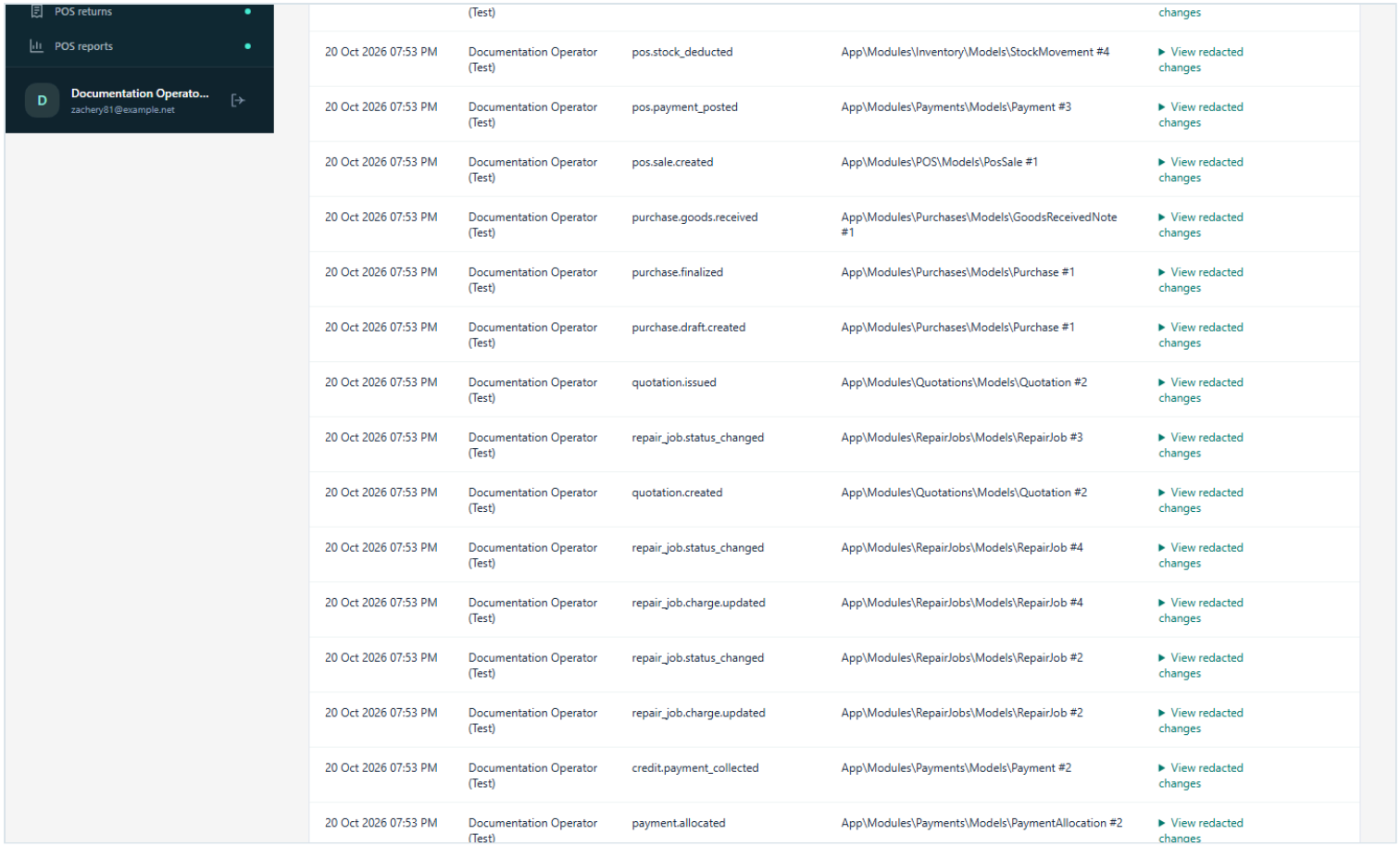
Actual Chrome screenshot | /audit-logs | isolated test fixtures | panel 1/3



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/audit.png. The capture is limited to the first 2600 vertical pixels; content below is not shown.

Figure 45.2 - Audit log

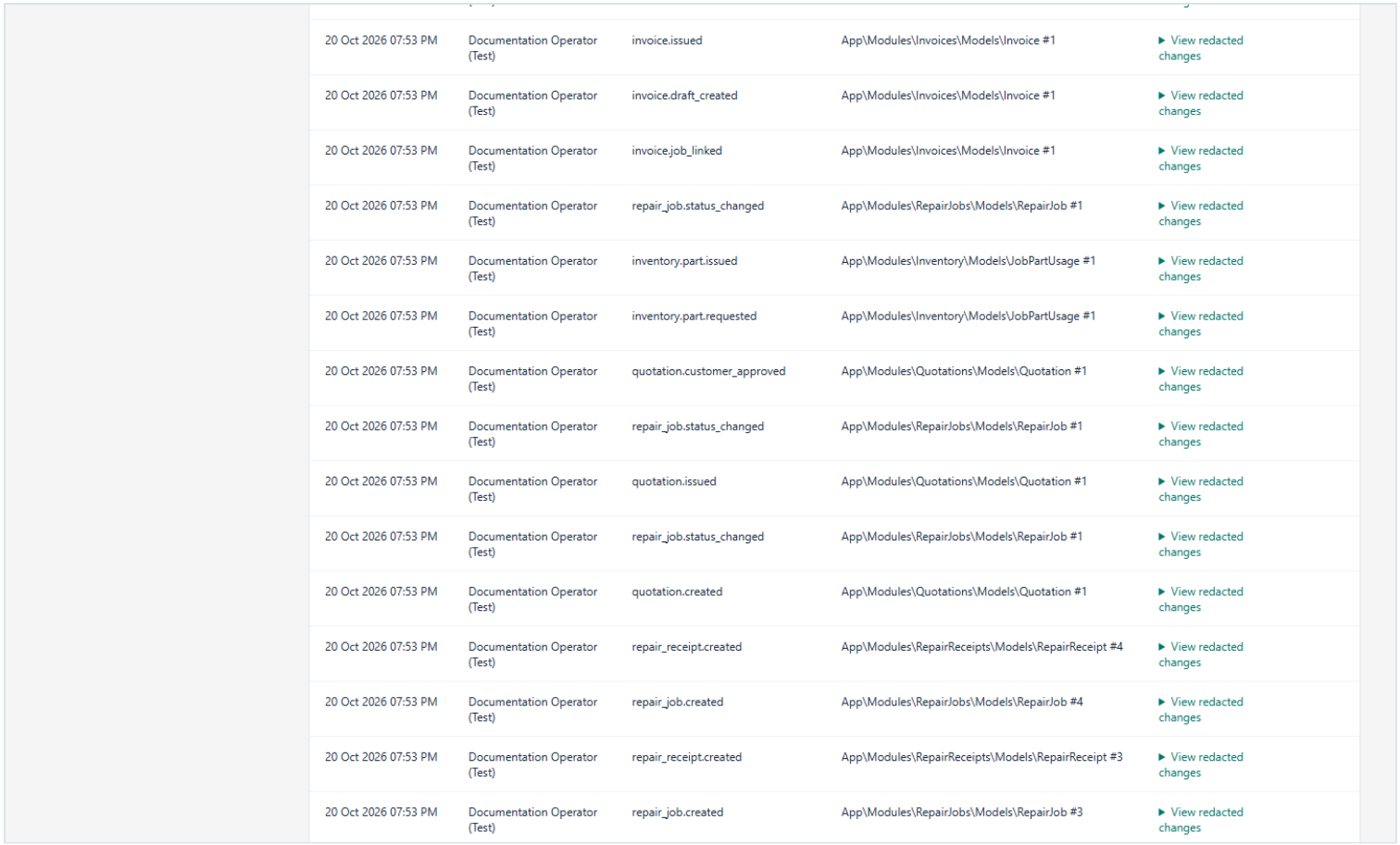
Actual Chrome screenshot | /audit-logs | isolated test fixtures | panel 2/3



Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/audit.png. The capture is limited to the first 2600 vertical pixels; content below is not shown.

Figure 45.3 - Audit log

Actual Chrome screenshot | /audit-logs | isolated test fixtures | panel 3/3



20 Oct 2026 07:53 PM	Documentation Operator (Test)	invoice.issued	App\Modules\Invoices\Models\Invoice #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	invoice.draft_created	App\Modules\Invoices\Models\Invoice #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	invoice.job_linked	App\Modules\Invoices\Models\Invoice #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_job.status_changed	App\Modules\RepairJobs\Models\RepairJob #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	inventory.part.issued	App\Modules\Inventory\Models\JobPartUsage #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	inventory.part.requested	App\Modules\Inventory\Models\JobPartUsage #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	quotation.customer_approved	App\Modules\Quotations\Models\Quotation #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_job.status_changed	App\Modules\RepairJobs\Models\RepairJob #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	quotation.issued	App\Modules\Quotations\Models\Quotation #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_job.status_changed	App\Modules\RepairJobs\Models\RepairJob #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	quotation.created	App\Modules\Quotations\Models\Quotation #1	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_receipt.created	App\Modules\RepairReceipts\Models\RepairReceipt #4	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_job.created	App\Modules\RepairJobs\Models\RepairJob #4	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_receipt.created	App\Modules\RepairReceipts\Models\RepairReceipt #3	▸ View redacted changes
20 Oct 2026 07:53 PM	Documentation Operator (Test)	repair_job.created	App\Modules\RepairJobs\Models\RepairJob #3	▸ View redacted changes

Continuation panels collectively preserve the captured screen. Original PNG is in documentation/screenshots/audit.png. The capture is limited to the first 2600 vertical pixels; content below is not shown.